

Localization costs and information growth for exact Gaussian observations

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Abstract

For the image of a uniform cube under a spherical coordinate map, we prove that finite messages from t blocks of $\Theta(d)$ exact Gaussian measurements reveal only $O_A(dt)$ information when each message has at most $\exp(Ad^2)$ values, for fixed A . The same bound holds when each message is supplemented with a nested cell that restores the required geometric spread.

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1 Introduction

An exact linear observation can convey arbitrarily many bits about a real-valued signal. A finite message computed from several observations need not convey that much, but its information is not controlled by the number of observations alone: the prior distribution matters. If earlier messages have concentrated the signal in a small region, the next message may exploit that concentration.

This paper studies how to restore a useful geometric spread condition while accounting for the information revealed in doing so.

The observation model is noiseless Gaussian regression. The unknown signal is a unit vector $S \in S^{d-1}$. A row $x \sim N(0, I_d)$, independent of the signal and previous rows, arrives together with its exact label $\langle x, S \rangle$. A learner processes these pairs in order. Between observations it retains one of at most 2^M states; arbitrary computation and fresh randomness are allowed during a transition. It stops by a deterministic finite horizon T , and its unit-vector output uses its terminal state, stopping index, and fresh randomness. The precise measurable model, including data-independent shared randomness, is given in Section 2. Angular accuracy ϵ means $\arccos\langle S, \hat{S} \rangle \leq \epsilon$.

Our first information theorem concerns a prior for which the localization cost can be seen directly. Put $n = d - 1$, take Z uniformly in the cube

$$[-1/(2\sqrt{n}), 1/(2\sqrt{n})]^n, \quad S = \phi(Z) = (Z, \sqrt{1 - \|Z\|^2}),$$

and group observations into blocks of $k = \lfloor n/16 \rfloor$ rows. A message from a block may depend on all its rows and labels and on preceding messages. If each message has at most $\exp(Ad^2)$ possible values, with A fixed, then the first t messages contain at most $C_A dt$ nats of mutual information about Z . Theorem 3.2 proves a stronger statement: the same bound holds after each message is supplemented with a nested dyadic cell containing Z that restores the required spread condition. The constant depends on A , and the theorem gives the exact alphabet condition. The prior is the image of uniform cube volume, rather than uniform area on the spherical patch. This theorem serves as a complete first example; each later route establishes its own information estimate from its stated assumptions.

1.1 Why localization can pay for itself

Suppose the current posterior is supported in a dyadic parent cell. It is sufficiently spread for our projection estimate when each descendant at relative depth j has probability at most $B2^{-nj/2}$. A fresh block then carries at most $Cd + \log B + (2/m + e^{-ck}) \log N$ information in a message with N possible values, where $m = \lfloor n/8 \rfloor$. The proof first inserts arbitrary L^2 weights into the posterior, bounds the exact projected density in L^m , and then uses duality to control the message likelihoods.

After the message, we reveal the deepest cell on the sampled point's dyadic chain whose posterior mass is at least $2^{-nj/2}$. At depth j there are at most $2^{nj/2}$ such cells. Naming the selected cell therefore costs about $nj/2$ bits, plus the entropy of its depth. In contrast, confinement to any depth- j cell certifies nj bits relative to the initial uniform cube. This difference pays for the extra revelations. The normalization is by the probability of selecting the cell, not merely the probability of lying in it. Controlling that selection probability is what also pays for the next value of $\log B$.

Section 3 completes this argument and its finite-state application. The next eight arguments change the localization or geometric input, with the distinct estimates in Table 1. The final two routes instead correct the information itself. Each route has its own hypotheses and parameters.

Construction	Estimate or information retained
<i>The cube-chart prior</i>	
Maximum score (Section 4)	A finite maximizing-cell selection, with a one-block information bound charging half the concentration slack.
Finite entropy (Section 5)	Regularization of a possibly unbounded density of finite relative entropy, with finite expected termination and the complete stopping transcript retained.
Bounded-density split (Section 6)	A finite partition and a depth–entropy balance; two independent block copies at the same signal control a squared state likelihood.
Likelihood truncation (Section 7)	A finite-horizon success bound from explicitly bounded exceptional mass and a pointwise density bound on the surviving histories.
<i>The uniform spherical prior</i>	
Grid refinement (Section 8)	Unconditional information in the full history, including the entire selected grid-cell label after each block.
Side projection (Section 9)	Information in the full state history conditional on one fixed Gaussian side observation; predicting adaptive cells bounds the conditional entropy of their labels.
Separated tuples (Section 10)	A terminating radius refinement and an equal-label tuple of controlled affine volume; a sliced-entropy recurrence for at most d blocks.
Predecessor spans (Section 11)	A selected posterior draw and a discrete smaller-ball code, paid by radius reduction, for possibly unbounded finite-entropy posteriors.

Table 1: Eight localization estimates following the dyadic proof. The group labels indicate the priors in the learner applications; the local estimates apply to the conditional laws specified in each section. The side-observation matrix is drawn independently of the learner’s observations and the signal. All auxiliary cells, projections, and posterior draws belong to the analysis; the learner retains only its prescribed finite state.

For the first information correction, let ν be a bounded-density probability on the sphere and put $p = 2\lfloor d/16 \rfloor$. Define

$$J_\nu(s) = \int \|u - s\|^{-p} d\nu(u), \quad \Phi(\nu) = D(\nu\|\sigma) - \int \log J_\nu(s) d\nu(s),$$

where σ is uniform spherical probability and D denotes relative entropy. Section 12 proves that Φ is convex, controls a fixed fraction of relative entropy up to $O(d)$, and grows by at most $O(d)$ per block in its stated width range. Convexity permits averaging away the entering state, so the iteration follows the current padded state, including its terminal label and stopping index. Section 13 instead keeps the complete boundary-state history and subtracts a correction built from the largest cell mass weighted by scale. Its geometric input bounds an inverse simplex volume under local ball bounds with no support restriction. The two corrections use different conditioning variables; neither is identified with the entropy cost of the dyadic revelations.

For $M(d) = o(d^2)$, the learner applications give

$$T(d) \geq cd \log(1/\epsilon(d)), \quad 0 < \epsilon(d) \leq 1/10,$$

with an absolute $c > 0$ and a dimension threshold that may depend on the memory sequence. The cube arguments assume success at least $2/3$ for every signal, which supplies both their cube prior and the full-sphere prior used at constant accuracy. The spherical arguments assume average

success at least $2/3$ for a uniform signal. Each application accounts for the stopping index and the constant-accuracy endpoint. For the controlled-horizon estimate, output counting first puts any putative shorter run within the required range.

1.2 Context and companion inputs

Memory constraints distinguish what a learner can extract from a sample from what it can retain for later samples. Steinhardt and Duchi [13] established memory-sample tradeoffs for noisy sparse regression. Steinhardt, Valiant, and Wager [14] formulated a parity-learning conjecture; Raz [10] proved a quadratic-memory versus exponential-sample separation and subsequently extended the method to broader finite learning problems [11]. For real-valued linear prediction, Dagan, Kur, and Shamir [3, Section 3.3, Theorem 10] proved a quadratic-memory lower bound for approximately solving a consistent linear system whose rows arrive once in random order. Their guarantee concerns residual error on that fixed system, whereas here the signal is fixed before independent Gaussian samples are drawn.

Sharan, Sidford, and Valiant [12] studied the statistical Gaussian-regression problem with small positive observation noise. Their Theorem 1 gives a sample lower bound of $\Omega(d \log r)$ for Euclidean error $\epsilon = d^{-r}$, memory at most $d^2/4$ bits, and $r \leq O(d/\log d)$. Their Section 7, Lemma 12 expands projection moments into independent signal copies and controls them by successive orthogonalization. This is a close methodological predecessor of the projection arguments below. We prove the exact multirow density estimates, their limiting passages, and the costs of restoring geometric spread locally. The separate condition-number conjecture in their Section 1.1 concerns exact labels and ill-conditioned Gaussian covariance at constant error; the covariance here is the identity and the lower bound tracks the required accuracy.

The information layer uses relative entropy on standard Borel spaces. Austin [1] gives the conditional chain rules and product decompositions that make these identities meaningful without a finite signal alphabet. The variational inequality used to charge selected laws is recorded in [5]; at each use we prove the necessary integrability or truncation step for the unbounded logarithm. The tuple arguments use total correlation in Watanabe’s terminology [15]. Spherical slicing is related to Federer’s coarea formula [6], and Section 10 gives the specific surface-area calculation. Its subset-entropy comparison credits the entropy covering argument of Chung, Graham, Frankl, and Shearer [2]; the conditional differential-entropy argument is proved here. The predecessor-span code uses Kraft’s inequality [7], with the countable form in [4, Theorem 2.4.2], followed by a local conditional entropy calculation.

Only two long analytic inputs are taken from companion articles. The finite-measure identity of [8, Theorem 3.3] supplies the exact equal-label reference in Section 11. We apply its uniform spherical case at two separate row counts and prove the finite-entropy extension locally. The mixed projection moment of [9, Theorem 3.1] supplies the positive-scale estimate in Section 12. We verify its two local-mass bounds after reweighting, change the actual-row law while the scale is positive, and then take the exact-label limit under the actual law. The remaining localization arguments are proved in this paper.

1.3 Reading the arguments

Sections 2 and 3 give a complete first proof: the model, measurable-rule reduction, information conventions, and endpoint, followed by dyadic localization and its learner application. The four other cube routes require this common model and the cube geometry of Lemma 3.1, but not the preceding cube proofs. The spherical routes can be read after the common model. In particular, the

side-projection argument uses the statement of Corollary 8.4, not the grid proof, and the two tuple arguments are independent of each other.

Only the predecessor-span and kernel routes use the companion inputs described above, stated locally in Proposition 11.1 and Theorem 12.4. The other geometric and localization estimates are proved here. Each route fixes its parameters at its outset, and its final application states the prior and stopping conventions used for the sample bound.

2 Finite memory, priors, and information

We specify the observation model once, before applying the different localization estimates. A signal $s \in S^{d-1}$ is fixed before sampling. At index j the learner receives

$$x_j \sim N(0, I_d), \quad y_j = \langle x_j, s \rangle,$$

with independent rows x_j . It cannot choose a row or revisit an earlier observation. Its persistent state at each index lies in a finite set of cardinality at most 2^M . Given that state and the entire current pair (x_j, y_j) , a transition may use unrestricted computation and fresh randomness. All information retained for later indices must be represented by the new state.

The learner stops by a prescribed deterministic horizon $T < \infty$, possibly before taking a sample. Its unit-vector output uses only the terminal state, the stopping index, and fresh randomness. The rules and output laws may also depend on data-independent shared randomness chosen before sampling, as well as on d , the prescribed accuracy, and the index; they cannot depend on the unknown signal. The shared rule randomness and the initial finite state are jointly independent of the signal and all sample rows. The initial state may depend on the shared randomness, but conditional on that randomness it carries no signal or sample information. In particular, the last observation affects the output only through the terminal state. This is a bit-state model, not a model with persistent real registers.

For a unit-vector estimate $\hat{S}$, angular success at accuracy ϵ is the event $\arccos\langle S, \hat{S} \rangle \leq \epsilon$. It implies $\|S - \hat{S}\| \leq \epsilon$, since the chord length at angle θ is $2\sin(\theta/2) \leq \theta$. Each application states whether success is averaged under a specified prior or required for every fixed signal.

As part of the randomized-learner model, the initial variables, sequential finite choices, and output form a jointly measurable experiment in the signal, the sample rows, and the randomness. Consequently prior mixing and conditioning on the shared randomness refer to measurable joint laws. The next regularity allowance concerns the coordinate kernels of that experiment at a fixed shared seed.

For each fixed realization of the shared rule randomness, the transition and stopping probabilities may be measurable in the completion of the Borel sigma field under

$$\lambda(dx, dy) = \gamma_d(dx) dy,$$

where γ_d is standard Gaussian probability on $\mathbb{R}^d$. The terminal output laws are Borel probability measures on the sphere. The next lemma explains why reference experiments may nevertheless evaluate state rules at every matrix and label.

Lemma 2.1 (Borel rules under an absolutely continuous prior). *Let P be a probability measure on S^{d-1} such that, for independent $S \sim P$ and $x \sim \gamma_d$, the pair $(x, \langle x, S \rangle)$ has law absolutely continuous with respect to λ . Fix a realization of the shared rule randomness and a finite horizon. Every completed-measurable finite-state learner has everywhere-defined Borel transition and stopping kernels with the same state sets and output laws and the same P -prior joint law of the signal, state path, stopping index, and output.*

Proof. At a fixed state and index, combine continuation and stopping choices into one finite set of tagged destinations. Choose Borel versions of their completed-measurable coordinate probabilities. Such versions exist: measurable simple approximations may have each of their countably many level sets replaced by a Borel set modulo a null set. Outside a Borel λ -null set the versions agree with the original coordinates and form a probability vector. On the exceptional set use a fixed point mass. This gives nonnegative Borel coordinates whose sum is one everywhere.

There are finitely many states and indices, so a single Borel λ -null set contains every modification. Pregenerate all sample pairs through the horizon, including pairs ignored after stopping. Each of their unconditional marginals avoids this null set by the hypothesis on P . A finite union over indices therefore shows that no modified pair occurs almost surely. This argument uses the unconditional pair law; it does not require a positive density after conditioning on a state selected from past samples.

Couple the old and new finite kernels by the same fresh uniforms. Induction gives identical state paths and stopping decisions almost surely. At an identical terminal state and index the output laws are unchanged and can be coupled identically. This proves the joint-law claim. $\square$

Lemma 2.2 (Sphere and regular cube-chart priors). *The hypothesis of Lemma 2.1 holds for uniform probability σ on S^{d-1} , $d \geq 2$. It also holds for $S = \phi(Z)$ when Z has a density relative to Lebesgue measure on a bounded q -dimensional cube, $q \geq 1$, and ϕ is C^1 on a neighborhood of its interior with $\text{rank } D\phi(z) \geq 1$ for almost every z .*

Proof. For fixed $x \neq 0$, rotational invariance makes $\langle x, S \rangle / \|x\|$ the first coordinate of a uniform spherical point. Its density on $(-1, 1)$ is proportional to $(1 - t^2)^{(d-3)/2}$; this also is integrable for $d = 2$. For example the formula follows by writing the point as a normalized standard Gaussian and separating its first coordinate from the remaining squared length. Thus the conditional label law has a Lebesgue density for every nonzero x . Fubini, discarding $x = 0$, proves the first assertion.

For a chart put $f_x(z) = \langle x, \phi(z) \rangle$. At every regular z , the set of x with $\nabla f_x(z) = D\phi(z)^\top x = 0$ is a proper linear subspace and has Gaussian measure zero. Fubini shows that for Gaussian-almost every x , the critical set of f_x has Lebesgue measure zero in the cube. Around every other interior point, choose a coordinate in which a partial derivative is nonzero. The inverse function theorem makes f_x , together with the remaining coordinates, a C^1 local coordinate system. In each such neighborhood the preimage of a Lebesgue-null subset of $\mathbb{R}$ has zero q -dimensional volume, by Fubini and change of variables. A countable cover of the regular set proves absolute continuity of the label pushforward. The critical set and cube boundary are null. Integrating over x proves the second assertion. $\square$

When a proof fixes randomness under a chosen prior, the order matters. If the original success is at least p and $q < p$, first fix the shared rule randomness with prior success greater than q . Apply Lemma 2.1 for that fixed rule and that prior. Then, the conditional initial-state law remains independent of the signal and rows. If a deterministic program is useful, sample that initial state, sample the finite choices by uniforms indexed by state and index, and sample the output law for each terminal state and index. These variables are independent of the inputs. Averaging fixes them with prior success at least q .

Alternatively, when the same numerical success bound is proved at each fixed shared seed, apply the Borel replacement separately there and transfer the bound back to the original conditional prior law. The original jointly measurable experiment makes those conditional success probabilities measurable, so their bound may be integrated over the seed. Neither argument requires a jointly chosen family of Borel replacements. A replacement for one prior preserves that prior experiment, rather than every individual signal.

Blocks and the stopping index. Starting from a known state, composition of the everywhere Borel kernels gives a Borel probability kernel on the complete data of a block. A block of b rows can report its continuing state, or its terminal state and a stopping offset in $\{0, \dots, b\}$, or a dummy value if it was already inactive. Thus

$$N_b \leq (b+2)2^M + 1$$

is a valid local alphabet bound. A proof may retain a larger displayed bound from its own convention. The sequence of block indices and local offsets determines the global stopping index. Pregenerating unused suffixes keeps every new block independent of the past. Auxiliary locations or projections introduced below are part of the analysis; they are not supplied to the learner.

Information conventions. All logarithms are natural unless a dyadic scale is explicitly written with $\log_2$. For probabilities $P \ll Q$ on a standard Borel space,

$$D(P\|Q) = \int \log(dP/dQ) dP,$$

with value $+\infty$ without absolute continuity. For a discrete variable W , $H(W) = -\sum_w P(W=w) \log P(W=w)$, where $0 \log 0 = 0$. Mutual information is $I(U;W) = D(P_{U,W}\|P_U \otimes P_W)$; conditional quantities average the corresponding conditional relative entropies. We use the chain rule and data processing on standard Borel spaces in the conditional-relative-entropy formulation of [1, Lemma 3.6 and Sections 3.3–3.4].

Two elementary consequences will be used repeatedly. If an event has probability b under P and b_0 under Q , then

$$D(P\|Q) \geq b \log(1/b_0) - \log 2. \quad (1)$$

Indeed data processing gives binary relative entropy, and the sum of $b \log b + (1-b) \log(1-b)$ is at least $-\log 2$, while the remaining term $-(1-b) \log(1-b_0)$ is nonnegative. If $P \ll Q$, $a > 0$, and $F > 0$, then

$$\mathbb{E}_P \log F \leq \frac{D(P\|Q) + \log \mathbb{E}_Q F^a}{a} \quad (2)$$

whenever the left side is defined and the right side is finite. For a bounded logarithm, normalize $F^a Q$ and use nonnegativity of relative entropy. Apply this argument to clipped logarithms and use monotone truncation of the positive part and integrability of the negative part for the displayed form. This is the entropy variational inequality; the bounded-measurable probability version is recorded in [5, Equations (1.1)–(1.2)].

Relative entropy to a finite nonzero measure η means $\int \log(dP/d\eta) dP$. If $c = \eta(\Omega)$, it equals $D(P\|\eta/c) - \log c$. Therefore it contracts under a measurable map: normalization adds the same constant before and after the map. This observation is used only where the comparison measure and all relevant logarithmic integrals are specified.

Lemma 2.3 (Spherical caps and a full-data endpoint). *Write $n = d - 1$. Uniform spherical probability satisfies*

$$\sigma(B(u, r)) \leq r^n \quad (u \in S^{d-1}, r > 0), \quad \sigma(B(z, r)) \leq (2r)^n \quad (z \in \mathbb{R}^d, r > 0).$$

If $S \sim \sigma$, $0 \leq T < d$ independent Gaussian rows and all their exact labels are revealed, and an estimate uses these data and independent randomness, then for every $0 < \epsilon < r_0 < 1$,

$$\mathbb{P}\{\|\hat{S} - S\| \leq \epsilon\} \leq \frac{1}{2} + \frac{1}{2} \mathbb{P}\{\|P_{\ker X} S\| \leq r_0\} \leq \frac{1}{2} + \frac{T}{2d(1-r_0^2)}. \quad (3)$$

Proof. For $0 < r \leq 1$, a chordal cap about u has angular radius $\theta \leq \pi/3$, $\sin \theta \leq r$, and $\cos \theta \geq 1/2$. Its normalized area is

$$\frac{\int_0^\theta \sin^{n-1} v \, dv}{\int_0^\pi \sin^{n-1} v \, dv}.$$

The numerator is at most $2r^n/n$ after substituting $w = \sin v$; the denominator is at least $2/n$, since $\int_0^{\pi/2} \sin^{n-1} v \, dv \geq \int_0^{\pi/2} \sin^{n-1} v \cos v \, dv = 1/n$. For $r \geq 1$ the same bound is trivial. An ambient ball meeting the sphere is contained, on the sphere, in a radius- $2r$ ball about any point of its intersection, proving the second bound.

The row span of X has dimension T almost surely. Its labels specify the row-space projection of S . Conditional on the rows and labels, the remaining component is uniform on its sphere in $\ker X$, with radius $R = \|P_{\ker X} S\|$. To see this, write S as a normalized standard Gaussian and separate its independent row-space and nullspace components; the direction of the latter is independent of both lengths and of the former component.

When $R > r_0 > \epsilon$, the set of residual directions that place the signal within ϵ of a fixed estimate cannot contain both members of any antipodal pair. Its conditional mass is at most $1/2$, including after averaging the estimate's independent randomness. Finally, $\mathbb{E}\|P_{\text{row } X} S\|^2 = T/d$ by rotational invariance. Markov's inequality applied to $\|P_{\text{row } X} S\|^2 = 1 - R^2$ proves (3). $\square$

The constants used later follow directly from this proof. For success at least $2/3$ and $\epsilon \leq 1/10$, the same calculation with the strict event $R < 1/2$ gives $T > d/4$: if $T \leq d/4$, then $\mathbb{P}(R < 1/2) < 1/3$ and success is less than $2/3$. The weaker $T > d/8$ follows by taking $r_0 = 1/\sqrt{2}$, which gives success at most $5/8$ when $T \leq d/8$. For a fixed program with success at least $3/5$, take the event $R \leq \epsilon$ directly. Its probability is at most $T/[d(1 - \epsilon^2)]$, so $T \leq d/20$ would give success strictly below $3/5$. These statements grant the entire stream and therefore include early stopping.

3 Dyadic localization and its information cost

We first prove the repeated-localization estimate that motivates the paper. The prior is the image of a uniform cube under a spherical coordinate map, not uniform surface measure on the patch.

Put

$$n = d - 1, \quad k = \lfloor n/16 \rfloor, \quad m = \lfloor n/8 \rfloor,$$

and assume d is sufficiently large. Let μ_0 be normalized Lebesgue measure on the half-open cube

$$Q_0 = [-1/(2\sqrt{n}), 1/(2\sqrt{n})]^n, \quad \phi(z) = (z, \sqrt{1 - \|z\|^2}).$$

Thus $\phi(Q_0) \subset S^{d-1}$. Half-open cells give each point a unique cell at every level.

Lemma 3.1 (Geometry of the cube chart). *The map ϕ is 2-Lipschitz, and a level- J dyadic cell has side $2^{-J}/\sqrt{n}$ and μ_0 -mass 2^{-nJ} . For every fixed $v \in S^{d-1}$ and $\epsilon > 0$,*

$$\mu_0\{z : \arccos\langle \phi(z), v \rangle \leq \epsilon\} \leq (\sqrt{2\pi e} \epsilon)^n \leq (5\epsilon)^n.$$

Proof. On Q_0 , $\|z\| \leq 1/2$, so the gradient of $\sqrt{1 - \|z\|^2}$ has norm at most $1/\sqrt{3}$. This proves the Lipschitz bound; the cell formulas follow from the side length and normalized cube volume. Angular error at most ϵ implies chordal error at most ϵ , so the first n coordinates lie in the radius- ϵ ball about the projection of v . Its normalized cube volume is at most $n^{n/2} v_n \epsilon^n$. The Gaussian integral on the radius- $\sqrt{n}$ ball gives $v_n \leq (2\pi e/n)^{n/2}$, proving the claim. $\square$

Draw $Z \sim \mu_0$. In block i , independently draw a standard Gaussian $k \times d$ matrix X_i and observe $(X_i, X_i \phi(Z))$. A message W_i takes at most N values. Its conditional law may be any measurable function of that exact block and the preceding messages; independent randomization is allowed. More precisely, for each fixed preceding message history the rule is an everywhere-defined Borel probability kernel on $\mathbb{R}^{k \times d} \times \mathbb{R}^k$: its coordinates are nonnegative and sum to one at every (X, y) . Euclidean spaces and the sphere carry their Borel sigma fields. Neither the matrices nor their labels are automatically retained in the message history.

Information and relative entropy use the conventions of Section 2.

Theorem 3.2 (Repeated localization). *There is an absolute constant $c > 0$ with the following property. For any fixed $C_0 > 0$, consider the experiment above with $t \geq 0$ blocks and $N \geq 1$ possible message values, and suppose*

$$(2/m + e^{-ck}) \log N \leq C_0 d. \quad (4)$$

One can reveal nested dyadic cells $Q_0 \supset Q_1 \supset \dots \supset Q_t$ containing Z , where Q_i is a finite-valued measurable function of Z , the preceding revelations, and $W_1, \dots, W_i$, so that the augmented transcript

$$\Pi_t = (W_1, Q_1, \dots, W_t, Q_t)$$

satisfies

$$I(Z; \Pi_t) \leq C dt. \quad (5)$$

Here C is a constant depending only on C_0 . If J_t is the level of Q_t , the construction also gives $\mathbb{E} J_t \leq Ct$.

For fixed A , an alphabet of size at most $\exp(Ad^2)$ satisfies (4) with a constant depending only on A . The theorem allows arbitrary rules inside a block; it restricts only what a block passes to the next one. Since the original message history is a function of Π_t , it inherits the same information bound.

The proof has three steps. Section 3.1 shows that if a posterior assigns little mass to every small dyadic cell, exact Gaussian projection has controlled density even after an L^2 weight is inserted. Section 3.2 turns that estimate into a bound on the information in one finite message. Section 3.3 selects a deepest cell whose posterior mass is large relative to its scale and bounds the entropy of that selection. The information chain rule then closes the iteration in Section 3.4. Section 3.5 gives a complete application to finite-state learners, including bounded stopping.

3.1 A weighted density for exact projections

We first work inside one dyadic cell and put $\alpha = 1/2$. The hypothesis below measures concentration at every smaller scale. Its role is to prevent independent points from accumulating near a low-dimensional affine subspace.

The weight in the coming estimate has a specific information-theoretic purpose. A message's probability, after restricting labels to a bounded ball, is a function of the signal. Testing that function against every L^2 weight of norm at most one bounds its L^2 norm by duality. Jensen's inequality then controls the message's relative-entropy contribution using this likelihood norm. Section 3.2 will carry out these two steps. We therefore need a projection-density bound that is uniform over the weights, not just a density bound for the unweighted posterior.

Let Q be a level- J cell and let μ be a probability measure supported in Q . For $B \geq 1$, assume

$$\mu(R) \leq B 2^{-\alpha n l} \quad \text{for every level-}(J+l) \text{ descendant } R \text{ of } Q \text{ and } l \geq 0. \quad (6)$$

Write $r = 2^{-J}$, $c_Q = \phi(\text{center } Q)$, and

$$a(z) = \frac{\phi(z) - c_Q}{r}.$$

Then $\|a(z)\| \leq 2$. Let P_X denote the law of a standard Gaussian $k \times d$ matrix.

Lemma 3.3 (Weighted exact projection). *Under (6), for every measurable h with $\|h\|_{L^2(\mu)} \leq 1$, put $\nu = |h|\mu$. The image of $P_X(dX)\nu(dz)$ under $(X, z) \mapsto (X, Xa(z))$ has a density f_ν relative to $P_X(dX)du$, where du is Lebesgue measure on $\mathbb{R}^k$, and*

$$\|f_\nu\|_{L^m(P_X du)} \leq C^d B^{1/2} \quad (7)$$

for an absolute constant C .

Proof. We first derive a tube bound from (6), then estimate a smoothed projection by expanding its m th moment. Finally we pass to the exact measure. The replica expansion and Gaussian orthogonalization have a close predecessor in [12, Section 7, Lemma 12]; the weighted multirow estimate and the exact-measure passage are proved here.

For every affine q -plane $V \subset \mathbb{R}^d$ with $q \leq m - 2$, we claim

$$\mu\{z : \text{dist}(a(z), V) \leq \delta\} \leq BC^d \delta^{\alpha n - q} \quad (0 < \delta \leq 1). \quad (8)$$

Projection onto the first n coordinates shows that every such z lies within δr of the projection of $c_Q + rV$. Only the part of that projected plane within $2r$ of the center of Q is needed. A maximal δr -separated subset of that part has at most $(C/\delta)^q$ points, by volume comparison in the plane. Balls of radius $2\delta r$ about these points cover all relevant z .

Choose $l \geq 0$ so that $2^{-l} \leq \delta < 2^{-l+1}$. Each covering ball meets at most C^n level- $(J+l)$ cells. Indeed those cells are disjoint, have side greater than $\delta r/(2\sqrt{n})$, and lie inside a ball of radius $3\delta r$. If v_n is the volume of the unit ball in $\mathbb{R}^n$, the estimate $v_n \leq (2\pi e/n)^{n/2}$ gives the claimed count; the powers of $\sqrt{n}$ cancel. Each such cell has mass at most $B2^{-\alpha nl} \leq B\delta^{\alpha n}$, proving (8).

By Cauchy-Schwarz, ν has mass at most one. For $\eta > 0$ define

$$F_\eta(X, u) = \int (2\eta)^{-k} \mathbf{1}_{\{\|u - Xa(z)\|_\infty \leq \eta\}} d\nu(z).$$

This is smoothing in the label variable only. Expanding its m th power introduces points $z_1, \dots, z_m$. Put $a_i = a(z_i)$, define D_i for $2 \leq i \leq m$, and set

$$D_i = \text{dist}(a_i, \text{aff}(a_1, \dots, a_{i-1})), \quad D = [a_2 - a_1, \dots, a_m - a_1], \quad V_*^2 = \det(D^T D).$$

The exponent in (8) is positive. Thus each affine span in this display has zero μ -mass and zero ν -mass. Almost every tuple is affinely independent, and Gram-Schmidt gives $V_* = \prod_{i=2}^m D_i$.

The intersection of the m smoothing boxes has volume at most $(2\eta)^k$ and is empty unless $\|X(a_i - a_1)\|_\infty \leq 2\eta$ for every $i \geq 2$. For a fixed affinely independent tuple, one row of these differences is Gaussian with covariance $D^T D$ and maximum density $(2\pi)^{-(m-1)/2} V_*^{-1}$. The k rows are independent. Bounding the probability of the difference box by its volume times this maximum density, and using Tonelli's theorem, yields

$$\int F_\eta^m dP_X du \leq C^{k(m-1)} \int V_*^{-k} d\nu^{\otimes m}. \quad (9)$$

All powers of η have canceled. We now bound the right-hand side uniformly over the preceding points in each affine span.

Put $\gamma = \alpha n - (m - 2)$. Cauchy–Schwarz and (8) give, for each fixed preceding tuple,

$$\nu\{D_i \leq \delta\} \leq (BC^d)^{1/2} \delta^{\gamma/2} \quad (0 < \delta \leq 1).$$

The chosen k, m satisfy $\gamma/2 - k \geq n/8$. Integrating the tail of D_i^{-k} therefore gives

$$\begin{aligned} \int D_i^{-k} d\nu &\leq 1 + k \int_0^1 \nu\{D_i \leq \delta\} \delta^{-k-1} d\delta \\ &\leq 1 + \frac{k(BC^d)^{1/2}}{\gamma/2 - k} \leq C^d B^{1/2}. \end{aligned}$$

Integrate $z_m, z_{m-1}, \dots, z_2$ in this order in (9); the remaining z_1 integral has mass at most one. Taking the m th root gives

$$\|F_\eta\|_m \leq [C^{k(m-1)} (C^d B^{1/2})^{m-1}]^{1/m} \leq C^d B^{1/2}.$$

To recover exact labels, let ψ be continuous and compactly supported in the variables (X, u) . Averaging ψ over a shrinking box and applying bounded convergence shows

$$\int \psi F_\eta dP_X du \longrightarrow \int \psi(X, Xa(z)) dP_X d\nu(z).$$

Hölder's inequality bounds this limiting functional by $C^d B^{1/2} \|\psi\|_{m/(m-1)}$. Continuous compactly supported functions are dense in $L^{m/(m-1)}(P_X du)$. Duality supplies an L^m density with that bound. Agreement on continuous compactly supported tests identifies the two locally finite Borel measures. This proves (7) for the exact joint law, so it can be tested against arbitrary measurable message rules. $\square$

3.2 The information in one message

The weighted estimate allows us to control the probability of each message as a function of the signal. That is the link between the geometric hypothesis and information.

Lemma 3.4 (One block). *Let μ, Q, B satisfy (6). Draw $Z \sim \mu$ and a standard Gaussian $k \times d$ matrix X independently. Suppose W has at most N values and conditional probabilities $\pi_w(X, X\phi(Z))$. Assume each π_w is Borel on $\mathbb{R}^{k \times d} \times \mathbb{R}^k$, with $\pi_w \geq 0$ and $\sum_w \pi_w(X, y) = 1$ for every (X, y) . Then*

$$I_\mu(Z; W) \leq Cd + \log B + (2/m + e^{-ck}) \log N \quad (10)$$

for absolute constants $C, c > 0$.

Proof. Keep the normalization $X\phi(z) = Xc_Q + rXa(z)$ from Section 3.1. Let $L = \{u \in \mathbb{R}^k : \|u\| \leq 4\sqrt{k}\}$. Since $\|a(z)\| \leq 2$,

$$\mathbb{P}_X\{Xa(z) \notin L\} \leq e^{-ck} \quad (11)$$

uniformly in z . For example, if G is standard Gaussian in $\mathbb{R}^k$, then $\mathbb{E}e^{\|G\|^2/4} = 2^{k/2}$, and Markov's inequality at $\|G\| > 2\sqrt{k}$ proves this with $c = 1 - \frac{1}{2} \log 2$. Also $|L| = v_k(4\sqrt{k})^k \leq C^k$.

Define the subprobabilities

$$p_w(z) = \mathbb{E}_X[\pi_w(X, X\phi(z)) \mathbf{1}_{\{Xa(z) \in L\}}]$$

For a reference experiment, keep X Gaussian and replace the normalized exact label $Xa(z)$ by an independent uniform point in L . Its message probabilities are

$$b_w = |L|^{-1} \int_L \mathbb{E}_X \pi_w(X, Xc_Q + ru) du.$$

They satisfy $\sum_w b_w = 1$. This uses the probability-kernel identity at every label, including labels outside the support of the observation law. Test Lemma 3.3 with $\pi_w(X, Xc_Q + ru)\mathbf{1}_L(u)$. Because $0 \leq \pi_w \leq 1$, Hölder's inequality gives, for every $\|h\|_2 \leq 1$,

$$\int |h| p_w d\mu \leq C^d B^{1/2} (|L| b_w)^{1-1/m}.$$

Duality in $L^2(\mu)$ consequently gives

$$\|p_w\|_2 \leq A b_w^{1-1/m}, \quad A = C^d B^{1/2} \geq 1. \quad (12)$$

We next turn this into information without assuming that the posterior is uniform. Adjoin the bit $G = \mathbf{1}_{\{Xa(Z) \in L\}}$ to W , which can only increase mutual information. Put $P_w = \int p_w d\mu$ and $P_g = \sum_w P_w$. For $P_w > 0$, Jensen's inequality under the probability measure $p_w \mu / P_w$ gives

$$\int p_w \log(p_w / P_w) d\mu \leq 2P_w \log(\|p_w\|_2 / P_w).$$

Thus the contribution from $G = 1$ to $I(Z; W, G)$ is at most

$$2 \log A - (2 - 2/m) \sum_w P_w \log(P_w / b_w) + (2/m) \sum_w P_w \log(1 / P_w).$$

If $b_w = 0$, then (12) gives $P_w = 0$, and the corresponding terms vanish. The log-sum inequality and the entropy bound for N outcomes give

$$\sum_w P_w \log(P_w / b_w) \geq P_g \log P_g, \quad \sum_w P_w \log(1 / P_w) \leq \log N - P_g \log P_g.$$

The good contribution is therefore at most $2 \log A + (2/m) \log N - 2P_g \log P_g$.

For $G = 0$, let $q_w(z)$ be the corresponding conditional subprobability and $Q_w = \int q_w d\mu$. Since $q_w \leq 1$,

$$\sum_w \int q_w \log(q_w / Q_w) d\mu \leq \sum_w Q_w \log(1 / Q_w) \leq P_b \log N - P_b \log P_b,$$

where $P_b = 1 - P_g \leq e^{-ck}$ by (11). The functions $-P_g \log P_g$ and $-P_b \log P_b$ are bounded by absolute constants. Finally $2 \log A \leq Cd + \log B$, proving (10). $\square$

3.3 Selecting a cell and paying for its entropy

After a message, (6) may fail with a useful value of B . We restore it by selecting a cell using the current posterior. The selection must be based on the event of choosing that cell, rather than on the larger event that the signal merely lies in it.

Fix a finite-valued history H and a value h of probability $p > 0$. Suppose the posterior $\lambda = \mathcal{L}(Z \mid H = h)$ is supported in a level- J dyadic cell Q . A descendant R at relative depth j is called *heavy* when

$$\lambda(R) \geq 2^{-\alpha n j}.$$

For Z drawn from λ , let U be its deepest heavy ancestor inside Q , and write $j(U)$ for its relative depth.

Lemma 3.5 (Deepest heavy cell). *The selection U is measurable and finite-valued. Put $p_R = \mathbb{P}(U = R \mid H = h)$ for each selected cell of positive probability, and define*

$$B_R = \max\{1, 2^{-\alpha n j(R)} / p_R\}.$$

The conditional law $\mathcal{L}(Z \mid H = h, U = R)$ satisfies (6) with parent R and constant B_R . Moreover,

$$H(U) \leq H(j(U)) + \alpha n \log 2 \mathbb{E}j(U), \quad (13)$$

$$H(j(U)) \leq \log 2 (\mathbb{E}j(U) + 1), \quad (14)$$

$$\mathbb{E} \log B_U \leq H(j(U)) + 1. \quad (15)$$

Here every probability, entropy, and expectation is conditional on $H = h$.

Proof. For every measurable set A , the posterior satisfies $\lambda(A) \leq \mu_0(A)/p$. A heavy cell at relative depth j therefore satisfies

$$2^{-\alpha n j} \leq p^{-1} 2^{-n(J+j)}.$$

Since $1 - \alpha > 0$, this bounds j . The parent cell is heavy, so each path has a deepest heavy cell. There are only finitely many cells at the relevant levels, and all selection events are measurable.

Fix a selected cell R at relative depth j , and let R' be a strict descendant at further depth l . If R' is heavy, the event $\{U = R\} \cap R'$ is empty: any point of R' has a deeper heavy ancestor. If R' is not heavy, then $\lambda(R') < 2^{-\alpha n(j+l)}$. In either case,

$$\mathbb{P}(Z \in R' \mid H = h, U = R) \leq \frac{2^{-\alpha n j}}{p_R} 2^{-\alpha n l} \leq B_R 2^{-\alpha n l}.$$

At depth zero the bound follows from $B_R \geq 1$. Notice that this uses the probability p_R of *selection*; replacing p_R by $\lambda(R)$ would condition on a different event.

At depth j , there are at most $K_j = 2^{\alpha n j}$ heavy cells, because their posterior masses sum to at most one. The entropy chain rule and the bound on the number of choices at each depth prove (13). Comparing the depth distribution with the geometric distribution $2^{-(j+1)}$, nonnegativity of relative entropy gives (14).

For the remaining estimate, write $\log^+ x = \max\{0, \log x\}$ for $x > 0$. Let $P_j = \mathbb{P}(j(U) = j)$ and, on a level of positive mass, set $\rho_R = p_R / P_j$. There are at most K_j possible cells on that level. Consequently, for $u \geq 0$,

$$\mathbb{P}\left\{\log^+ \frac{1}{K_j \rho_U} > u \mid j(U) = j\right\} \leq e^{-u}.$$

Indeed every cell in this event has conditional mass less than e^{-u}/K_j , and there are at most K_j of them. Integrating the tail shows that this positive logarithm has conditional expectation at most one. Since

$$\log B_R = \log^+ \frac{1}{K_j P_j \rho_R} \leq \log(1/P_j) + \log^+ \frac{1}{K_j \rho_R},$$

averaging over the levels proves (15). $\square$

The three estimates serve different purposes. The first pays for the cell itself. The second pays for the scale used to name it. The third bounds the extra concentration factor that enters the next application of Lemma 3.4. All three costs are governed by the same expected increase in depth.

3.4 Closing the information bound

We can now construct the cells in Theorem 3.2. The proof compares two bounds for information: an upper bound that charges the messages and revelations, and a lower bound supplied by the volume of the final cell.

Proof of Theorem 3.2. Start with Π_0 empty, Q_0 as above, $J_0 = 0$, and $B_0 = 1$. After observing W_i , put $\Pi_i^- = (\Pi_{i-1}, W_i)$. Apply Lemma 3.5 to this history and the parent Q_{i-1} . Let Q_i be the selected cell, j_i its relative depth, and $J_i = J_{i-1} + j_i$. Set $\Pi_i = (\Pi_i^-, Q_i)$, and let B_i be the constant supplied by that lemma.

Only positive-probability histories matter. Each history has a finite range by induction: the message range is finite, and each intermediate history permits finitely many selected cells by Lemma 3.5. Thus every entropy below is finite. Fresh rows remain independent of Z conditional on the past, because all past messages and revelations depend only on Z , past data, and independent past randomization. The rule for W_i , conditional on this enlarged history, is an admissible kernel for Lemma 3.4; the extra cells are analysis variables and do not change the original rule.

Let $h_i = H(j_i \mid \Pi_i^-)$, including the average over intermediate histories. The alphabet hypothesis and Lemmas 3.4 and 3.5 give

$$I(Z; W_i \mid \Pi_{i-1}) \leq Cd + \mathbb{E} \log B_{i-1}, \quad \mathbb{E} \log B_i \leq h_i + 1.$$

The selection is deterministic from (Z, Π_i^-) , so

$$\begin{aligned} I(Z; Q_i \mid \Pi_i^-) &= H(Q_i \mid \Pi_i^-) \\ &\leq h_i + \alpha n \log 2 \mathbb{E} j_i, \quad h_i \leq \log 2 (\mathbb{E} j_i + 1). \end{aligned}$$

Summing the information chain rule, using $B_0 = 1$ and $\sum_i j_i = J_t$, yields

$$I(Z; \Pi_t) \leq Cdt + (\alpha n + 2) \log 2 \mathbb{E} J_t. \tag{16}$$

The 2 in the depth coefficient accounts for the depth entropy at selection and for its use in the next block's concentration factor.

For a fixed terminal history, its posterior λ is supported in its level- J_t cell Q_t . Writing $\mu_0(\cdot \mid Q_t)$ for normalized restriction to this cell gives the exact identity

$$D(\lambda \parallel \mu_0) = nJ_t \log 2 + D(\lambda \parallel \mu_0(\cdot \mid Q_t)).$$

Nonnegativity and averaging imply

$$I(Z; \Pi_t) \geq n \log 2 \mathbb{E} J_t. \tag{17}$$

Combining (16) and (17), and using $(1 - \alpha)n - 2 \geq n/4$ for large n , proves $\mathbb{E} J_t \leq Ct$. Substitution into (16) proves (5). $\square$

The strict inequality $\alpha < 1$ is what makes this comparison close. Revealing depth J_t certifies $nJ_t \log 2$ information, while the heavy-cell count charges only $\alpha n J_t \log 2$, apart from the two scale-entropy terms. No uniformity assumption on the posteriors is used.

3.5 Application to finite-state Gaussian regression

The information theorem applies to the finite-state model of Section 2. This application uses the original every-signal success premise to obtain success under the cube prior.

Corollary 3.6. *Suppose $M(d) = o(d^2)$ and $0 < \epsilon(d) \leq 1/10$. If a family of learners in the model of Section 2 satisfies*

$$\mathbb{P}\{\arccos\langle \hat{s}, s \rangle \leq \epsilon(d)\} \geq 2/3 \quad \text{for every } s \in S^{d-1},$$

then $T(d) \geq cd \log(1/\epsilon(d))$ for an absolute $c > 0$ and all sufficiently large d . The dimension threshold may depend on the memory sequence.

Proof of Corollary 3.6. First average the success guarantee over $s = \phi(Z)$, $Z \sim \mu_0$. Fix the shared rule-selection randomness, if present, with cube-prior average success greater than $3/5$; such a choice exists because the original average is at least $2/3$. The chart has full-rank derivative because its first n coordinates are z , so Lemma 2.2 applies. Apply Lemma 2.1 to these fixed rules. Now expose all remaining independent randomness and fix a realization with prior-average success at least $3/5$. Finite transition and stopping kernels can be sampled using independent uniforms indexed by state and sample index. Output laws can likewise be sampled in advance for each terminal state and stopping index. The resulting deterministic learner has the same state bound, and its output is a function of its terminal state and stopping index.

Divide its horizon into $t = \lceil T/k \rceil$ blocks. Pregenerate all k rows in every block, including an unused final suffix. A block message records one of the following: the continuing state; a terminal state together with the local stopping offset; or a dummy value if the learner was already inactive. Allowing offsets at the block endpoints and immediate stopping, it suffices to use

$$N \leq (k+3)2^M + 1. \quad (18)$$

The transcript determines the global stopping index from the block index and local offset. Conditional on its past, the starting state is known, and composing the transitions gives a measurable rule of the whole exact block. Thus Theorem 3.2 applies. Eventually $M \leq d^2$, so (18) gives $\log N \leq Cd^2$ and hence (4) with an absolute C_0 .

We next lower-bound the information needed for accurate estimation. For a fixed unit vector v , angular error at most ϵ implies $\|\phi(Z) - v\| \leq \epsilon$, and hence $\|Z - P_{\mathbb{R}^n} v\| \leq \epsilon$. The normalized cube volume of this event is at most

$$n^{n/2} v_n \epsilon^n \leq (5\epsilon)^n. \quad (19)$$

Let $p \geq 3/5$ be the success probability for the fixed learner and the cube prior. Under the product of the marginal laws of $(Z, \hat{s})$, the same success event has probability $p_0 \leq (5\epsilon)^n$. Relative-entropy data processing to its binary indicator gives

$$\begin{aligned} I(Z; \hat{s}) &\geq p \log(p/p_0) + (1-p) \log((1-p)/(1-p_0)) \\ &\geq p \log(1/p_0) - \log 2 \\ &\geq pn \log(1/(5\epsilon)) - \log 2. \end{aligned}$$

For $\epsilon \leq 1/10$,

$$\log(1/(5\epsilon)) \geq \left(1 - \frac{\log 5}{\log 10}\right) \log(1/\epsilon).$$

Since the output is determined by the messages, data processing and Theorem 3.2 imply $t \geq c \log(1/\epsilon)$ for large d .

It remains to remove the ceiling in $t = \lceil T/k \rceil$. Return to the original randomized learner and average its every-signal success over a uniform spherical signal. The $T > d/8$ instance of Lemma 2.3 applies to this same learner and prior; it grants the entire stream and therefore includes early stopping. This step does not infer cube-prior success from a sphere-average premise. Thus $T/k \geq 1$, $\lceil T/k \rceil \leq 2T/k$, and $k \geq c'd$ for large d . The block lower bound gives $T \geq c''d \log(1/\epsilon)$, as required. $\square$

4 Maximum-score localization

We now select a cell by maximizing its posterior mass weighted by depth. The selected posterior again satisfies a dyadic mass bound, with a slack that accounts for conditioning on the selection event. We first construct the selection and estimate its entropy. We then prove a block-information estimate that charges one half of the incoming slack. The iteration closes because the cell volume certifies more information per level than the leading cost of naming the cell; the depth entropy controls the remaining slack.

Put $n = d - 1$, let τ be uniform probability on

$$Q = [-1/(2\sqrt{n}), 1/(2\sqrt{n})]^n, \quad s(z) = (z, \sqrt{1 - \|z\|^2}).$$

Assign dyadic boundaries consistently to obtain a nested Borel partition. The proof of Lemma 3.1 applies on this closure: a depth- i cell has side $2^{-i}/\sqrt{n}$ and τ -mass 2^{-ni} , the graph map is 2-Lipschitz, and its first n coordinates recover z . Set

$$k = \ell = \lfloor n/8 \rfloor, \quad \beta = n/2,$$

and take n sufficiently large that $k = \ell \geq 2$. For a probability μ supported on a depth- i cell C , the mass bound with slack $a \geq 0$ is

$$\mu(B) \leq e^a 2^{-\beta j} \quad \text{for every depth-}(i+j) \text{ subcell } B \subseteq C, \quad j \geq 0. \quad (20)$$

4.1 Selecting a scale and paying for its label

After a block, we restore this bound by examining all depths on the cell chain of the sampled point. We reveal the first maximizing depth and its cell. The resulting law is conditioned on the selection event inside that cell, which is generally a proper subset.

Lemma 4.1 (Maximum-score refinement). *Let a positive-probability discrete history, of probability $P > 0$, have posterior π for $Z \sim \tau$, supported on a depth- i cell C . Thus $\pi \leq \tau/P$. For $z \in C$, let $C_{i+j}(z)$ be its depth- $(i+j)$ cell. Choose the first maximizer $J \geq 0$ of*

$$2^{\beta j} \pi(C_{i+j}(Z)), \quad j \geq 0,$$

and reveal (J, B) , where $B = C_{i+J}(Z)$. This is a measurable finite selection and

$$J \leq \frac{\log(1/P)}{(n - \beta) \log 2}.$$

For an outcome (j, B) of conditional probability $p > 0$, the resulting posterior satisfies (20) in B with slack $a' = \log(\pi(B)/p) \geq 0$. Conditionally on the original history,

$$\mathbb{E}a' \leq H(J) \leq (1 + \mathbb{E}J) \log 2, \quad (21)$$

$$H(J, B) \leq (\beta + 1)(\log 2) \mathbb{E}J + \log 2. \quad (22)$$

Proof. The score at depth zero is one. Every score at depth j is at most

$$P^{-1}2^{-n(i+j)}2^{\beta j} = P^{-1}2^{-ni}2^{-(n-\beta)j},$$

which tends to zero uniformly in z . Hence a maximum is attained among finitely many depths. The first such depth is measurable because each score is constant on Borel cells and there are countably many score comparisons. Comparing its score with one proves the stated bound on J .

Let $E_{j,B} \subseteq B$ be the actual selection event, of π -mass p . Comparison with the depth-zero score gives $\pi(B) \geq 2^{-\beta j}$. If a further subcell $B' \subseteq B$ of relative depth m has positive posterior mass after selection, it meets $E_{j,B}$. The maximizing inequality at such a point implies $\pi(B') \leq 2^{-\beta m}\pi(B)$. Its new posterior mass is therefore at most $p^{-1}\pi(B') \leq e^{a'}2^{-\beta m}$; if it has zero selected mass, the same bound holds. Since $p \leq \pi(B)$, the slack is nonnegative.

For fixed j , let p_B be the probabilities of its cell outcomes and $p_j = \sum_B p_B$. These B 's are disjoint at one grid level, so $\sum_B \pi(B) \leq 1$. The log-sum inequality then gives

$$\sum_B p_B \log \frac{\pi(B)}{p_B} \leq p_j \log(1/p_j).$$

Summing over j proves $\mathbb{E}a' \leq H(J)$. Comparison of the law of J with the probability vector $(2^{-j-1})_{j \geq 0}$, by nonnegativity of relative entropy, gives $H(J) \leq (1 + \mathbb{E}J) \log 2$. These quantities are finite because J has the displayed finite bound. Finally

$$\log(1/p) = \log(1/\pi(B)) + a' \leq \beta j \log 2 + a'.$$

Averaging proves (22). □

The refinement has two costs: the entropy of its label and the slack in the selected posterior. The lemma bounds both in terms of the expected depth increase. It remains to control the information in a fresh block under (20). The next estimate retains the mass of a restricted event, so that it can subsequently be applied to sets where a message creates a large posterior density.

4.2 Projection moments of a restricted measure

The inverse-distance calculation is related to the Gaussian projection method of Sharan, Sidford, and Valiant [12]. We give the full estimate for exact block observations and arbitrary measurable state rules. Restrictions are left unnormalized throughout.

Lemma 4.2 (Restricted projection moment). *Let μ be a probability measure supported on a depth- i cell $C \subseteq Q$ satisfying (20) with slack $a \geq 0$. Set $r = 2^{-i}$. There are absolute constants $C_2, C_3 \geq 1$ such that, with $K = C_2^n e^a$, the following assertions hold. For every μ -measurable $F \subseteq C$ of mass $f = \mu(F)$ and every affine subspace $P \subseteq \mathbb{R}^d$ of dimension at most ℓ ,*

$$\int_F \text{dist}(s(z), P)^{-k} d\mu(z) \leq 2r^{-k} K^{1/3} f^{2/3}. \quad (23)$$

Let $\gamma_{k,d}$ be the law of a k -by- d standard Gaussian matrix A . The joint image of $\gamma_{k,d}(dA) \mu|_F(dz)$ under $(A, z) \mapsto (A, As(z))$ has a nonnegative density $v_F(A, b)$ relative to $\gamma_{k,d}(dA) db$, and

$$\int v_F(A, b)^\ell d\gamma_{k,d}(A) db \leq f \left[C_3^k r^{-k} K^{1/3} f^{2/3} \right]^{\ell-1}. \quad (24)$$

In particular, for almost every A , $v_F(A, \cdot)$ is a density of the image of the unnormalized measure $\mu|_F$ under As . When $f = 0$, all these measures and integrals are zero.

Proof. Let z_c be the center of C and $s_c = s(z_c)$. The radius of C is $r/2$, so its graph image lies in $B(s_c, r)$. For $0 < h \leq r$, choose $j \geq 0$ with $r2^{-j} \leq h < 2r2^{-j}$. A depth- $(i+j)$ cell meeting the first-coordinate projection of an ambient ball of radius h lies in the concentric n -ball of radius $3r2^{-j}$. These cells have disjoint interiors and volume $(r2^{-j}/\sqrt{n})^n$. The unit-ball volume estimate $v_n \leq (2\pi e/n)^{n/2}$ therefore bounds their number by C^n , for an absolute C , by the Gaussian-integral proof in Lemma 3.1. By (20),

$$\mu\{z : \|s(z) - x\| \leq h\} \leq C^n e^a (h/r)^\beta. \quad (25)$$

After increasing C , this bound also holds trivially when $h > r$.

If P has dimension $m \leq \ell$, the projection of $B(s_c, r)$ onto P is contained in an m -ball of radius r . A maximal h -separated set in that ball has at most $(1 + 2r/h)^m$ points: its disjoint radius- $h/2$ balls fit inside the concentric ball of radius $r + h/2$. Ambient balls of radius $2h$ about these centers cover points of $B(s_c, r)$ at distance at most h from P . Applying the ball bound and absorbing the covering constants gives, for C_2 absolute,

$$\mu\{z : \text{dist}(s(z), P) \leq h\} \leq K(h/r)^{\beta-m} \quad (0 < h \leq r). \quad (26)$$

The estimate also shows that P has zero μ -mass.

Suppose $f > 0$, and put $\theta = k/(\beta - m)$. Since $m \leq \ell = k \leq n/8$, we have $\theta \leq 1/3$. For $t \geq 1$, the μ -mass within F where $(r/\text{dist}(s(z), P))^k > t$ is at most $\min\{f, Kt^{-1/\theta}\}$. Integrating this tail and splitting at $t = (K/f)^\theta \geq 1$ gives

$$\int_F \left(\frac{r}{\text{dist}(s(z), P)} \right)^k d\mu \leq \frac{K^\theta f^{1-\theta}}{1-\theta} \leq 2K^{1/3} f^{2/3}.$$

Here $K \geq 1$, $0 < f \leq 1$, and $\theta \leq 1/3$ justify the last inequality. This proves (23).

Let ν_A be the image of $\mu|_F$ under $z \mapsto As(z)$. If F is measurable only in the completion, replace it by a Borel representative modulo μ , which leaves the restricted measure unchanged. For $h > 0$, write v_k for the volume of the unit ball in $\mathbb{R}^k$ and define its ball-average density

$$v_{F,h}(A, b) = \frac{\nu_A(B(b, h))}{v_k h^k}.$$

Expand the ℓ th power using $z_1, \dots, z_\ell$ integrated against $(\mu|_F)^{\otimes \ell}$, and put $s_j = s(z_j)$. The intersection of the balls $B(As_j, h)$ has volume at most $v_k h^k$, and is empty unless $\|A(s_j - s_1)\| \leq 2h$ for all $2 \leq j \leq \ell$. For a fixed tuple, condition successively on the preceding difference images. Orthogonal decomposition of each Gaussian row shows that the conditional image at step j has an independent centered residual with covariance $\text{dist}(s_j, P_j)^2 I_k$, where

$$P_j = s_1 + \text{span}\{s_t - s_1 : 2 \leq t < j\}.$$

The subspace bound makes all zero-distance tuples null. On every remaining tuple, the conditional probability of a radius- $2h$ ball is at most

$$v_k (2h)^k (2\pi)^{-k/2} \text{dist}(s_j, P_j)^{-k}.$$

Multiplying these bounds, dividing by $(v_k h^k)^\ell$, and including the intersection volume cancels every h and v_k factor. Integrating $z_\ell, \dots, z_2$ in this order and applying (23) at each step leaves the factor $\mu(F) = f$ for z_1 . Absorbing the numerical Gaussian and radius factors into C_3^k proves

$$\int v_{F,h}^\ell d\gamma_{k,d} db \leq f [C_3^k r^{-k} K^{1/3} f^{2/3}]^{\ell-1}$$

uniformly for $h > 0$.

For every continuous compactly supported test function on the matrix and label space, integrals against these ball averages converge as $h \downarrow 0$ to its integral against the exact joint image measure. This follows by averaging translations of the test function over a shrinking ball and then using bounded convergence. Hölder's inequality bounds the limiting functional on $L^{\ell/(\ell-1)}(\gamma_{k,d}(dA) db)$ by the ℓ th root of the last display. Continuous compactly supported tests are dense for this locally finite reference measure. Extension and L^p duality therefore give a jointly measurable representing density with (24); equality on these tests identifies it with the exact image measure and makes it nonnegative. Finally, testing the joint identity on a countable algebra of rational label boxes and arbitrary measurable matrix sets identifies the conditional density for almost every A . The averaging operation has only proved a statement about exact labels; it is absent from the observation rule. $\square$

4.3 The one-half cost of slack

Lemma 4.3 (Information from one block). *In the setting of Lemma 4.2, let $Z \sim \mu$ be independent of A . Suppose a variable U , taking at most $W \leq (k+2)2^{d^2}$ values, is generated from the exact data $(A, As(Z))$ by a measurable probability kernel. There is an absolute C_0 such that, for all sufficiently large d ,*

$$I(Z; U) \leq C_0 d + \frac{a}{2}. \quad (27)$$

Proof. Write $q_u(A, b)$ for the kernel probabilities, so that $q_u \geq 0$ and $\sum_u q_u = 1$. With the center s_c used above, define

$$E = \{\|As(Z) - As_c\| \leq 3r\sqrt{k}\}, \quad V = v_k(3r\sqrt{k})^k \leq (3\sqrt{2\pi e}r)^k.$$

For each fixed Z , the vector $A(s(Z) - s_c)$ is Gaussian with covariance at most $r^2 I_k$. Hence

$$\Pr(E^c \mid Z) \leq e^{-9k/4} 2^{k/2} \leq e^{-k},$$

by Markov's inequality and $\mathbb{E} \exp(\|G\|^2/4) = 2^{k/2}$ for standard Gaussian $G \in \mathbb{R}^k$. Define comparison probabilities

$$\lambda_u = \mathbb{E}_A \frac{1}{V} \int_{B(As_c, 3r\sqrt{k})} q_u(A, b) db, \quad \sum_u \lambda_u = 1.$$

These probabilities use an auxiliary label uniform on the displayed ball, independently conditional on A .

For a measurable F of mass f , apply Hölder to its exact joint density v_F and the function $q_u(A, b) \mathbf{1}_{\{\|b - As_c\| \leq 3r\sqrt{k}\}}$. Since $0 \leq q_u \leq 1$, its conjugate norm is at most $(V\lambda_u)^{1-1/\ell}$. Lemma 4.2 therefore gives

$$\Pr(Z \in F, U = u, E) \leq (V\lambda_u)^{1-1/\ell} \{f[C_3^k r^{-k} K^{1/3} f^{2/3}]^{\ell-1}\}^{1/\ell}.$$

The r powers cancel. The exact exponent of f is

$$\frac{1 + (2/3)(\ell - 1)}{\ell} = \frac{2}{3} + \frac{1}{3\ell}.$$

Because $f \leq 1$, it may be weakened to $2/3$. The exponent of K is $(\ell - 1)/(3\ell) \leq 1/3$. Thus, for an absolute $C_4 \geq 1$,

$$\Pr(Z \in F, U = u, E) \leq L_0 \lambda_u^{1-1/\ell} f^{2/3}, \quad L_0 = C_4^n e^{a/3}. \quad (28)$$

Put $p_u = \Pr(U = u, E)$. For $p_u > 0$, let $\nu_u = \mathcal{L}(Z \mid U = u, E)$ and $f_u = d\nu_u/d\mu$. Applying (28) with $F = C$ shows that

$$D_u^* = \frac{L_0 \lambda_u^{1-1/\ell}}{p_u} \geq 1;$$

in particular $\lambda_u > 0$. Since $\mu\{f_u > e^t\} \leq e^{-t}$, a second application to that set gives

$$\nu_u\{\log f_u > t\} \leq \min\{1, D_u^* e^{-2t/3}\} \quad (t \geq 0).$$

Integrating this upper bound gives $\frac{3}{2}(\log D_u^* + 1)$. Relative entropy is no greater than the expected positive part of the log density, so

$$D(\nu_u \parallel \mu) \leq \frac{3}{2}(\log D_u^* + 1). \quad (29)$$

This calculation is why the $a/3$ in L_0 becomes $a/2$.

Let $p_E = \sum_u p_u$. The log-sum inequality and the entropy bound for at most W letters give

$$\sum_u p_u \log \frac{\lambda_u}{p_u} \leq p_E \log(1/p_E), \quad \sum_u p_u \log(1/p_u) \leq p_E \log(1/p_E) + p_E \log W.$$

The identity

$$\log \frac{\lambda_u^{1-1/\ell}}{p_u} = \left(1 - \frac{1}{\ell}\right) \log \frac{\lambda_u}{p_u} + \frac{1}{\ell} \log(1/p_u)$$

therefore bounds the weighted sum of these log ratios by $1 + (\log W)/\ell$. All zero terms are omitted. For the outcomes (U, E^c) , their weighted divergences to μ sum to at most

$$p_{E^c} \log(1/p_{E^c}) + p_{E^c} \log W.$$

Indeed, conditioning first on E^c multiplies the density by at most $1/p_{E^c}$, and revealing U afterwards adds at most its conditional entropy. This bound is zero when $p_{E^c} = 0$.

The mutual information of the finite pair $(U, \mathbf{1}_E)$ is the sum of these good and bad weighted divergences. Combining the last bounds with (29) gives

$$I(Z; U) \leq I(Z; U, \mathbf{1}_E) \leq Cn + \frac{a}{2} + C \frac{\log W}{\ell} + e^{-k} \log W + C.$$

Here $(\log W)/\ell = O(d)$ and $e^{-k} \log W = O(1)$ in the stated width range. Absorbing the absolute terms proves (27). $\square$

4.4 Iteration and the accuracy bound

Use the finite-state observation model of Section 2, with an integer horizon $T \geq 0$.

Proposition 4.4 (Maximum-score information bound). *Suppose $M \leq d^2$. For $N = \lceil T/k \rceil$, the learner under $Z \sim \tau$ has an augmented discrete history $\mathcal{H}_N$ satisfying*

$$I(Z; \mathcal{H}_N) \leq C_6 d N, \quad I(Z; \hat{S}) \leq C_6 d N \quad (30)$$

for an absolute C_6 and all sufficiently large d . If a data-independent shared seed selects the rules, the first inequality holds conditional on that seed with the same constant; the displayed output bound holds after averaging over it. The statement holds for every finite T .

Proof. First fix a realization of a data-independent shared seed, if present. The graph derivative contains the identity in its first n rows, so Lemma 2.2 applies to the cube prior. For this fixed seed, Lemma 2.1 supplies everywhere-defined Borel rules with the same prior law. We do this separately for each seed value under consideration; no jointly chosen family of Borel versions is needed. Use the initial finite state as $\mathcal{H}_0$. By the joint independence of the shared seed and initial state from the signal and all rows in Section 2, conditional on the fixed seed this state has zero information about Z , and the future rows remain independent of both. At an active block end record the continuing state or the terminal state and its stopping position within the block, including immediate stopping. Given the preceding history there are at most $(k+2)2^M$ such destinations. After a stop, later messages may be dummy messages because the earlier history already records the terminal state and index. Padding the last block by ignored Gaussian rows is harmless. This count depends only on the block length, not the total horizon.

Start at depth $i_0 = 0$ with slack $a_0 = 0$. At step t , apply Lemma 4.3 to its outgoing message U_t . Then apply Lemma 4.1 to the posterior given $\mathcal{H}_t^- = (\mathcal{H}_{t-1}, U_t)$, and append its selection (J_t, B_t) to form $\mathcal{H}_t$. The new depth is $i_t = i_{t-1} + J_t$, and its slack is a_t . The added labels use only Z and the history already observed. Future Gaussian rows remain independent of that joint collection. Given the starting state, the message is a kernel of the fresh block data, so the block lemma applies on every positive-probability history.

Every such history of probability P has posterior at most τ/P . All entropies used below are finite. Indeed a block message has a finite alphabet, and the depth bound in Lemma 4.1 is integrable after averaging over the preceding history because it is bounded by a constant times $\log(1/P)$. The label-charge inequality then preserves finite history entropy by induction.

The information chain rule, followed by (21) and (22), gives

$$\begin{aligned} I(Z; \mathcal{H}_N) &\leq \sum_{t=1}^N \left(C_0 d + \frac{1}{2} \mathbb{E} a_{t-1} + H(J_t, B_t \mid \mathcal{H}_t^-) \right) \\ &\leq C_5 d N + (\beta + \frac{3}{2})(\log 2) \sum_{t=1}^N \mathbb{E} J_t \end{aligned}$$

for an absolute C_5 . On the other hand the final posterior is supported on a depth- i_N cell, of prior mass 2^{-ni_N} . Relative entropy to τ is therefore at least $ni_N \log 2$, since the conditional divergence to the uniform law of that cell is nonnegative. Averaging yields

$$I(Z; \mathcal{H}_N) \geq n(\log 2) \mathbb{E} i_N = n(\log 2) \sum_{t=1}^N \mathbb{E} J_t.$$

The ratio $(\beta + 3/2)/n = 1/2 + 3/(2n)$ is bounded away from one for large n . Substituting the last bound into the preceding upper bound and moving that term to the left proves $I(Z; \mathcal{H}_N) \leq C_6 d N$.

The history records the terminal state and stopping index. Conditional on those data, the output uses only fresh randomness, so data processing gives $I(Z; \hat{S}) \leq I(Z; \mathcal{H}_N)$. For each fixed seed, Lemma 2.1 preserved the original conditional prior law, so the output-information bound just proved holds for that original law. Integrating the original measurable conditional information values over a shared seed R , whose law is independent of Z , gives

$$I(Z; \hat{S}) \leq I(Z; \hat{S}, R) = \mathbb{E}_R I(Z; \hat{S} \mid R) \leq C_6 d N.$$

When $T = 0$, the same argument has no blocks and the output is independent of Z . This proves the stated zero upper bound as well. $\square$

For a fixed unit output, angular success implies that Z lies within ϵ of its first n coordinates. The ball-volume estimate used above and the cube volume $n^{-n/2}$ bound this prior mass by $(\sqrt{2\pi e}\epsilon)^n \leq (5\epsilon)^n$. If the learner succeeds with probability at least $2/3$ for every unit signal, its cube success has the same lower bound. Applying (1) to the success event under the joint and product laws, and then using (30), gives

$$\frac{2}{3}n \log \frac{1}{5\epsilon} - \log 2 \leq C_6 d \lceil T/k \rceil. \quad (31)$$

For $L = \log(1/\epsilon) \geq \log 10$,

$$L - \log 5 \geq \left(1 - \frac{\log 5}{\log 10}\right) L.$$

Thus $L \leq C(T/k + 1)$ for large d , uniformly in $0 < \epsilon \leq 1/10$. If $L \geq 2C$, then $T \geq k(L/C - 1) \geq kL/(2C) \geq cdL$.

For the remaining values $L < 2C$, the original learner needs $T > d/10$ samples. Average its every-signal guarantee over a uniform signal S on the full sphere, and grant it all T pregenerated rows, including unused rows. If $T \leq d/10$, Lemma 2.3 applies to this original randomized learner. Writing V for the row span and letting r_0 decrease to ϵ in (3) gives

$$\Pr(\text{success}) \leq \frac{1}{2} + \frac{1}{2} \Pr\{\|P_{V^\perp} S\| \leq \epsilon\} \leq \frac{1}{2} + \frac{T}{2d(1 - \epsilon^2)}. \quad (32)$$

For $T \leq d/10$ and $\epsilon \leq 1/10$, the right side is less than $2/3$. Combining this linear bound with the large- L bound proves $T = \Omega(d \log(1/\epsilon))$. The constant is absolute, and $M(d) = o(d^2)$ ensures $M \leq d^2$ eventually. No restriction on T or on the rate of growth of L was used.

5 Regularization from finite relative entropy

This argument restores a dyadic mass bound after every block. The input to its refinement procedure is any posterior of finite relative entropy in its current cell. The procedure may take an unbounded number of steps, but its expected number of attempts and expected increase in depth are finite. These bounds pay for the complete refinement label, including every decision to remain outside the selected descendants. Finite relative entropy supplies this expected termination bound even when no uniform bound on the posterior density is available.

Use the half-open cube Q_0 , uniform probability μ_0 , and graph map ϕ of Section 3. In this section write $Q_{\text{ho}} = Q_0$, $\mu = \mu_0$, $s = \phi$, and $n = d - 1$. The dyadic cells remain half-open. For a cell C , let μ_C be uniform probability on C . A descendant of relative depth h has μ_C -mass 2^{-nh} .

For a probability $P \ll \mu_C$ of finite relative entropy in a depth- j cell, the change of reference measure gives

$$D(P\|\mu) = D(P\|\mu_C) + nj \log 2, \quad (33)$$

because $d\mu_C/d\mu = 2^{nj}$ on C . The depth term is the information certified by confinement to the cell; the remaining divergence measures concentration within it. This identity holds even when P is supported on a proper subset of C , so complement decisions remain represented in the current-cell divergence. The refinement below bounds the expected remaining divergence and expected depth increase by a single budget: the incoming divergence plus an absolute constant. That budget also includes decisions that keep the current cell.

Definition 5.1. Put $\alpha = 1/2$. A probability P supported on a cell C is regular on C if every descendant E of relative depth $h \geq 1$ satisfies

$$P(E) \leq 22^{-\alpha nh}.$$

5.1 The refinement and its termination

Lemma 5.2 (Finite expected regularization). *Suppose $P \ll \mu_C$ and $K_0 = D(P\|\mu_C) < \infty$. For all sufficiently large n , a countable label determined by $W \sim P$ specifies a descendant C' and a conditional law P' regular on C' . The label is the whole terminating history in the construction below, including every complement decision; P' is conditioned on this label and need not be $P(\cdot | C')$.*

Let H be the increase in cell depth and A_ the number of attempts. Then*

$$\mathbb{E}H < \infty, \quad \mathbb{E}A_* \leq 2(1 + \mathbb{E}H) < \infty,$$

and, with $b = (1 - \alpha)(\log 2)/2$,

$$\mathbb{E}D(P'\|\mu_{C'}) + bn \mathbb{E}H \leq K_0 + 2/e. \quad (34)$$

The hypothesis imposes no boundedness on $dP/d\mu_C$.

Proof. At an attempt, consider the current conditional law in its current cell. A strict descendant E of relative depth $h \geq 1$ is heavy if its current mass m_E exceeds $2^{-\alpha nh}$. Select the heavy descendants with no heavy strict ancestor of positive relative depth. Every heavy cell has such a first heavy ancestor, because its chain of ancestors back to the current cell is finite. The selected cells are a countable disjoint family.

If W is in one of these cells, reveal which cell, condition on it, replace the current cell by it, and begin another attempt. Otherwise reveal that W is in the complement, whose current probability is q , condition on that complement, and keep the current cell. Stop if $q \geq 1/2$; if $0 < q < 1/2$, begin another attempt. Zero-probability branches are omitted. These operations are Borel and the label records their finite sequence if the procedure stops.

At a stop, a descendant that was heavy immediately before the last complement decision lies in a selected cell, so it has zero posterior mass. Every other descendant of depth h had mass at most $2^{-\alpha nh}$, and conditioning on a complement of probability at least $1/2$ increases this by at most two. The resulting law is therefore regular.

It remains to prove that a stop occurs and that its cost satisfies (34). Let f be the density of the current law relative to the uniform probability on the current cell. On a heavy transition of depth h , its value at the sampled point changes to

$$f(W) \frac{2^{-nh}}{m_E}.$$

Its logarithm therefore decreases by at least $(1 - \alpha)nh \log 2$. On a complement transition it increases by $\log(1/q)$. These are exact density updates on every positive-probability branch, with all previous complement exclusions retained in the conditional law.

Truncate after $N \geq 1$ attempts, freezing a path once it has stopped. Let H_N be accumulated heavy depth, J_N the sum of complement logarithmic increases, and A_N the number of attempts actually made. If f_N is the resulting conditional density, the pathwise updates give

$$\log f_N(W) + (1 - \alpha)n(\log 2)H_N \leq \log f_0(W) + J_N. \quad (35)$$

For an active attempt the conditional expected complement increase, counted as zero on a heavy branch, is $q \log(1/q) \leq 1/e$. Thus

$$\mathbb{E}J_N \leq \mathbb{E}A_N/e \leq N/e.$$

We justify taking expectations of the logarithms before using (35). For any density f relative to a probability measure λ ,

$$\int_{\{f < 1\}} f \log(1/f) d\lambda \leq 1/e,$$

because $x \log(1/x) \leq 1/e$ for $0 < x < 1$. This bounds the expected negative log part at the sampled point, conditionally on each positive-probability history. Finite K_0 then makes the positive part of $\log f_0(W)$ integrable. The path inequality bounds the positive part of $\log f_N(W)$ by $(\log f_0(W))^+ + J_N$. Consequently $\mathbb{E} \log f_N(W)$ is well defined and is the expected conditional relative entropy, hence is nonnegative. Taking expectations in (35) first shows $\mathbb{E} H_N < \infty$.

An attempt after the first follows a heavy transition or a continuing complement. The number of heavy transitions is at most H_N , because each adds at least one unit of depth. At each active attempt, the conditional probability of a continuing complement is at most $1/2$. Counting such predecessors and taking expectations gives

$$\mathbb{E} A_N \leq 1 + \mathbb{E} H_N + \frac{1}{2} \mathbb{E} A_N, \quad \mathbb{E} A_N \leq 2(1 + \mathbb{E} H_N). \quad (36)$$

Substituting $\mathbb{E} J_N \leq \mathbb{E} A_N/e$ into the expected path inequality now yields

$$\mathbb{E} \log f_N(W) + ((1 - \alpha)n \log 2 - 2/e) \mathbb{E} H_N \leq K_0 + 2/e. \quad (37)$$

For sufficiently large n , the coefficient is at least $bn = (1 - \alpha)n(\log 2)/2$. Hence the expectations of H_N and, by (36), A_N , are bounded uniformly in N .

Both sequences increase on every path. Monotone convergence gives finite expectations for their limits H, A_* , and preserves $\mathbb{E} A_* \leq 2(1 + \mathbb{E} H)$. A path that never stops has $A_N \rightarrow \infty$, so nontermination has probability zero. The possible terminating histories form a countable set of finite sequences.

The limiting complement sum J also satisfies, by monotone convergence,

$$\mathbb{E} J \leq \mathbb{E} A_*/e \leq (2/e)(1 + \mathbb{E} H) < \infty.$$

At termination the positive log part of the final density is integrable by the path inequality, and its negative log part is integrable by the conditional bound above. We may therefore take expectations directly in the terminating path inequality. This gives

$$\mathbb{E} D(P' \| \mu_{C'}) + (1 - \alpha)n(\log 2) \mathbb{E} H \leq K_0 + (2/e)(1 + \mathbb{E} H),$$

which implies (34). In particular, no passage of a signed logarithm through a limit is being used. $\square$

5.2 Local mass and exact projection densities

We next bound the information from a block that starts with a regular law. The projection estimate is related to the independent-draw Gaussian projection method of Sharan, Sidford, and Valiant [12, Section 7, Lemma 12]. The estimate below is proved for exact multirow observations and for finite restrictions with their density multiplier visible.

Lemma 5.3 (Ambient ball bound). *Suppose $d \geq 3$ and P is regular on a depth- j cell. Let z be the graph image of its center and set $R = 2 \cdot 2^{-j}$. Then the image of P is supported in $B(z, R)$. With $\beta = 1/3$, there is an absolute $C_0 \geq 1$ such that*

$$P\{w : \|s(w) - x\| \leq r\} \leq C_0^d (r/R)^{\beta d} \quad (x \in \mathbb{R}^d, 0 < r \leq R). \quad (38)$$

Proof. The cell diameter is 2^{-j} ; the 2-Lipschitz property gives the stated support radius. For $r \leq R/2$, choose $h \geq 1$ with $r \leq R2^{-h} < 2r$. The depth- $(j+h)$ cells have side between $r/(2\sqrt{n})$ and $r/\sqrt{n}$, and diameter at most r . A ball test for the graph image restricts w to a ball of radius r in its first n coordinates. Every cell meeting that ball lies in the concentric ball of radius $2r$. Comparing disjoint cell volumes and using the unit-ball estimate from Lemma 3.1 gives at most

$$v_n(2r)^n(2\sqrt{n}/r)^n \leq (4\sqrt{2\pi e})^n$$

such cells. Each has P -mass at most $22^{-\alpha nh}$. Since $2^{-h} < 2r/R$ and $\alpha n = (d-1)/2 \geq d/3 = \beta d$, all factors are bounded by $C_0^d(r/R)^{\beta d}$ for an absolute C_0 . For $R/2 < r \leq R$, enlarge C_0 and use total mass one. $\square$

Lemma 5.4 (Projection with a density multiplier). *Fix $\beta = 1/3$, put $\gamma = \beta/2 = 1/6$, and let $p = \lfloor \beta d/2 \rfloor \geq 2$. Let $1 \leq k \leq \gamma d/2$, and let ρ be the law of a k -by- d standard Gaussian matrix. Suppose that ν is a finite positive Borel measure of mass at most one, supported in $B(z, R)$ for $R > 0$, and that for some $a \geq 0$,*

$$\nu(B(x, r)) \leq e^{ad} C_0^d (r/R)^{\beta d} \quad (x \in \mathbb{R}^d, 0 < r \leq R).$$

The image of $\rho(dA)\nu(ds)$ under $(A, s) \mapsto (A, As)$ has a nonnegative density g relative to $\rho(dA) dy$ with

$$\int g(A, y)^p d\rho(A) dy \leq \left[2R^{-k} e^{(a+C_1)k/\gamma} \right]^{p-1}. \quad (39)$$

Here $C_1 \geq 0$ depends only on C_0 , uniformly over all the other parameters in the stated ranges.

Proof. Let F be an affine subspace of dimension $m \leq p-2$. For $0 < r \leq R$, a maximal r -separated set in the projection of $B(z, R)$ onto F has at most $(5R/r)^m$ points, by packing disjoint radius- $r/2$ balls. Ambient balls of radius $2r$ at these centers cover the part of $B(z, R)$ at distance at most r from F . If $2r > R$, the trivial mass bound can replace the local bound at that radius, whose right side would already be at least one. Since $\beta d - m \geq \gamma d$, absorbing the covering constants gives

$$\nu\{s : \text{dist}(s, F) \leq r\} \leq \min\{1, e^{(a+C_1)d} (r/R)^{\gamma d}\}. \quad (40)$$

In particular F has zero ν -mass. Set $r_0 = Re^{-(a+C_1)/\gamma}$. Layer-cake integration, using total mass at most one above the cutoff r_0 , gives

$$\begin{aligned} \int \text{dist}(s, F)^{-k} d\nu(s) &\leq r_0^{-k} + k r_0^{-\gamma d} \int_0^{r_0} r^{\gamma d - k - 1} dr \\ &= r_0^{-k} \left(1 + \frac{k}{\gamma d - k} \right) \leq 2R^{-k} e^{(a+C_1)k/\gamma} =: B. \end{aligned} \quad (41)$$

The last inequality uses $k \leq \gamma d/2$.

Choose a continuous nonnegative probability density φ_η supported in $B(0, \eta) \subseteq \mathbb{R}^k$, and define

$$g_\eta(A, y) = \int \varphi_\eta(y - As) d\nu(s).$$

For $\nu^{\otimes p}$ -almost every tuple $s_1, \dots, s_p$, the differences $s_i - s_1$, $2 \leq i \leq p$, are linearly independent: at the i th step, the preceding affine span has dimension at most $i-2 \leq p-2$, so it is null by (40). For such a tuple put $D = [s_2 - s_1, \dots, s_p - s_1]$. The joint Gaussian vector of its images under A has density at most

$$(2\pi)^{-k(p-1)/2} \det(D^\top D)^{-k/2} \leq \prod_{i=2}^p \text{dist}(s_i, \text{aff}(s_1, \dots, s_{i-1}))^{-k}.$$

The determinant factorization is Gram–Schmidt.

Expand g_η^p , shift the label integration variable by As_1 , and integrate over the Gaussian difference images. Bounding their density by the displayed supremum leaves a product of p probability kernels. Its integral over the $p - 1$ difference variables and then the shifted label variable is one. Tonelli's theorem and successive integration of $s_p, \dots, s_2$, using (41), therefore give

$$\int g_\eta^p d\rho dy \leq B^{p-1};$$

the last variable contributes $\nu(\mathbb{R}^d) \leq 1$.

For a continuous compactly supported test function $v(A, y)$, integration against g_η converges to $\int v(A, As) d\rho(A) d\nu(s)$ as $\eta \downarrow 0$. Shrinking support and bounded convergence prove this for the finite measure ν . Hölder bounds the limiting functional on $L^{p/(p-1)}(\rho(dA) dy)$ by $B^{(p-1)/p}$. Density of continuous compactly supported tests and L^p duality give a representing density with that norm. Equality on those tests identifies it with the exact image measure, so a nonnegative version may be used. This proves (39). The auxiliary kernels have been removed before any message is formed. $\square$

5.3 Information in one finite message

Lemma 5.5 (Regular block information). *There is an absolute $c_* > 0$ such that the following holds for $k = \lfloor c_* d \rfloor$. Let W have any law P regular on one cell of Q_{ho} , let A be an independent k -by- d standard Gaussian matrix, and put $Y = As(W)$. Suppose U takes at most m_d values and is conditionally independent of W given (A, Y) . If $\log m_d = o(d^2)$, then*

$$I(W; U) \leq 2d \tag{42}$$

for all sufficiently large d , uniformly over these laws and channels. The threshold may depend on the alphabet sequence, while c_ and the coefficient 2 are absolute.*

Proof. For each positive-probability letter let $p_u = \Pr(U = u)$ and $\ell_u(w) = \Pr(U = u \mid W = w)$. Its posterior density relative to P is ℓ_u/p_u . Set

$$B_u = \{w : \ell_u(w)/p_u > e^d\}, \quad \lambda = e^{-d}.$$

Since this density integrates to one, $P(B_u) \leq \lambda$. Outside $W \in B_u$, the log posterior density is at most d . We will show that $\Pr(W \in B_u)$ is exponentially small in d , and then bound its contribution using the finite alphabet.

Use z, R from Lemma 5.3. The exact joint law of (A, Y) has a density $D(A, y)$ by Lemma 5.4 with $a = 0$. The graph image of $\lambda^{-1}P|_{B_u}$ has mass at most one and satisfies the ball hypothesis with multiplier e^d , namely with $a = 1$. Let $g_u(A, y)$ be its exact projection density from Lemma 5.4. Under the original experiment,

$$b_u(A, Y) := \Pr(W \in B_u \mid A, Y) = \frac{\lambda g_u(A, Y)}{D(A, Y)}$$

almost surely. This is the Radon–Nikodym ratio of the restricted joint measure to the original one; points where $D = 0$ have zero original probability.

To control this ratio, choose an absolute $C_2 > 2\sqrt{2\pi e}$ and define

$$G = \{D(A, Y) \geq (C_2 R)^{-k}\}.$$

For every fixed s in the support, $A(s - z)$ is centered Gaussian with covariance at most $R^2 I_k$. Markov's inequality with exponential parameter $1/4$ gives

$$\Pr\{\|Y - Az\| > 2\sqrt{k}R\} \leq e^{-k}2^{k/2}.$$

The label ball of radius $2\sqrt{k}R$ has volume at most $(2\sqrt{2\pi e}R)^k$. The probability of points inside it where $D < (C_2 R)^{-k}$, after integrating over A , is at most $(2\sqrt{2\pi e}/C_2)^k$. Thus

$$\Pr(G^c) \leq 2e^{-c_2 k} \quad (43)$$

for some absolute $c_2 > 0$.

Let $p = \lfloor \beta d/2 \rfloor$. On G , the density denominator is bounded below, so

$$\begin{aligned} \mathbb{E}[\mathbf{1}_G b_u(A, Y)^p] &\leq \lambda^p (C_2 R)^{k(p-1)} \int g_u(A, y)^p d\rho(A) dy \\ &\leq \exp\{-dp + C_3 k(p-1)\}, \end{aligned} \quad (44)$$

where

$$C_3 = \log C_2 + (1 + C_1)/\gamma + \log 2$$

is valid because $k \geq 1$. Choose the fixed $c_* > 0$ so that

$$c_* \leq \gamma/2, \quad C_3 c_* \leq 1/4.$$

Then the projection lemma applies and the last moment is at most $e^{-3dp/4}$ for all sufficiently large d .

Given (A, Y) , the conditional independence assumption lets us average the choice of U separately from the event $W \in B_u$. Consequently

$$\Pr(W \in B_U, G) \leq \mathbb{E}[\mathbf{1}_G \max_u b_u(A, Y)].$$

The p th power of the maximum is at most the sum of the p th powers. Since $\log m_d \leq dp/4$ eventually, Hölder and (44) bound this probability by $e^{-d/2}$. Also $k \geq c_* d/2$ eventually, so (43) yields

$$q_0 := \Pr(W \in B_U) \leq e^{-c_3 d}$$

after decreasing an absolute $c_3 > 0$ and increasing the dimension threshold.

For $q_u = \Pr(U = u, W \in B_U) \leq p_u$, the contribution of the event $W \in B_U$ to $\mathbb{E} \log(\ell_U(W)/p_U)$ is at most

$$\sum_u q_u \log(1/p_u) \leq \sum_u q_u \log(1/q_u) \leq q_0 \log(m_d/q_0).$$

The last bound is the entropy bound for at most m_d letters after normalization by q_0 ; its value is zero if $q_0 = 0$. The exponential bound on q_0 and $\log m_d = o(d^2)$ make this $o(1)$. Outside the event, the log posterior density is at most d . Hence $I(W; U) \leq d + o(1) \leq 2d$, as claimed. $\square$

5.4 Iteration in the finite-state model

Use the finite-state model of Section 2, now with $M(d) = o(d^2)$. Suppose the original learner has angular success at least $2/3$ for every signal at $0 < \epsilon \leq 1/10$. For the cube experiment, first fix the data-independent shared choice of rules with average success greater than $3/5$, which is possible by averaging the original success $2/3$. The derivative of the graph map contains the identity in its first n rows, so Lemma 2.2 and then Lemma 2.1 supply Borel representatives for this fixed choice without changing its cube law. Now implement the finite-output transition kernels by independent

uniforms indexed by state and layer, and sample the output law independently for each terminal state and index. Include the initial state among these remaining choices; its conditional law given the fixed shared seed is independent of the signal and rows by the joint independence in Section 2. Averaging these choices fixes a deterministic program with cube success at least $3/5$. This reduction is only for the cube prior.

Use $L_* = \lceil T/k \rceil$ blocks, padding the final block with ignored independent rows. Pregenerate rows after any early stop. An active block message records its continuing state, or its terminal state and within-block stopping offset, including immediate stopping. Use a dummy message on all later blocks. Thus the alphabet per block has size at most

$$m_d = (k+2)2^M + 1, \quad \log m_d = o(d^2). \quad (45)$$

The block number is already known, so no factor depending on T is needed. The messages determine the terminal state, global stopping index, and fixed output.

Retain the messages in an analysis history. Immediately after each message, append the entire regularization label from Lemma 5.2. Initially the law is μ , which is regular. On every positive-probability history the current law is regular on its indicated cell. That history uses only W , past rows, and the fixed rules; future independent rows remain independent of it and of W . Given the entering state, the next message is a channel of the fresh block data. Hence Lemma 5.5 applies conditionally; a dummy message has zero information.

Let K_i be the expected conditional divergence to the uniform law on the current cell after i messages and regularizations, and let j_i be the expected current depth. Both start at zero. Conditioning a current density f on a message u multiplies it by ℓ_u/p_u , where $0 \leq \ell_u \leq 1$ and $p_u > 0$ on that branch. Taking logarithms and averaging shows that the expected divergence to the old cell grows by exactly the conditional mutual information of the message. Finite entropy is preserved on each branch: its positive log part grows by at most $\log(1/p_u)$, and its expectation under the new law is finite because that law has density at most $1/p_u$ relative to the old one. Its negative part has the universal integrable bound used above. The average increase is at most $2d$ by the block lemma. Applying the regularization budget conditionally then gives

$$K_{i+1} + bnj_{i+1} \leq K_i + bnj_i + 2d + 2/e. \quad (46)$$

This also proves inductively that the next finite-entropy hypothesis holds.

Let $\mathcal{H}$ be the full history, including every complement decision. The bound just proved ensures finite expected divergence and depth. Histories are countable, so averaging (33) for their conditional laws and using the defining Radon–Nikodym identity for mutual information gives

$$\begin{aligned} I(W; \mathcal{H}) &= K_{L_*} + n(\log 2)j_{L_*} \\ &\leq \max\{1, \log 2/b\}(2d + 2/e)L_* \leq C_4 d L_* \end{aligned} \quad (47)$$

for an absolute C_4 . This calculation includes the information in all auxiliary labels; it does not assume a bound on their individual Shannon entropies.

The deterministic output is known from $\mathcal{H}$. Lemma 3.1 bounds the angular success probability of any fixed output by $(\sqrt{2\pi e}\epsilon)^n$. Under the product law of W and $\mathcal{H}$, the same bound holds; under the joint law success is at least $3/5$. Equation (1) gives the exact endpoint

$$I(W; \mathcal{H}) \geq \frac{3}{5}n \log \frac{1}{\sqrt{2\pi e}\epsilon} - \log 2. \quad (48)$$

For $L = \log(1/\epsilon) \geq \log 10$,

$$L - \log \sqrt{2\pi e} \geq \left(1 - \frac{\log \sqrt{2\pi e}}{\log 10}\right) L,$$

with a positive absolute coefficient. Thus, for sufficiently large d , (48) is at least cdL , uniformly in the accuracy. Combining it with (47) gives $\lceil T/k \rceil \geq c'L$. When $L \geq 2/c'$, $T \geq k(c'L - 1) \geq c'dL$, because $k \geq c_*d/2$ eventually. When $T = 0$, the zero upper bound and positive lower bound already contradict success.

For the bounded remaining range of L , the original learner satisfies $T > d/100$. Take a uniform full-sphere signal S and grant the learner all $T \leq d/100$ pregenerated rows. Let V be their span and $p_V = P_V S$. Conditional on the rows and p_V , the residual is uniform on the sphere of radius $\sqrt{1 - \|p_V\|^2}$ in $V^\perp$. This follows by writing S as a normalized Gaussian: the direction of its orthogonal Gaussian component is independent of both component lengths and of the component in V , hence also of the row-space projection after normalization. Isotropy gives

$$\mathbb{E}\|p_V\|^2 \leq T/d, \quad \Pr\{\|p_V\| > 1/2\} \leq 4T/d.$$

Conditional also on the learner's independent randomness, its unit output is fixed. The inner product of that output with the residual has mean zero and variance at most $1/(d - T)$. On $\|p_V\| \leq 1/2$, angular success requires this inner product to be at least 0.4, since $\cos(\epsilon) \geq 0.9$. Chebyshev's inequality therefore bounds success by

$$\frac{4T}{d} + \frac{1}{0.4^2(d - T)} < \frac{2}{3} \tag{49}$$

for sufficiently large d , a contradiction. Hence the bounded range also has $T = \Omega(dL)$. All lower-bound constants are absolute; the eventual dimension threshold may depend on the rate $M(d)/d^2 \rightarrow 0$.

6 A finite split for bounded posterior densities

We partition a bounded-density posterior into finitely many components, each supported in a dyadic cell and satisfying a bound on mass at every smaller scale. Revealing the component has an entropy cost. An exact change of reference from the parent cell to the selected cell compares this cost with the information supplied by the cell's depth. We first construct the partition, then prove the block estimate, and finally combine them into a finite-block information bound and a learner lower bound.

Put $n = d - 1$, and use the closed cube

$$Q = [-1/(2\sqrt{n}), 1/(2\sqrt{n})]^n, \quad s(z) = (z, \sqrt{1 - \|z\|^2}).$$

Let τ be its uniform probability and assign dyadic boundaries consistently. The geometric proof of Lemma 3.1 applies on this closure: s is 2-Lipschitz, a depth- h cell has side $2^{-h}/\sqrt{n}$ and mass 2^{-nh} , and the first n graph coordinates recover z . For a cell A , let τ_A be uniform probability on it. Throughout this section set

$$k = \lfloor n/64 \rfloor, \quad J = 2k, \quad p = \lfloor n/8 \rfloor,$$

and take n sufficiently large that $k \geq 1$ and $p \geq 2$. Here k is the block length, J is the number of rows in two stacked blocks, and p is the projection moment order. Constants denoted by C, c are positive and absolute and may change between estimates. The independent-draw, span-distance projection calculation is related to that of Sharan, Sidford, and Valiant [12]; the exact-label density and block estimates are proved below.

6.1 A finite partition and its entropy identity

Lemma 6.1 (Bounded-density split). *Let μ be a probability supported on a dyadic cell A , with bounded density relative to τ_A . There is a finite measurable partition into positive-probability labels i , with probabilities p_i , descendant cells A_i of relative depths j_i , and numbers $q_i \in (0, 1]$. Writing $\mu_i = \mu(\cdot | i)$ and $Q_i = \log(1/q_i)$, each μ_i is supported on A_i and satisfies*

$$\mu_i(D) \leq q_i^{-1} 2^{-n\ell/2} \quad (50)$$

for every relative-depth- ℓ descendant D of A_i , and

$$-\log p_i \leq \frac{1}{2}n(\log 2)j_i + Q_i, \quad \mathbb{E}_i Q_i \leq 1 + \mathbb{E}_i j_i. \quad (51)$$

$$\mathbb{E}_i D(\mu_i \| \tau_{A_i}) = D(\mu \| \tau_A) + H(i) - n(\log 2)\mathbb{E}_i j_i. \quad (52)$$

The descendant bound includes $\ell = 0$, and $\mathbb{E}_i$ means expectation with weights p_i .

Proof. At a node cell C , use the original law μ conditioned on C . Select all inclusion-maximal strict dyadic descendants D whose conditional mass exceeds $2^{-n\ell/2}$, where ℓ is depth relative to C . They are disjoint: two dyadic cells are disjoint or nested, and nesting would contradict maximality. A point in a selected cell moves to that cell. Every other point stops in the remainder of the current cell. Repeat at the selected cells. The label records the terminal node and its remainder.

Let $K = \|d\mu/d\tau_A\|_\infty < \infty$; necessarily $K \geq 1$. Along a nontrivial path reaching relative depth j , the product of the selected conditional masses exceeds $2^{-nj/2}$. No complement decision occurs before stopping, so reaching the path cell is exactly membership in it. The same product is its original μ -mass, which is at most $K2^{-nj}$. Hence

$$j < \frac{2 \log K}{n \log 2}.$$

This uniformly bounds every nontrivial path depth and leaves finitely many grid cells. At an individual node, its bounded conditional density likewise rules out heavy descendants at unbounded depth, so each selected inclusion-maximal descendant exists. The construction is therefore a finite tree and defines a finite partition.

Let q_i be the conditional probability of the terminal remainder at its node, omitting zero-probability remainders. A strict descendant that exceeded its threshold before the remainder decision is contained in a selected cell and has zero remainder mass. Every other descendant has its conditional mass increased by at most q_i^{-1} . This proves (50); the case $\ell = 0$ is immediate. The product of masses on the path, multiplied by q_i , is p_i , so its lower bound proves $-\log p_i \leq \frac{1}{2}n(\log 2)j_i + Q_i$.

For the expected remainder cost, a node reached with probability r and having remainder probability q contributes $rq \log(1/q)$ to $\mathbb{E}_i Q_i$. This is at most r , since $q \log(1/q) \leq 1$, with value zero at $q = 0$. Summing over the finite tree bounds $\mathbb{E}_i Q_i$ by the expected number of visited nodes. Each move increases depth by at least one, so this number is at most $1 + \mathbb{E}_i j_i$.

On the part with label i , the density relative to τ_{A_i} is the original density relative to τ_A , multiplied by $p_i^{-1} 2^{-nj_i}$. Taking its logarithm, integrating on that part, and averaging over i proves (52). All terms are finite because the partition is finite and each conditional density is bounded. $\square$

The entropy identity credits $n(\log 2)\mathbb{E}_i j_i$ for the change of cell. Naming the component uses at most half of this credit, plus $\mathbb{E}_i Q_i$. The remaining task is to bound the next block's information in component i by $Cn + CQ_i$; the bound $\mathbb{E}_i Q_i \leq 1 + \mathbb{E}_i j_i$ will then pay for its dependence on the remainder probability.

6.2 A centered exact projection density

Squaring a state likelihood introduces two independent k -row matrices at the same signal. We will therefore need a density estimate for their $J = 2k$ stacked rows. The p independent signal draws in the proof below serve a different purpose: they expand the p th moment of that density.

Consider any probability μ on a depth- h cell A that satisfies (50) with $q \in (0, 1]$. Put $Q_* = \log(1/q)$, $R = 2^{-h}$, and let s_0 be the graph image of the cell center. The cell has radius $R/2$, so $\|s(z) - s_0\| \leq R$. The following estimates use only this dyadic mass hypothesis; the bounded-density premise was used to construct the finite split.

Lemma 6.2 (Local density for exact labels). *Let μ be a probability on a depth- h dyadic cell A such that, for some $q \in (0, 1]$, every relative-depth- ℓ descendant D satisfies $\mu(D) \leq q^{-1}2^{-n\ell/2}$. Put $Q_* = \log(1/q)$, $R = 2^{-h}$, and $s_0 = s(\text{center } A)$, with J, p as fixed above. Let X' have J independent standard Gaussian rows, independently of $z \sim \mu$, and put $b = X'(s(z) - s_0)$. On the open region $\Omega = \{b \in \mathbb{R}^J : \|b\| < 3\sqrt{J}R\}$, the joint law of (X', b) has a density f relative to $dP_{X'} db$. If h_R is the density of $N(0, R^2 I_J)$, then*

$$\|\mathbf{1}_\Omega f / h_R\|_{L^p(dP_{X'}, h_R(b) db)} \leq C^J e^{C(J/n)Q_*}, \quad (53)$$

where $\mathbf{1}_\Omega f / h_R$ is extended by zero outside Ω .

Proof. We first use the dyadic mass bound to control inverse Gram determinants of independent centered signal vectors. These determinants bound the moment of a smoothed projection density. We then remove the smoothing and compare the exact density with the Gaussian reference.

For every $y \in \mathbb{R}^d$ and $0 < r \leq 1$,

$$\Pr_\mu\{\|s(z) - y\| \leq rR\} \leq C^m e^{Q_*} r^{n/2}. \quad (54)$$

Indeed, choose $\ell \geq 0$ with $2^{-\ell-1} < r \leq 2^{-\ell}$. A relative-depth- ℓ cell meeting the projection of this ball onto the first n coordinates lies in the concentric ball of radius $2 \cdot 2^{-\ell} R$. Its side is $2^{-\ell} R / \sqrt{n}$. The unit-ball volume estimate in Lemma 3.1 therefore bounds the number of such disjoint cells by C^m . Multiply by their mass bound $e^{Q_*} 2^{-n\ell/2}$ and use $2^{-\ell} < 2r$.

If $U \subseteq \mathbb{R}^d$ is a linear subspace of dimension $m \leq n/8$, a maximal rR -separated set in its radius- R ball has at most $(1 + 2/r)^m$ points. Its radius- rR balls cover that ball, by maximality. Because $\|s - s_0\| \leq R$, ambient balls of radius $2rR$ about the translated centers cover the event $\text{dist}(s - s_0, U) \leq rR$. Using (54), and the trivial mass bound for $r > 1/2$, gives

$$\Pr_\mu\{\text{dist}(s - s_0, U) \leq rR\} \leq C^m e^{Q_*} r^{n/2-m} \quad (0 < r \leq 1). \quad (55)$$

For $m \leq p-1$, put $a = n/2 - m \geq 3n/8$ and $D_U = R^{-1} \text{dist}(s - s_0, U)$. Then $0 < D_U \leq 1$ almost surely. Let $\mathcal{A} = C^m e^{Q_*} \geq 1$. Since $J \leq a/2$, layer-cake integration of $\Pr(D_U < r) \leq \min\{1, \mathcal{A} r^a\}$, split at $\mathcal{A}^{-1/a}$, gives

$$\begin{aligned} \mathbb{E}_\mu D_U^{-J} &= 1 + J \int_0^1 r^{-J-1} \Pr(D_U < r) dr \\ &\leq \frac{a}{a-J} \mathcal{A}^{J/a} \leq 2 \mathcal{A}^{J/a} \leq (C e^{CQ_*/n})^J. \end{aligned} \quad (56)$$

For independent $z_1, \dots, z_p \sim \mu$, put $s_i = s(z_i)$ and let G be the Gram matrix of $e_i = s_i - s_0$. Conditional on its predecessors, each e_i has positive distance from their span almost surely by (55). Gram-Schmidt and successive conditioning in (56) yield

$$\mathbb{E}(\det G)^{-J/2} \leq (C e^{CQ_*/n} / R)^{Jp}. \quad (57)$$

Add an independent uniform vector in $[-t, t]^J$, $t > 0$, to the exact label b . Its joint density is

$$f_t(X', b) = (2t)^{-J} \mathbb{E}_\mu \mathbf{1}_{\{\|X'(s-s_0)-b\|_\infty \leq t\}}.$$

Expand its integer p th power using independent $e_1, \dots, e_p$. For fixed b , Tonelli's theorem and independence of the J rows give

$$\int f_t(X', b)^p dP_{X'} = (2t)^{-Jp} \mathbb{E}_{e_1, \dots, e_p} \prod_{j=1}^J \Pr_x \{|\langle x, e_i \rangle - b_j| \leq t \text{ for all } i \leq p\}.$$

For almost every tuple the inner-product vector for one row is Gaussian with nonsingular covariance G . Its density is bounded by $(2\pi)^{-p/2}(\det G)^{-1/2}$. Each box probability is at most $(2t)^p$ times this supremum. The $2t$ powers cancel, and (57) gives, uniformly in t, b ,

$$\int f_t(X', b)^p dP_{X'} \leq (Ce^{CQ_*/n}/R)^{Jp}.$$

Integrating on Ω therefore bounds the local p th power integral by $|\Omega|(Ce^{CQ_*/n}/R)^{Jp}$.

For a continuous compactly supported test function on the product of the matrix space and Ω , its integral against the jittered law converges to its integral against the exact law as $t \downarrow 0$. Extend the test by zero outside a compact subset of Ω and use bounded convergence. Hölder bounds these limiting integrals by the local L^p bound times the test's $L^{p/(p-1)}(dP_{X'}, db)$ norm. Density of such tests and L^p duality give a density for the exact law on this open region, with

$$\int_\Omega \int f(X', b)^p dP_{X'} db \leq |\Omega|(Ce^{CQ_*/n}/R)^{Jp}. \quad (58)$$

Equality on the tests identifies the represented measure with the restricted exact law. This asserts a density on Ω , which is the only region used below.

On Ω , the Gaussian reference satisfies $h_R(b) \geq (CR)^{-J}$. Also the ball-volume estimate gives $|\Omega| \leq (CR)^J$. In the integral of $f^p h_R^{1-p}$, the powers of R from (58) are

$$R^{-Jp} R^J R^{J(p-1)} = 1.$$

Taking the p th root gives (53). The auxiliary jitter has been removed before the exact observation rule is applied. $\square$

6.3 Two copies of the block and the state information

Lemma 6.3 (Diffuse block information). *Let μ be a probability on a depth- h dyadic cell A such that, for some $q \in (0, 1]$, every relative-depth- ℓ descendant D satisfies $\mu(D) \leq q^{-1}2^{-n\ell/2}$, and put $Q_* = \log(1/q)$. Suppose $\log W \leq n^2$, with $k = \lfloor n/64 \rfloor$ as above. From a fixed entering state, let u be any variable with at most W values obtained by a measurable probability kernel from a k -row standard Gaussian matrix X and exact labels $Xs(z)$, where X is independent of $z \sim \mu$. Then*

$$I_\mu(z; u) \leq Cn + CQ_*. \quad (59)$$

Proof. Put $R = 2^{-h}$ and $s_0 = s(\text{center } A)$. Write the kernel probabilities as $g_u(X, Y) \in [0, 1]$, with $\sum_u g_u(X, Y) = 1$ for every matrix-label pair (X, Y) . Define

$$\mathcal{G} = \{\|X(s - s_0)\| \leq 2\sqrt{k}R\}, \quad K_u(s) = \mathbb{E}_X[\mathbf{1}_{\mathcal{G}} g_u(X, Xs)],$$

and define a comparison probability vector by

$$\rho_u = \mathbb{E}_{X,Z} g_u(X, X s_0 + RZ), \quad Z \sim N(0, I_k) \text{ independently.}$$

In $\mathbb{E}_\mu K_u(s)^2$, use two independent block matrices at the same signal s . Their stack has $J = 2k$ rows. When both blocks satisfy $\mathcal{G}$, the stacked residual norm is at most $2\sqrt{J}R < 3\sqrt{J}R$, inside Ω . Apply Lemma 6.2 and Hölder's inequality to this exact joint law. Under the Gaussian reference, the two matrices and the two residual vectors are independent between blocks. The product of the two g_u 's lies in $[0, 1]$ and has reference expectation ρ_u^2 . Dropping the good indicators in its conjugate norm gives

$$\mathbb{E}_\mu K_u(s)^2 \leq C^J e^{C(J/n)Q_*} \rho_u^{2(1-1/p)}. \quad (60)$$

Thus $\rho_u = 0$ implies $K_u = 0$ almost surely.

For every fixed s , the probability of $\mathcal{G}^c$ is at most

$$e^{-k} 2^{k/2} = e^{-ck} =: \delta < 1$$

for an absolute $c > 0$, by the Gaussian exponential moment at $1/4$. Put $g_0(s) = \sum_u K_u(s) \geq 1 - \delta$ and $P_g = \mathbb{E}_\mu g_0(s)$. Including the good indicator with u can only increase information. The indicator costs at most $\log 2$, and the weighted bad conditional information costs at most $\delta \log W$.

The weighted good conditional information is at most

$$\mathbb{E}_\mu \sum_u K_u(s) \log \frac{K_u(s)}{g_0(s) \rho_u}. \quad (61)$$

Indeed, replacing the actual conditional marginal $\Pr(u | \mathcal{G})$ by ρ adds $P_g D(\Pr(u | \mathcal{G}) \| \rho) \geq 0$. Zero states cause no issue by (60). The contribution of $-\log g_0(s)$ is at most $\log(1/(1 - \delta))$. For the remaining term, Jensen's inequality under the probability measure with weights $\mu(ds) K_u(s)/P_g$ gives

$$\begin{aligned} \mathbb{E}_\mu \sum_u K_u(s) \log \frac{K_u(s)}{\rho_u} &\leq P_g \log \left(\frac{1}{P_g} \sum_{u: \rho_u > 0} \frac{\mathbb{E}_\mu K_u(s)^2}{\rho_u} \right) \\ &\leq P_g \log \left(\frac{C^J e^{C(J/n)Q_*} W^{2/p}}{P_g} \right). \end{aligned}$$

The last inequality uses (60) and $\sum_u \rho_u^{1-2/p} \leq W^{2/p}$, by concavity for $p \geq 2$. All sums are finite. Combining the terms gives

$$I_\mu(z; u) \leq Cn + CQ_* + \frac{2}{p} \log W + e^{-ck} \log W.$$

Since k, p are proportional to n and $\log W \leq n^2$, the last two terms are $O(n)$. This proves the lemma. $\square$

6.4 The finite-block information bound

The split identity and the block estimate can now be combined without an accuracy or stopping assumption. We keep every split label in the analysis history, so its entropy must be included in the bound.

Proposition 6.4 (Information after finite splitting). *Let $z \sim \tau$, and let $B \geq 0$ and $W \geq 1$ be integers with $\log W \leq n^2$. Independently of z , draw independent standard Gaussian $k \times d$ matrices $X_1, \dots, X_B$, with k as fixed above. Consider a computation with a fixed initial state u_0 and at most W states at each block boundary. At block b , the outgoing state u_b is an everywhere-defined Borel function of $(u_{b-1}, X_b, X_b s(z))$.*

Before each block one can reveal a finite partition label i_b , determined by z and the preceding analysis history, such that the complete history

$$\mathcal{H} = (i_1, u_1, \dots, i_B, u_B)$$

has finite range and satisfies $I(z; \mathcal{H}) \leq 4C'nB$ for an absolute constant $C' > 0$ and all sufficiently large n . The labels indicate nested dyadic cells containing z ; they do not alter the computation's state rules.

Proof. Before each block, apply Lemma 6.1 to the law conditional on the analysis transcript, and record its label. Also record the outgoing boundary state after the block. Initially the cell is the whole cube and the law is τ . Every positive-probability posterior has bounded density relative to its current cell's uniform law: splitting preserves this property, and conditioning on a state multiplies the density by a likelihood in $[0, 1]$ and divides by a positive state probability. Every split is finite, so each stage has finitely many histories. Future Gaussian rows remain independent of the signal and these histories, which depend only on the signal and past blocks. The block from a known entering state is a measurable kernel of its exact data. Lemma 6.3 therefore applies in each selected component.

For a current cell of depth h and conditional law μ , define

$$\Phi = \frac{1}{4}n(\log 2)h + D(\mu \| \tau_A). \quad (62)$$

It is nonnegative. By (52), the expected change under a split is

$$H(i) - \frac{3}{4}n(\log 2)\mathbb{E}_i j_i \leq -\frac{1}{4}n(\log 2)\mathbb{E}_i j_i + \mathbb{E}_i Q_i.$$

Within component i , conditioning on its outgoing state keeps the cell fixed and increases the expected relative entropy by exactly $I_{\mu_i}(z; u)$. This follows by multiplying the density by $\Pr(u | z, i) / \Pr(u | i)$, taking logarithms, and averaging. The block lemma bounds this increase by $Cn + CQ_i$. Thus the combined conditional drift is at most

$$Cn - \frac{1}{4}n(\log 2)\mathbb{E}_i j_i + (C + 1)\mathbb{E}_i Q_i \leq Cn + (C + 1) + (C + 1 - \frac{1}{4}n \log 2)\mathbb{E}_i j_i \leq C'n \quad (63)$$

for large n , using $\mathbb{E}_i Q_i \leq 1 + \mathbb{E}_i j_i$. Only this expected bound on Q_i is used.

The initial potential is zero, so after B blocks $\mathbb{E}\Phi \leq C'nB$. For the final transcript $\mathcal{H}$, its posterior in a depth- h cell satisfies

$$D(P(z | \mathcal{H}) \| \tau) = n(\log 2)h + D(P(z | \mathcal{H}) \| \tau_A) \leq 4\Phi.$$

Averaging over the finite history gives

$$I(z; \mathcal{H}) \leq 4C'nB. \quad (64)$$

□

6.5 Application to the learner

Use the finite-state model of Section 2, with $M(d) = o(d^2)$, integer horizon T , and every-signal angular success at least $2/3$ at $0 < \epsilon \leq 1/10$. We will prove $T \geq cd \log(1/\epsilon)$, with an absolute constant and a dimension threshold that may depend on the memory sequence. First record the route's separate sample bound for the original randomized learner:

$$T \geq d/16. \quad (65)$$

Take a uniform full-sphere signal and pregenerate all T rows, including unused rows. Let V be their span. Conditional on the rows and $P_V S$, the residual has mean zero; this remains true after conditioning on the learner's independent random tape. All labels and the output are functions of the conditioned variables. Consequently

$$\mathbb{E}\langle S, \hat{S} \rangle \leq \mathbb{E}\|P_V S\| \leq \sqrt{\mathbb{E}\|P_V S\|^2} \leq \sqrt{T/d}.$$

The last inequality uses isotropy and independence of the rows and signal. Success gives

$$\mathbb{E}\langle S, \hat{S} \rangle \geq (2/3) \cos(1/10) - 1/3 > 1/4,$$

which proves (65).

Put $L = \log(1/\epsilon)$. In the cube experiment, first fix shared rule randomness with success greater than $1/2$. The graph derivative has full rank, so Lemmas 2.2 and 2.1 give Borel representatives for that fixed choice. Then fix the remaining transition and output randomness, together with the initial state, by averaging. Conditional on the shared seed these choices are independent of the signal and rows, by the joint independence in Section 2. This gives a deterministic program of cube success at least $1/2$. Its terminal state and stopping index determine its output, giving at most $(T+1)2^M$ possible outputs. Lemma 3.1 bounds the cube success mass of each fixed output by $(5\epsilon)^n$. Hence

$$n(L - \log 5) \leq M \log 2 + \log(2(T+1)). \quad (66)$$

We may restrict to $T < dL$, since the complementary case already has the required lower bound. For $L \geq \log 10$, put $a_0 = \log 2 / \log 10 > 0$; then $L - \log 5 \geq a_0 L$. Also $\log(2(T+1)) \leq C + \log d + \log L \leq C + \log d + L$. Thus

$$(a_0 n - 1)L \leq M \log 2 + C + \log d,$$

so $L = o(d)$. Pad the deterministic program after an early stop, retaining its terminal state and stopping index and ignoring later samples. The width

$$W = (T+2)2^M \quad (67)$$

is enough for this padded program. Since $T < dL$, $L = o(d)$, and $M = o(d^2)$, it satisfies $\log W = o(n^2)$, hence $\log W \leq n^2$ eventually. Use $B = \lceil T/k \rceil$ blocks, adding ignored fresh rows to complete the last block.

The fixed initial state, independent blocks, and Borel transition rules now satisfy Proposition 6.4. Let $\mathcal{H}$ be its augmented history. The padded state retains the terminal state and stopping index, so the output is a function of $\mathcal{H}$.

Under the joint law, the output's first n coordinates are within ϵ of z with probability at least $1/2$; under the product law the probability is at most $(5\epsilon)^n$ by the coordinate-ball volume calculation in Lemma 3.1. Equation (1) yields

$$I(z; \mathcal{H}) \geq \frac{1}{2}n(L - \log 5) - \log 2 \geq cnL \quad (68)$$

for sufficiently large n , uniformly for $L \geq \log 10$. Combining (64) and (68) gives $B \geq c_1 L$. Since $B \leq T/k + 1$,

$$T \geq k(c_1 L - 1).$$

For $L \geq 2/c_1$, this is $\Omega(dL)$. For $\log 10 \leq L < 2/c_1$, the separate bound (65) gives the same order. The previously excluded case $T \geq dL$ is immediate. This proves the worst-signal precision lower bound for every stated accuracy sequence, with absolute constants and a dimension threshold that may depend on the memory sequence.

7 Likelihood truncation and selected dyadic cells

This section gives a forward proof of the precision lower bound using a cube chart of the sphere. At a fixed incoming state, the likelihood of a block-ending state at a signal is the probability that the fresh block ends in that state when the signal is fixed. A block may concentrate the conditional signal law, but a Gaussian projection estimate limits the increase in a typical state likelihood. We then select a dyadic cell for analysis and condition on the event that this cell was selected. The cost of that conditioning and the depth of the cell enter one density bound. Iterating the bound prevents a small number of blocks from localizing the signal to the accuracy required by the output.

The use of high Gaussian projection moments has a methodological precedent in Sharan, Sidford, and Valiant [12, Section 7, Lemma 12]. We prove the exact-label estimate used here, including its dependence on the mass of a restricted set. The lower bound in this section starts from success for every signal: that hypothesis supplies both the cube-prior average used by the forward argument and the uniform-sphere average used at constant accuracy.

7.1 The learner, the cube prior, and measurable rules

We use the finite-state learner of Section 2, including its jointly measurable experiment, joint independence of the shared rule seed and initialization from the signal and rows, and permitted per-seed completed-measurability. The deterministic horizon is $T \geq 0$, and the output uses the terminal state, stopping index, and fresh randomness.

Theorem 7.1 (Precision lower bound from likelihood truncation). *Let $M(d) = o(d^2)$ be a nonnegative integer-valued memory bound, and let $0 < \epsilon(d) \leq 1/10$. Suppose a family of learners in the model above satisfies*

$$\mathbb{P}\{\arccos\langle \hat{s}, s \rangle \leq \epsilon(d)\} \geq \frac{2}{3} \quad \text{for every } s \in S^{d-1}.$$

There is an absolute $c > 0$ such that, for all sufficiently large d ,

$$T(d) \geq c d \log_2 \frac{1}{\epsilon(d)}.$$

The eventual dimension threshold may depend on the memory sequence.

Put $p = d - 1$ and consider the normalized cube law

$$Q_* = \left[-\frac{1}{2\sqrt{p}}, \frac{1}{2\sqrt{p}} \right]^p, \quad \mu = \text{Unif}(Q_*), \quad s(z) = (z, \sqrt{1 - \|z\|^2}). \quad (69)$$

The parameter law has density $p^{p/2}$ on the cube. This is the cube chart of Lemma 3.1, with n renamed p . That lemma proves that s is 2-Lipschitz and, for every fixed unit output v ,

$$\mu\{z : \arccos\langle v, s(z) \rangle \leq \epsilon\} \leq (\sqrt{2\pi e} \epsilon)^p \leq (5\epsilon)^p. \quad (70)$$

Write V_h for the volume of the unit ball in $\mathbb{R}^h$. The same Gaussian-integral proof gives $V_h \leq (\sqrt{2\pi e}/\sqrt{h})^h < (5/\sqrt{h})^h$, which will also be used below.

The following reduction puts the learner in the form required by the block argument. It makes the order of the two operations explicit: first fix the shared rule-selection randomness, then choose Borel versions for the resulting cube-prior experiment.

Lemma 7.2 (A deterministic Borel learner for the cube experiment). *Fix $d \geq 2$, M , and a finite horizon T . If a learner in the stated model has average success at least $2/3$ under (69), then there is a deterministic learner with Borel transition and stopping rules, the same state and horizon bounds, and a fixed unit output at each terminal state/index label, whose average success under that prior is at least $3/5$.*

Proof. Choose a shared-seed value with conditional cube-prior success greater than $3/5$. The joint experiment in Section 2 makes this conditioning meaningful, and the conditional initial-state law remains independent of the signal and rows. The derivative of the chart has the identity in its first p rows, so Lemma 2.2 proves absolute continuity of its one-pair law. Lemma 2.1 therefore provides everywhere Borel finite kernels preserving this fixed-seed cube-prior joint law of the signal, path, stopping index, and output.

Use the finite-randomness reduction following that lemma. Sample the conditional initial state, independent uniforms for all finite choices indexed by state and index (including an input-free index-zero stopping choice), and one output from each terminal state/index law. These finitely many choices are independent of the inputs. Fixing a realization by averaging retains cube-prior success at least $3/5$. The resulting state rules are Borel, and the state and horizon bounds are unchanged. $\square$

7.2 The projection estimate inside a regular cell

We next control one block, starting from a parameter law with a bound on concentration at every smaller dyadic scale. For sufficiently large d , set

$$n = \lfloor d/32 \rfloor, \quad k = \lfloor d/8 \rfloor, \quad \lambda = p/2, \quad Q = \lambda - k, \quad \alpha = \frac{n}{Q}. \quad (71)$$

Here n is the number of rows in a block and k is the moment order. For example, $d \geq 64$ ensures $n \geq 1$, $k \geq 2$, $Q \geq d/3$, and $\alpha \leq 1/4$.

Subdivide Q_* dyadically in every coordinate, using half-open cells so that every parameter belongs to one cell at each depth. A cell C at depth l has side length $R/\sqrt{p}$, where $R = 2^{-l}$. A descendant at relative depth j is a depth- $l+j$ cell inside C . Call a Borel probability measure P supported in C regular when

$$P(V) \leq 2^d 2^{-\lambda j} \quad \text{for every descendant } V \text{ at relative depth } j \geq 0. \quad (72)$$

We need a projection estimate that retains the probability of a restricted parameter set. In the likelihood cutoff, this set will consist of signals for which a particular state is unusually likely. The factor $\rho^{1-\alpha}$ below lets the small mass ρ of that set survive both the projection and the choice of a state.

Lemma 7.3 (Restricted mass and exact projection moments). *There are absolute constants $C_0, C_1, C_2 \geq 1$ such that the following holds with (71) for all sufficiently large d . Let P be regular in a cell C of scale R . For every $x \in \mathbb{R}^d$, $u > 0$, and affine subspace $H \subseteq \mathbb{R}^d$ of dimension at most k ,*

$$P\{z : s(z) \in B(x, u)\} \leq C_0^d (u/R)^\lambda, \quad (73)$$

$$P\{z : \text{dist}(s(z), H) \leq u\} \leq (C_1 u/R)^Q. \quad (74)$$

For a Borel set $E \subseteq C$ of mass $\rho = P(E) > 0$,

$$\int_E \text{dist}(s(z), H)^{-n} dP(z) \leq C_2^d R^{-n} \rho^{1-\alpha}. \quad (75)$$

Let $\gamma_{n,d}$ denote standard Gaussian probability on $\mathbb{R}^{n \times d}$ and put $\beta(dA, dy) = \gamma_{n,d}(dA) dy$. For $\eta > 0$, let φ_η be the centered Gaussian density on $\mathbb{R}^n$ with covariance $\eta^2 I_n$, and define

$$J_\eta(A, y) = \int_E \varphi_\eta(y - As(z)) dP(z).$$

Then

$$\int J_\eta^k d\beta \leq (C_2^d R^{-n})^{k-1} \rho^{1+(k-1)(1-\alpha)}. \quad (76)$$

For every Borel joint set $B \subseteq \mathbb{R}^{n \times d} \times \mathbb{R}^n$ with $\beta(B) < \infty$, the exact observations satisfy

$$\mathbb{P}_{z \sim P, A \sim \gamma_{n,d}}\{z \in E, (A, As(z)) \in B\} \leq \beta(B)^{1-1/k} (C_2^d R^{-n})^{1-1/k} \rho^{1-\alpha}. \quad (77)$$

The same assertions involving E are trivial when $\rho = 0$.

Proof. For $0 < u < R$, choose an integer $j \geq 1$ with $R2^{-j} \leq u < 2R2^{-j}$, and write $\delta = R2^{-j}$. Project $B(x, u)$ onto the first p coordinates. Each relative-depth j cell meeting the projected ball has diameter δ , so the union of those cells lies in a ball of radius $u + \delta < 3\delta$. The cells have disjoint interiors and volume $(\delta/\sqrt{p})^p$. The unit-ball estimate used in (70) bounds their number by

$$\frac{V_p(3\delta)^p}{(\delta/\sqrt{p})^p} \leq 15^p.$$

Using (72) for each cell gives $P\{s(z) \in B(x, u)\} \leq 15^p 2^d 2^{-\lambda j}$. Since $2^{-j} \leq u/R$, this proves (73) after choosing C_0 absolutely. For $u \geq R$ the bound follows from $P(C) = 1$.

Let s_0 be the lift of the center of C . The cell has Euclidean radius $R/2$ in parameter space, so the Lipschitz bound implies $\|s(z) - s_0\| \leq R \leq 2R$ for $z \in C$. Orthogonal projection onto H is nonexpansive. Thus the projections of all these lifted points lie in the radius- $2R$ ball in H about the projection of s_0 . Disjoint ball packing covers that ball by at most $(1 + 4R/u)^k$ balls of radius u . A point at distance at most u from H whose projection lies in one of these balls is within $2u$ of its center. For $u \leq R$, applying (73) to the cover gives

$$P\{\text{dist}(s(z), H) \leq u\} \leq (1 + 4R/u)^k C_0^d (2u/R)^\lambda \leq C^d (u/R)^{\lambda-k}.$$

Here and below C denotes an absolute constant that may increase. The identity $Q = (d-1)/2 - \lfloor d/8 \rfloor \geq d/3$ for large d lets us absorb C^d into C_1^Q . For $u > R$, enlarge C_1 so that the right side is at least one. This proves (74). In particular every affine subspace of the stated dimension has P -mass zero, by letting u decrease to zero.

To prove (75), apply the layer-cake formula to $(\text{dist}(s(z), H)/R)^{-n} \mathbf{1}_E$. At height $t > 0$ its tail has mass at most $\min\{\rho, C_1^Q t^{-Q/n}\}$. With $t_0 = C_1^n \rho^{-n/Q}$,

$$\begin{aligned} \int_0^\infty \min\{\rho, C_1^Q t^{-Q/n}\} dt &\leq \rho t_0 + C_1^Q \int_{t_0}^\infty t^{-Q/n} dt \\ &= \frac{C_1^n}{1-\alpha} \rho^{1-\alpha}. \end{aligned}$$

Because $\alpha \leq 1/4$ and $n \leq d$, an absolute C_2 bounds the prefactor by C_2^d . Restoring R^{-n} proves the assertion.

We turn from affine distances to the projection moment. Tonelli's theorem expands $\int J_\eta^k d\beta$ over $z_1, \dots, z_k \in E$. Write $s_i = s(z_i)$. For fixed A and this tuple, integration of $\prod_{i=1}^k \varphi_\eta(y - As_i)$ over the common label y is the density at zero of the $k-1$ differences from the first of the independently mollified labels $As_i + \eta Z_i$. After averaging over A , the differences are centered Gaussian, independently across rows, with per-row covariance

$$G + \eta^2(I + \mathbf{1}\mathbf{1}^\top), \quad G = (\langle s_i - s_1, s_j - s_1 \rangle)_{i,j=2}^k. \quad (78)$$

Their density at zero is

$$(2\pi)^{-n(k-1)/2} \det(G + \eta^2(I + \mathbf{1}\mathbf{1}^\top))^{-n/2} \leq \det(G)^{-n/2},$$

where the upper bound is interpreted as infinity when G is singular. For nonsingular G , the inequality follows by factoring $G^{1/2}$ and observing that all eigenvalues of $I + \eta^2 G^{-1/2}(I + \mathbf{1}\mathbf{1}^\top)G^{-1/2}$ are at least one. Gram-Schmidt gives

$$\det(G)^{-n/2} = \prod_{i=2}^k \text{dist}(s_i, \text{aff}(s_1, \dots, s_{i-1}))^{-n}.$$

The affine spaces here have dimension at most $k-2$. The null-mass consequence of (74), applied successively, shows that degenerate tuples have product P -measure zero. Integrate the last display from z_k down to z_2 using (75), then integrate z_1 with mass ρ . The result is

$$\rho(C_2^d R^{-n} \rho^{1-\alpha})^{k-1},$$

which is (76).

Hölder's inequality now bounds the mollified measure of B by

$$\beta(B)^{1-1/k} (C_2^d R^{-n})^{1-1/k} \rho^{1-\alpha+\alpha/k}.$$

Since $\rho \leq 1$, this implies the weaker power of ρ in (77). For an open B , couple the mollified labels as $As(z) + \eta Z$ using one independent standard Gaussian Z . As η decreases to zero, membership of $(A, As(z))$ in B implies eventual membership of the approximants. Fatou's lemma therefore passes the same bound to the exact observation measure. The Gaussian-Lebesgue measure β is a locally finite regular Borel measure on the Euclidean joint space. Its outer regularity extends the bound from open sets to every Borel B of finite measure. In particular the exact measure vanishes on β -null sets, so the same bound extends to the completion of β as well. The mollifier was used only in this calculation; the observation in (77) is the exact label. $\square$

We now apply the exact-set estimate to the signals for which one state's likelihood exceeds a multiple of its marginal probability. The proof restricts labels to a ball about the projected cell center. The resulting sets of observations producing different states are disjoint, and their total reference volume will control the sum over states.

Lemma 7.4 (A regular-cell likelihood cutoff). *There are absolute $a, c_1 > 0$ and a dimension threshold such that the following holds. Let P be regular in a cell C , let $A \sim \gamma_{n,d}$ be independent of $z \sim P$, and let*

$$W = f(A, As(z))$$

for a Borel map f with at most N values. Suppose $N^{1/k} \leq 2^d$, and set

$$q_w(z) = \mathbb{P}_A\{f(A, As(z)) = w\}, \quad b_w = \int q_w dP.$$

Then

$$\mathbb{P}\{q_W(z) > 2^{ad}b_W\} \leq 2^{-c_1d}. \quad (79)$$

Proof. The functions q_w are Borel by integration of the Borel routing indicators. For $b_w > 0$ define $E_w = \{z : q_w(z) > 2^{ad}b_w\}$. Markov's inequality gives $P(E_w) \leq 2^{-ad}$. A state with $b_w = 0$ is reached with probability zero and contributes nothing.

Retain the lift s_0 of the center of C used in the preceding proof. Conditional on z , the vector $A(s(z) - s_0)$ is distributed as $\|s(z) - s_0\|Z$ for a standard Gaussian $Z \in \mathbb{R}^n$. Because $\|s(z) - s_0\| \leq 2R$,

$$\mathbb{P}\{\|As(z) - As_0\| > 4R\sqrt{n}\} \leq \mathbb{P}\{\|Z\|^2 > 4n\} \leq 2^{n/2}e^{-n}. \quad (80)$$

The last inequality uses $\mathbb{E}e^{\|Z\|^2/4} = 2^{n/2}$. Since $n \geq d/64$ for large d , the right side is exponentially small in d .

Define the joint domain

$$\mathcal{D} = \{(A, y) : \|y - As_0\| \leq 4R\sqrt{n}\}$$

and let $B_w = \mathcal{D} \cap f^{-1}(\{w\})$. These Borel sets are disjoint, and the volume bound gives

$$\sum_w \beta(B_w) \leq \beta(\mathcal{D}) = V_n(4R\sqrt{n})^n \leq (20R)^n.$$

Apply (77) to E_w and B_w , and sum over w . Hölder's inequality on the finite state set yields

$$\begin{aligned} \sum_w \mathbb{P}\{z \in E_w, (A, As(z)) \in B_w\} &\leq (C_2^d R^{-n})^{1-1/k} 2^{-ad(1-\alpha)} N^{1/k} \left(\sum_w \beta(B_w) \right)^{1-1/k} \\ &\leq C_4^d N^{1/k} 2^{-ad(1-\alpha)} \end{aligned} \quad (81)$$

for an absolute $C_4 \geq 1$. The powers of R cancel exactly. Since $1 - \alpha \geq 3/4$ and $N^{1/k} \leq 2^d$, choose a so that $3a/4 \geq \log_2 C_4 + 3$. The last display is then at most 2^{-2d} . Adding (80) and reducing an absolute $c_1 > 0$ if necessary proves (79) for large d . $\square$

Remark 7.5. The projection estimate also verifies the cube-prior absolute continuity used in Lemma 7.2. At the root, $\mu(V) = 2^{-pj}$ for a depth- j cell, so μ is regular. Taking $P = \mu$ and $E = Q_*$ in (77) shows that the exact joint block law is absolutely continuous with respect to β . For the one-pair reference measure $\lambda_d(dx, dy) = \gamma_d(dx) dy$ from Section 2, the inverse image of a λ_d -null set under one-row projection is β -null by the product structure and σ -finiteness. The one-pair marginal is therefore absolutely continuous as well.

7.3 Restoring regularity by a selection event

Fix a deterministic Borel learner, padded to m blocks of n rows. At each block boundary let its state alphabet have size at most N , with $N^{1/k} \leq 2^d$. A halted path keeps a tag recording its terminal state and index; the quantitative application below supplies this padding. For a fixed starting state, the next boundary state is a Borel function of the whole exact block $(A, As(z))$. Trailing rows after a halt are generated and ignored.

For the proof only, a history records the boundary states, selected dyadic cells, and survival of the deletions defined below. At time t it is an event determined by the parameter and the first t blocks. We condition only on histories of positive probability. Future rows remain independent standard Gaussian rows conditional on such a history and the parameter, since those rows are independent of the parameter and all preceding blocks.

Let μ_C be uniform probability on a cell C , and put $D = a + 2$, with a from Lemma 7.4. We will keep the following two properties after t blocks: the history fixes a cell C of depth l , its parameter law P is regular in C , and

$$P \leq K\mu_C, \quad K = 2^{Ddt-(p-\lambda)l}. \quad (82)$$

The measure inequality means that it holds on every Borel set. Initially $C = Q_*$, $P = \mu$, $t = l = 0$, so both properties hold.

Lemma 7.6 (One block and one selected cell). *Suppose a positive-probability history after t blocks has a fixed state and a conditional law satisfying (72) and (82) in a cell C of depth l . One can discard conditional probability at most*

$$3 \cdot 2^{-c_1 d} + (\lfloor J_t \rfloor + 1)2^{-d}, \quad J_t = \frac{ad + 1 + Ddt}{p - \lambda}, \quad (83)$$

and partition the remaining outcomes into finitely many positive-probability histories satisfying both properties at time $t + 1$.

Proof. Run the fresh next block, and let $\mathcal{E} = \{q_W(z) > 2^{ad}b_W\}$ be the event in (79). Write $\delta = \mathbb{P}(\mathcal{E}) \leq 2^{-c_1 d}$. Discard $\mathcal{E}$. For $b_w > 0$, let $r_w = \mathbb{P}(\mathcal{E}^c \mid W = w)$, and also discard every state with $r_w < 1/2$. The mass of these states is at most 2δ :

$$\sum_{w:r_w < 1/2} b_w \leq 2 \sum_{w:r_w < 1/2} \mathbb{P}(\mathcal{E} \cap \{W = w\}) \leq 2\delta.$$

The two deletions therefore cost at most 3δ .

For a remaining state w , condition on $W = w$ and $\mathcal{E}^c$. Its parameter law P' is given by

$$P'(B) = \frac{1}{b_w r_w} \int_B \mathbf{1}_{\{q_w(z) \leq 2^{ad}b_w\}} q_w(z) dP(z)$$

for Borel B . Since $r_w \geq 1/2$,

$$P' \leq 2^{ad+1}P \leq 2^{ad+1}K\mu_C. \quad (84)$$

For $z \in C$, let $U_j(z)$ be the cell containing z at relative depth j . Select the least $j \geq 0$ maximizing the score

$$2^{\lambda j} P'(U_j(z)). \quad (85)$$

The score at $j = 0$ is one. Since $\mu_C(U_j(z)) = 2^{-pj}$, (84) gives

$$2^{\lambda j} P'(U_j(z)) \leq 2^{ad+1} K 2^{-(p-\lambda)j} \leq 2^{ad+1+Ddt-(p-\lambda)j}.$$

This upper bound tends to zero. Thus the maximum is attained, and every selected depth is at most J_t , since a selected score is at least one. The least-maximizer rule is unambiguous. It is Borel: there are finitely many possible depths, the cell at each depth is a Borel function of z , and each score is a fixed number on that cell.

For each possible selected cell U , define the actual selection event

$$F_U = \{z : \text{the rule in (85) selects } U\} \subseteq U. \quad (86)$$

These Borel events form a finite partition of C . Discard those for which $P'(F_U) < 2^{-d}P'(U)$. At a fixed depth the cells partition C , so their P' -masses sum to one. Summing over the at most $\lfloor J_t \rfloor + 1$ depths bounds this deletion by the second term of (83). Events of zero mass need no conditional law.

Fix a remaining event F_U of positive mass, and let j be the relative depth of U . Put $P'' = P'(\cdot | F_U)$. If a descendant V of U at further depth i meets F_U , choose $z \in V \cap F_U$. Maximality on this parameter's cell chain gives

$$2^{\lambda j} P'(U) \geq 2^{\lambda(j+i)} P'(V).$$

Consequently

$$P''(V) \leq \frac{P'(V)}{P'(F_U)} \leq 2^d 2^{-\lambda i},$$

using $P'(F_U) \geq 2^{-d}P'(U)$. If $V \cap F_U$ is empty, its P'' -mass is zero. This proves regularity in U .

The selected score is at least one, so $P'(U) \geq 2^{-\lambda j}$ and $P'(F_U) \geq 2^{-d-\lambda j}$. The cell reference measures satisfy $\mu_C|_U = 2^{-pj}\mu_U$. Therefore

$$\begin{aligned} P'' &\leq 2^{ad+1} K \frac{2^{-pj}}{P'(F_U)} \mu_U \\ &\leq 2^{(a+1)d+1} K 2^{-(p-\lambda)j} \mu_U \leq 2^{Dd} K 2^{-(p-\lambda)j} \mu_U. \end{aligned} \quad (87)$$

The last inequality uses $(a+1)d+1 \leq (a+2)d = Dd$ for $d \geq 1$. Substituting the value of K yields

$$P'' \leq 2^{Dd(t+1)-(p-\lambda)(l+j)} \mu_U,$$

which is (82) at the new time and total depth. The normalization throughout is by $P'(F_U)$, including when F_U is a proper subset of U . $\square$

Proposition 7.7 (The finite-horizon cube success bound). *For sufficiently large d , consider a deterministic Borel computation padded to $m \geq 0$ blocks of n rows as above. Suppose its boundary alphabet has size at most N , where $N^{1/k} \leq 2^d$, and its final boundary state fixes its unit output. Its average success under (69) is at most*

$$\Delta_m + 2^{2Ddm} (5\epsilon)^p, \quad \Delta_m = 3m2^{-c_1 d} + 2^{-d} \sum_{t=0}^{m-1} (\lfloor J_t \rfloor + 1). \quad (88)$$

The sum is empty when $m = 0$. In particular, $\Delta_m = o(1)$ for any sequence $m = o(d)$.

Proof. Apply Lemma 7.6 at each surviving history and at each block. There are finitely many histories at every time. The conditional deletion bounds are uniform, so their unconditional sum is at most Δ_m : at each time, the probabilities of the surviving histories sum to at most one. Also $p - \lambda = (d-1)/2$, and hence $J_t = O(t+1)$ with an absolute constant depending only on the already fixed a . Thus

$$\Delta_m \leq 3m2^{-c_1 d} + C(m^2 + m)2^{-d},$$

which tends to zero when $m = o(d)$.

At a surviving terminal history, integrating (82) forces $K \geq 1$. Hence $(p - \lambda)l \leq Ddm$. Since $\mu(C) = 2^{-pl}$, the bound relative to the original cube probability becomes

$$P \leq 2^{Ddm + \lambda l} \mu \leq 2^{2Ddm} \mu. \quad (89)$$

The last inequality uses $\lambda = p - \lambda$. The history fixes the final boundary state and therefore its output. Equation (70) bounds conditional success by $2^{2Ddm} (5\epsilon)^p$. Average over surviving histories and add the deleted mass to obtain (88). For $m = 0$, the same argument is the fixed-output bound at the root. $\square$

The finite-horizon estimate (88) holds for every m . We will obtain $m = o(d)$ only on the hypothetical short-run sequence in the proof of Theorem 7.1.

7.4 The constant-accuracy endpoint and the precision contradiction

The forward estimate pays a fixed amount at the first block. A separate full-data argument supplies the sample bound at constant accuracy. Even after all labels are revealed, the signal has an undetermined direction on a residual sphere. Its directional second moment will bound the success probability.

Lemma 7.8 (A residual-sphere sample bound). *For $0 < \epsilon \leq 1/10$ and all sufficiently large d , any learner with success at least $2/3$ for every unit signal must have $T > d/4$. This holds even if the learner is given all T observations before producing its output.*

Proof. Let S be uniform on S^{d-1} , independently of the T pregenerated Gaussian rows. The original jointly measurable experiment has prior success equal to the integral of its fixed-signal successes. For each admissible fixed shared seed, Lemmas 2.2 and 2.1 give Borel rules with the same uniform-prior law. The bound below is uniform over these rules, so it bounds the original conditional success before averaging over the seed. This returns to the original learner; it does not reuse the cube-selected program.

If $T \leq d/4$, the span U of the rows has dimension T almost surely. Given the rows, the labels determine $P_U S$, since the row map is injective on U . Conditional on the rows and $P_U S$, the residual $(I - P_U)S$ is uniform on the sphere in $U^\perp$ of radius $r = (1 - \|P_U S\|^2)^{1/2}$. This follows, for example, by representing S as a normalized standard Gaussian and using rotational invariance of its component in $U^\perp$.

Spherical symmetry gives $\mathbb{E}\|P_U S\|^2 = T/d$, so Markov's inequality yields

$$\mathbb{P}\{r < 1/2\} = \mathbb{P}\{\|P_U S\|^2 > 3/4\} \leq \frac{4T}{3d} \leq \frac{1}{3}.$$

Suppose $r \geq 1/2$ and condition on the rows and labels. Fix any possible output, and let w be its projection onto $U^\perp$. If $w = 0$, its residual error is $r > 1/10$. If $w \neq 0$, write Θ for the unit residual direction and $e = w/\|w\|$. When $|\langle \Theta, e \rangle| < 1/2$, the component of $r\Theta$ orthogonal to e has length greater than $\sqrt{3}r/2 > 1/10$, so the output cannot succeed. The second moment of $\langle \Theta, e \rangle$ for the uniform direction in the $(d - T)$ -dimensional space $U^\perp$ is $1/(d - T)$. Markov's inequality therefore bounds success for the fixed output by $4/(d - T)$. The learner's conditional output law uses only the revealed data and independent randomness, so averaging over that law preserves the bound. Angular success implies the Euclidean error bound used here.

The uniform-prior success is consequently at most $1/3 + 4/(d - T)$, which is below $2/3$ for large d when $T \leq d/4$. The every-signal premise gives uniform-prior average at least $2/3$, a contradiction. $\square$

Proof of Theorem 7.1. Write $L = \log_2(1/\epsilon)$ and keep the absolute constants a, c_1, D already fixed. Choose

$$L_0 \geq \max\{32D, 2\log_2 5\}, \quad 0 < c \leq \min\left\{1, \frac{1}{4L_0}, \frac{1}{2048D}\right\}. \quad (90)$$

Suppose for contradiction that there are infinitely many dimensions with successful learners satisfying

$$T < c d L. \quad (91)$$

All asymptotic reductions below concern these dimensions only.

We first obtain the precision range forced by the finite set of terminal labels. For fixed shared rules, there are at most $(T+1)2^M$ terminal state/index labels. Let $\mathcal{O}_{t,u}$ be the output law at one such label. For each parameter, its success probability is at most the sum, over all terminal labels, of the success probabilities under their output laws: this drops the probability of reaching each label, which is at most one. Integrating over z and then over each $\mathcal{O}_{t,u}$, using (70), bounds cube-prior success by $(T+1)2^M(5\epsilon)^p$. This bound is uniform in the fixed shared rules, so it also bounds the original randomized learner. Its every-signal guarantee gives cube-prior average at least $2/3$. Thus

$$p(L - \log_2 5) \leq M + \log_2(T+1) + \log_2(3/2). \quad (92)$$

Because $L \geq \log_2 10$, the absolute number $c_0 = 1 - \log_2 5 / \log_2 10$ is positive and $L - \log_2 5 \geq c_0 L$. Under (91) and $c \leq 1$,

$$\log_2(T+1) \leq 1 + \log_2 d + \log_2 L \leq 1 + \log_2 d + L.$$

For large d , absorb the last L into $c_0 p L$ on the left of (92). It follows that

$$L = O\left(\frac{M + \log d}{d}\right) = o(d), \quad T = o(d^2). \quad (93)$$

The constants in this bound are absolute. In particular, for a fixed memory sequence it bounds every L arising in (91) by the same sequence $o(d)$.

At each such dimension, Lemma 7.2 supplies a deterministic Borel learner with cube-prior success at least $3/5$. Pad a halted path by retaining its terminal state and its terminal index. At any later layer, the continuing states together with all terminal tags have total width at most

$$N = (T+2)2^M. \quad (94)$$

Pad the last block with ignored rows and put $m = \lceil T/n \rceil$. By (93), $\log_2 N = o(d^2)$ and $m = o(d)$. Since $k = \Theta(d)$, $N^{1/k} \leq 2^d$ for all sufficiently large dimensions under consideration. Proposition 7.7 applies and gives

$$\frac{3}{5} \leq \Delta_m + 2^{2Ddm}(5\epsilon)^p = o(1) + 2^{2Ddm}(5\epsilon)^p. \quad (95)$$

The $o(1)$ is uniform over the short-run precisions for a fixed memory sequence, by the uniform bound on L just obtained.

If $L \leq L_0$, then (91) and $c \leq 1/(4L_0)$ give $T < d/4$. Apply Lemma 7.8 to the original learner and its uniform-sphere average, obtaining a contradiction for large d . This use of the original learner preserves the prior of that lemma.

It remains to consider $L > L_0$. For large d , $n \geq d/64$, so

$$m \leq 64cL + 1, \quad 2Ddm \leq 128Dc dL + 2Dd \leq dL/8.$$

The last inequality follows from $c \leq 1/(2048D)$ and $L \geq 32D$. Also $p = d - 1 \geq d/2$ and $L \geq 2 \log_2 5$, whence

$$p(L - \log_2 5) \geq dL/4.$$

Therefore the second term in (95) is at most $2^{-dL/8}$. It tends to zero, as does Δ_m , contradicting the fixed lower bound $3/5$. No infinite subsequence (91) exists. This proves the asserted sample bound with the absolute c in (90). $\square$

8 A spherical grid with an entropy balance

This argument records a cell containing the signal after each block of samples. The outgoing memory state can make the posterior concentrate inside that cell. We therefore choose a finer cell from the new posterior and include its entire label in the analysis history. The decrease in cell size pays for that additional information. A projection estimate then bounds the remaining information acquired in one block.

The use of independent points and inverse distances to their successive affine spans is related to the projection method of Sharan, Sidford, and Valiant [12, Section 7, Lemma 12]. We prove the estimates used here for exact Gaussian labels. The choice of cell below is a global maximum over all finer levels; it need not be a stopping time along the grid.

We use the information conventions of Section 2, and write h for differential entropy when it is finite. Let σ be uniform probability on S^{d-1} , and take d larger than an absolute constant. Set

$$k = p = \lfloor d/8 \rfloor, \quad q = d/2, \quad \bar{q} = q + 2. \quad (96)$$

In particular, $q < d - 1$ and $q - (p - 1) - k$ is a positive multiple of d . Constants denoted by C are positive and absolute and may increase.

The cap estimate in Lemma 2.3 gives, for $z \in S^{d-1}$ and $r > 0$,

$$\sigma(B(z, r)) \leq r^{d-1}, \quad (97)$$

where $B(z, r)$ is the closed Euclidean ball.

For every integer $j \geq 0$, let $\delta_j = 2^{-j}$ and partition $\mathbb{R}^d$ into half-open cubes of side $\delta_j/\sqrt{d}$, using the same origin at every level. These partitions are nested. Write $Q_j(s)$ for the unique cube containing s . Its diameter is δ_j , so (97), centered at any point of its intersection with the sphere, gives

$$\sigma(Q_j(s)) \leq 2^{-(d-1)j}. \quad (98)$$

At most $(C2^j)^d$ level- j cubes meet the sphere. Indeed, their union lies in the ball of radius $1 + \delta_j \leq 2$, and their interiors are disjoint. The volume v_d of the unit ball satisfies $v_d \leq (2\pi e/d)^{d/2}$: integrate $e^{-\|x\|^2/2}$ over the ball of radius $\sqrt{d}$ and compare with its full Gaussian integral. Dividing the volume of the radius-two ball by the cube volume now gives the asserted count.

Lemma 8.1 (A refinement whose label is charged). *Let $S \sim \sigma$, and let G, W be countable discrete variables with $H(G, W) < \infty$. Suppose G determines an integer $a \geq 0$ and a level- a cube containing S almost surely, and that $\mathbb{E}a < \infty$. For every pair (g, w) of positive probability, put $\mu = \mathcal{L}(S \mid G = g, W = w)$. For μ -almost every s , define*

$$J(s) = \max_{j \geq a} \{ \log \mu(Q_j(s)) + q(j - a) \log 2 \},$$

and let $b(s)$ be the least level attaining this maximum. In the joint law define

$$N = \lfloor J(S)/d \rfloor, \quad Z = (b(S), Q_{b(S)}(S)).$$

The choices are measurable and finite almost surely. They satisfy $\mathbb{E}b < \infty$, $H(G, W, Z) < \infty$, and

$$0 \leq N \leq b - a, \quad (99)$$

$$\mathbb{E}J \leq -H(Z | G, W) + H(b | G, W) + q(\log 2)\mathbb{E}(b - a), \quad (100)$$

$$H(b | G, W) + H(N | G, W) \leq 2(1 + \mathbb{E}(b - a)) \log 2. \quad (101)$$

Proof. Fix (g, w) with probability $P_{g,w} > 0$. Conditioning gives $\mu \leq \sigma/P_{g,w}$. Outside one μ -null set, all the cells $Q_j(s)$ have positive μ -mass: at each level there are only countably many cells, and a countable union over levels still has zero mass. We make arbitrary choices on that null set. On the remaining points, the score at level a is zero, whereas (98) bounds the score at level j by

$$\log(1/P_{g,w}) - (d - 1 - q)j \log 2 - qa \log 2.$$

This upper bound tends to $-\infty$ uniformly in s . Hence a maximum is attained, and the least maximizing index is measurable because the scores are measurable and indexed by the integers. The nonnegative score at the maximizer also gives

$$(d - 1 - q)b \log 2 \leq \log(1/P_{g,w}).$$

Averaging proves $\mathbb{E}b < \infty$. Since $\log \mu(Q_b(S)) \leq 0$, we have $J \leq q(b - a) \log 2 \leq d(b - a)$, proving (99).

For a nonnegative integer variable V of finite mean, relative entropy to the probability masses 2^{-i-1} gives

$$H(V) \leq (1 + \mathbb{E}V) \log 2.$$

Apply this conditionally to $b - a$ and to N . The value of a is known from G , so $H(b | G, W) = H(b - a | G, W)$; moreover $\mathbb{E}N \leq \mathbb{E}(b - a)$. This proves (101). Given $b = j$, the cell label has at most $(C2^j)^d$ possibilities. Hence $H(Q_b(S) | G, W, b) \leq d \log C + d(\log 2)\mathbb{E}b$, and $H(G, W, Z)$ is finite.

It remains to account for the selected cell label. Fix g, w and a value $b = j$ of positive conditional probability. Compare the conditional distribution of $Q_j(S)$ under this selection with the level- j cell probabilities of μ . Nonnegative relative entropy gives

$$\mathbb{E}[\log \mu(Q_j(S)) | g, w, b = j] \leq -H(Q_j(S) | g, w, b = j).$$

Averaging this inequality and using the defining formula for J gives (100). This argument conditions on the selected level; it uses no stopping-time property of that level. $\square$

The next lemma measures the information gained from one block. Its posterior may be irregular before the classes indexed by N are formed. The factor $1/p$ in the state-alphabet cost comes from a p th projection moment.

Lemma 8.2 (Information acquired inside a grid cell). *Let $S \sim \sigma$. Let G be countable with $H(G) < \infty$, and suppose that it determines an integer $a \geq 0$ and a level- a cube containing S , with $\mathbb{E}a < \infty$. Independently of (S, G) , draw a standard Gaussian k -by- d matrix A . Given (A, AS, G) , let a measurable probability kernel produce W , with at most e^m possible values for each $G = g$, where $m \geq 0$. Form J, N from (G, W) as in Lemma 8.1. Then*

$$I(S; W | G) \leq Cd + \frac{m}{p} + \mathbb{E}J + H(N | G, W). \quad (102)$$

Proof. Here $H(G, W) \leq H(G) + m < \infty$, so the refinement lemma applies. We first bound projection densities for each class, using a fresh Gaussian matrix. We then compare the actual matrix, selected through W , with that fresh law.

Fix g, w, n of positive probability. Put

$$\alpha = \mathbb{P}(N = n \mid g, w), \quad \mu_n = \mathcal{L}(S \mid g, w, n), \quad \delta = 2^{-a}.$$

A level- j cube of positive μ_n -mass contains a class point with $J < (n+1)d$. Since the score at that level is at most J , and $\mu_n \leq \mu/\alpha$, it follows that

$$\mu_n(Q_j) \leq \alpha^{-1} e^{(n+1)d} (2^{a-j})^q \quad (j \geq a). \quad (103)$$

For $0 < t \leq 1$, choose $j \geq a$ with $t\delta/2 \leq \delta_j \leq t\delta$. A ball of radius $t\delta$ meets at most C^d such cubes: their union lies in a ball of radius $2t\delta$, and the volume estimate used above cancels the factor $d^{d/2}$ in the inverse cube volume. Thus (103) bounds the mass of this ball by $C^d e^{(n+1)d} \alpha^{-1} t^q$.

Let V be any affine plane of dimension $r \leq p-1$. The projection of the current level- a cube onto V lies in a plane-ball of radius δ . A maximal separated set covers that ball by at most $(1 + 2/t)^r$ balls of radius $t\delta$. Points of the current cube within distance $t\delta$ of V then lie in the corresponding ambient balls of radius $2t\delta$. For $t \leq 1/2$ apply the ball estimate; for $t > 1/2$ enlarge C and use total mass one. We obtain

$$\mu_n\{u : \text{dist}(u, V) \leq t\delta\} \leq C^d e^{(n+1)d} \alpha^{-1} t^{q-r} \quad (0 < t \leq 1). \quad (104)$$

In particular every such plane has zero mass. The layer-cake formula, applied to $(\delta/\text{dist}(u, V))^k$, gives

$$\int \text{dist}(u, V)^{-k} d\mu_n(u) \leq \delta^{-k} C^d e^{(n+1)d} \alpha^{-1}. \quad (105)$$

Indeed the integral after multiplication by δ^k is at most $1 + Kk/(q-r-k)$ with $K = C^d e^{(n+1)d}/\alpha$, and $q-r-k$ is a positive multiple of d .

Fix any $s \in S^{d-1}$ and draw $U_1, \dots, U_p$ independently from μ_n . Let Γ_s be the Gram matrix of the vectors $U_i - s$. Gram-Schmidt expresses its determinant as the product of the squared distances from U_i to $s + \text{span}\{U_h - s : h < i\}$. These planes have dimension at most $p-1$, and (104) shows that the distances are positive almost surely. Successive integration using (105) yields

$$\int (\det \Gamma_s)^{-k/2} d\mu_n^{\otimes p} \leq (\delta^{-k} C^d e^{(n+1)d} \alpha^{-1})^p. \quad (106)$$

The center s is fixed in this estimate and need not have law μ_n .

We turn this inverse-volume bound into a p th moment of the class projection density. That moment will control its average logarithm under the actual matrix law; the relative-entropy comparison will charge at most m/p for selection through W .

Let ϕ_η be the $N(0, \eta^2 I_k)$ density and define

$$\rho_\eta(A, y) = \int \phi_\eta(y - Au) d\mu_n(u).$$

Write γ_k for standard Gaussian k -by- d matrix law. For a fresh $\tilde{A} \sim \gamma_k$, expansion of the p th power gives

$$\mathbb{E}_{\tilde{A}} \rho_\eta(\tilde{A}, \tilde{A}s)^p = (2\pi)^{-kp/2} \int \det(\Gamma_s + \eta^2 I_p)^{-k/2} d\mu_n^{\otimes p}. \quad (107)$$

For each row, the vector of projected differences has covariance Γ_s ; adding the p independent mollifying Gaussians adds $\eta^2 I_p$. Evaluating its density at zero and multiplying across independent rows proves the identity. The right side is bounded by (106).

We next select one density version that is valid on the true-label diagonal as well as under the actual selected rows. The projection of uniform sphere measure onto $k < d$ coordinates has density

$$\frac{\Gamma(d/2)}{\pi^{k/2}\Gamma((d-k)/2)}(1 - \|y\|^2)^{(d-k-2)/2}\mathbf{1}_{\{\|y\|<1\}}.$$

For completeness, write a uniform sphere point as a normalized standard Gaussian. The squared length of its first k coordinates is the ratio $X/(X + X')$ for independent chi-square variables of k and $d - k$ degrees of freedom. Changing variables in their gamma densities gives the beta density with parameters $k/2, (d - k)/2$; its direction is uniform and independent. Polar coordinates give the displayed density. Rotations and an invertible range map prove absolute continuity for every rank- k linear map, and the property passes from σ to $\mu_n \ll \sigma$.

Consequently the law of $(\tilde{A}, \tilde{A}U)$, with $U \sim \mu_n$ independent of $\tilde{A}$, has a density ρ_0 relative to $\gamma_k(dA) dy$. The functions ρ_η converge to ρ_0 in this joint L^1 space. To see this, for almost every A use continuity of translations in $L^1(dy)$ for its conditional density and average the Gaussian translations; the error is at most two, so dominated convergence permits averaging over A . Choose a sequence $\eta_j \downarrow 0$ along which convergence also holds almost everywhere, and define everywhere

$$\rho(A, y) = \liminf_{j \rightarrow \infty} \rho_{\eta_j}(A, y).$$

This is a version of ρ_0 . The sequence was chosen without a value of s . Since (107) was bounded for every fixed s , Fatou's lemma now gives, for this same version and for each fixed s ,

$$\left(\mathbb{E}_{\tilde{A}} \rho(\tilde{A}, \tilde{A}s)^p\right)^{1/p} \leq \delta^{-k} C^d e^{(n+1)d} \alpha^{-1}. \quad (108)$$

There are only countably many triples (g, w, n) , so this construction can be made for every occurring triple. Conditional on such a triple, the actual matrix law is absolutely continuous with respect to γ_k . Also, for almost every such matrix, the conditional signal law is absolutely continuous with respect to σ : conditioning the pre-block law by the discrete events $W = w, N = n$ only reweights it. Its image under a full-rank matrix therefore has a Lebesgue density. Thus the actual joint law of the matrix and its label is absolutely continuous with respect to $\gamma_k(dA) dy$. In particular the chosen version ρ is valid in expectations under that actual law.

We have obtained the fresh projection estimate. To convert it into information, set $Y = AS$. Fresh independence and the conditional Markov property $S \rightarrow Y \rightarrow W$ given (A, G) imply

$$\begin{aligned} I(S; W | G) &= I(Y; W | A, G) - I(A; S | G, W) \\ &= h(Y | A, G) - h(Y | A, G, W) - I(A; S | G, W). \end{aligned} \quad (109)$$

The first line follows by expanding $I(S; A, W | G)$ in two orders. It also gives $I(A; S | G, W) \leq I(S; W | A, G) \leq m$.

All the differential entropies in this argument, including after conditioning on N , are finite. Here are the needed integrability details. Because $d - k \geq 2$, the coordinate density displayed above is bounded by a finite constant depending on d, k . For a rank- k matrix the corresponding bound is multiplied by $(\det AA^\top)^{-1/2}$. Gram-Schmidt on Gaussian rows writes this determinant as a product of squared residual lengths, each with a chi-square law; the absolute logarithm is integrable from its gamma density. Thus $\mathbb{E}|\log \det AA^\top| < \infty$. Given $G = g$, the signal density is bounded by

$1/\mathbb{P}(G = g)$ relative to σ , whose expected logarithm is $H(G)$. Conditioning further on (W, N) can multiply a conditional label density by at most $1/\mathbb{P}(W = w, N = n \mid A, G)$; its expected logarithm is $H(W, N \mid A, G) \leq H(W, N \mid G) < \infty$. These facts control the positive parts of the log label densities. To control their negative parts, compare with a fixed standard Gaussian label density. For two densities f, g ,

$$\int_{\{f < g\}} f \log(g/f) \leq 1/e,$$

because $t \log(1/t) \leq 1/e$ for $0 \leq t \leq 1$. The remaining Gaussian logarithm is integrable since $\mathbb{E}\|Y\|^2 = k$. This proves the claimed finiteness and justifies the entropy identity.

Let c_G be a specified corner of the current cube, so $\|S - c_G\| \leq \delta = 2^{-a}$. Comparing the conditional label density with $N(Ac_G, \delta^2 I_k)$, and using independence of A to obtain $\mathbb{E}[\|A(S - c_G)\|^2 \mid G] \leq k\delta^2$, gives

$$h(Y \mid A, G) \leq k\mathbb{E} \log \delta + Ck. \quad (110)$$

For the other two terms in (109), first condition on N . Differential entropy decreases under this conditioning, and, since N is determined by (S, G, W) ,

$$I(A; S \mid G, W) = I(A; N \mid G, W) + I(A; S \mid G, W, N).$$

The latter conditional mutual information is the average divergence from $\mathcal{L}(S \mid A, G, W, N)$ to its marginal μ_N . Data processing under $s \mapsto As$, followed by the definition of relative entropy for the two label densities, gives

$$\begin{aligned} h(Y \mid A, G, W, N) + I(A; S \mid G, W, N) \\ \geq h(Y \mid A, G, W, N) + \mathbb{E}D(\mathcal{L}(Y \mid A, G, W, N) \parallel \rho_{G, W, N}(A, y) dy) \\ = -\mathbb{E} \log \rho_{G, W, N}(A, AS). \end{aligned}$$

Together with the preceding conditioning and chain-rule inequalities, this proves

$$h(Y \mid A, G, W) + I(A; S \mid G, W) \geq -\mathbb{E} \log \rho_{G, W, N}(A, AS). \quad (111)$$

This comparison is between finite quantities: the conditional mutual information is at most m , and the true log density was shown integrable. The projected log density ratio has integrable negative part by the same $1/e$ bound and finite positive part by data processing.

If $P \ll P_0$ and $f \geq 0$, relative entropy to the probability law with density proportional to f^p gives

$$\mathbb{E}_P \log f \leq \frac{1}{p} (D(P \parallel P_0) + \log \mathbb{E}_{P_0} f^p). \quad (112)$$

Truncating $\log f$ proves this form whenever the left side is defined and the right side is finite. In the present application the preceding paragraph gives integrability. Apply (112) conditionally on (s, g, w, n) to the actual matrix law, taking $P_0 = \gamma_k$ and $f(A) = \rho_{g, w, n}(A, As)$. The value of n is already determined by (s, g, w) , and before W the matrix is independent of (S, G) . The average divergence in this comparison is therefore

$$I(A; W \mid S, G) \leq H(W \mid G) \leq m.$$

The average of $-\log \alpha$ is $H(N \mid G, W)$. Using (108) in (112) and then averaging gives

$$\mathbb{E} \log \rho_{G, W, N}(A, AS) \leq -k\mathbb{E} \log \delta + Cd + \mathbb{E}[(N + 1)d] + H(N \mid G, W) + m/p.$$

Finally $(N + 1)d \leq J + d$. Substitute this bound and (110)–(111) into (109) to obtain (102). $\square$

Proposition 8.3 (Iteration of the grid information balance). *Let $S \sim \sigma$, and start from a fixed state. Let $B \geq 0$ be an integer. Consider B blocks, each consisting of k fresh standard Gaussian rows and their exact labels. In each block, the outgoing state is a measurable kernel of the incoming state and that block's data, and its alphabet has logarithm at most $m \geq 0$. There is a countable analysis history G_B of finite entropy that contains the final state and satisfies*

$$I(S; G_B) \leq C\{d + B(d + m/p)\}. \quad (113)$$

Proof. Start with $G_0 = Q_0(S)$ and depth $a_0 = 0$. Its entropy is at most Cd by the grid count. After every outgoing state W , append the pair $Z = (b, Q_b(S))$ from Lemma 8.1. The enlarged history contains the outgoing state and a cell containing S at depth b . Its entropy and expected depth remain finite. Since this refinement uses only S and completed blocks, the next Gaussian rows remain independent of (S, G) .

For a current history of depth a , define

$$F(G) = I(S; G) - \bar{q}(\log 2)\mathbb{E}a.$$

Because Z is a function of (S, G, W) ,

$$I(S; G, W, Z) - I(S; G) = I(S; W \mid G) + H(Z \mid G, W).$$

The block estimate, (100), and (101) give

$$F(G, W, Z) \leq F(G) + Cd + m/p.$$

Here the term $q \log 2$ pays the selected cell mass, and the remaining $2 \log 2$ per unit of depth pays both the depth and class entropies. Thus the entire label Z has been included in the balance.

For each history value g , its conditional signal law is supported on the specified level- a cube. Relative entropy to σ is at least the negative logarithm of the σ -mass of that cube: this follows by writing it as relative entropy to normalized σ on the cube minus the logarithm of the cube mass. By (98),

$$I(S; G) \geq (d - 1)(\log 2)\mathbb{E}a.$$

It follows that $F(G) \geq (1 - \bar{q}/(d - 1))I(S; G)$, with a positive coefficient bounded away from zero for large d . Iterating the upper bound from $F(G_0) \leq Cd$ proves (113). $\square$

Application to precision and bounded stopping.

Corollary 8.4 (Precision under the uniform spherical prior). *Let $M(d) = o(d^2)$ and $0 < \epsilon(d) \leq 1/10$. For learners in the finite-state model of Section 2, uniform-sphere average angular success at least $2/3$ requires*

$$T(d) \geq cd \log \frac{1}{\epsilon(d)}$$

for an absolute $c > 0$ and all sufficiently large d . The dimension threshold may depend on the memory sequence.

Proof. Use the reductions following Lemma 2.2 under the uniform prior. First fix the shared rule randomness with success greater than $3/5$, and apply Lemma 2.1. Then fix the independent initial state, the finite transition choices, and the outputs assigned to terminal state/index pairs by averaging. The resulting deterministic Borel learner has uniform-prior success at least $3/5$ and the same state and horizon bounds.

Put $L = \log(1/\epsilon)$, with $0 < \epsilon \leq 1/10$. The case $T \geq dL$ already has the required order. In the other case the fixed learner has at most $(T+1)2^M$ terminal outputs, counting every possible stopping index. Angular error at most ϵ implies Euclidean error at most ϵ , because $2\sin(\theta/2) \leq \theta$. Applying (97) to each output gives

$$\frac{3}{5} \leq (T+1)2^M \epsilon^{d-1}, \quad (d-2)L \leq M \log 2 + O(\log d). \quad (114)$$

For the second inequality, take logarithms in the first and use $T < dL$ and $\log(dL+1) \leq \log(d+1)+L$. Thus $M = o(d^2)$ implies $L = o(d)$ along these putative short runs, and eventually $T \leq d^2$. This is a consequence of the short-run assumption, not an added restriction on the accuracy sequence.

Preserve a stopped state together with its terminal index, and use ignored fresh samples after stopping. At any later index there are at most 2^M active states and $(T+1)2^M$ tagged terminal states. Padding the last block as well therefore uses width at most

$$(T+2)2^M, \quad m = M \log 2 + \log(T+2) = o(d^2). \quad (115)$$

Take $B = \lceil T/k \rceil$ blocks, padding with fewer than k ignored samples if necessary. The final augmented history recovers the output. Under the product of the signal and output marginals, angular success has probability at most ϵ^{d-1} by (97). For an event with probabilities u and v under two laws, data processing to the event gives relative entropy at least $u \log(1/v) - \log 2$: expand binary relative entropy and use binary entropy at most $\log 2$. Hence

$$I(S; G_B) \geq I(S; \hat{S}) \geq \frac{3}{5}(d-1)L - \log 2. \quad (116)$$

Since $m/p = o(d)$, (113) and (116) imply $L \leq C(1+B)$. With $B \leq T/k + 1$, this gives $T \geq cdL$ whenever L exceeds a fixed absolute constant.

We finish the bounded range of L with the route's linear sample bound. Grant the learner all $T < d$ pre-generated rows and exact labels, including any it did not use. Their row space V has dimension T almost surely. The data determine $P_V S$, and conditional on the data the residual $S - P_V S$ has uniform direction in $V^\perp$, with radius $\sqrt{1 - \|P_V S\|^2}$. This follows by splitting a normalized standard Gaussian into its independent row-space and nullspace components. Its conditional mean is zero and its covariance is $(1 - \|P_V S\|^2)P_{V^\perp}/(d-T)$. For every unit estimate determined by the data, also after averaging independent output randomness,

$$\mathbb{E}\langle S, \hat{S} \rangle^2 \leq \mathbb{E}\|P_V S\|^2 + \frac{1}{d-T} = \frac{T}{d} + \frac{1}{d-T}. \quad (117)$$

On angular success the squared inner product is at least $\cos^2(1/10)$. Since $(3/5)\cos^2(1/10) > 1/2$, average success $3/5$ excludes $T \leq d/2$ for large d . This linear bound absorbs the bounded range of L and completes the proof. $\square$

9 Adaptive scales with one Gaussian side projection

We now use a different information balance. A finite message carries little information when the prior has sufficiently small mass near lower-dimensional affine planes. For a general bounded-density prior, a maximizing radius splits it into cells with that property. An independent Gaussian projection lets us bound the conditional entropy of the cell label. We charge this revelation cost in each block's information estimate, keeping the same projection throughout the transcript argument. It is an analysis variable; the learner still receives only its prescribed exact samples.

All logarithms and entropies in this section use natural units. Let μ be uniform probability on S^{d-1} . For sufficiently large d , fix

$$k_* = p = \lfloor d/32 \rfloor, \quad q = p - 1, \quad a = d/8, \quad \alpha = d/4, \quad \ell = \lfloor d/3 \rfloor. \quad (118)$$

Every constant denoted by C is positive and absolute.

We will use Lemma 2.3 also in lower ambient dimensions. If μ_n is uniform probability on S^{n-1} , $n \geq 2$, that lemma gives, for $s \in S^{n-1}$,

$$\mu_n(B(s, r)) \leq r^{n-1} \quad (0 < r \leq 1/2), \quad \mu_n(B(s, r)) \leq (2r)^{n-1} \quad (r > 0). \quad (119)$$

Lemma 9.1 (A finite message from a locally diffuse prior). *Let $\nu \ll \mu$ be a probability measure supported in $B(c, R)$, where $c \in \mathbb{R}^d$ and $R > 0$. Suppose that for ν -almost every s , simultaneously for all $r > 0$,*

$$\nu(B(s, r)) \leq K(r/R)^\alpha, \quad K \geq 1. \quad (120)$$

Draw $S \sim \nu$ and an independent standard Gaussian k -by- d matrix A , where $1 \leq k \leq k_$. A measurable probability kernel of (A, AS) produces a message Z in an alphabet of cardinality $w \geq 1$. Then*

$$I(S; Z) \leq C(d + \log K) + \frac{\log w}{p}. \quad (121)$$

Proof. We first control the volume of the simplex spanned by independent points from ν . The use of Gaussian replicas and Gram–Schmidt factors is related to the projection argument of Sharan, Sidford, and Valiant [12, Section 7, Lemma 12]. The geometric moment and the exact-message information estimate needed here are proved below.

Let V be an affine subspace of dimension $m \leq p - 2$, and let $0 < u \leq 1$. Points of $B(c, R)$ within distance uR of V project into $V \cap B(c, 2R)$. A maximal separated set covers this set by at most $(C/u)^m$ balls in V of radius uR ; when $m = 0$, one ball suffices. The corresponding points in the original tube lie in ambient balls of radius $2uR$. Such a ball of positive ν -mass contains a point where (120) holds, and it is contained, on the support of ν , in a radius- $4uR$ ball about that point. Hence

$$\nu\{s : \text{dist}(s, V) \leq uR\} \leq KC^d u^{\alpha-m}. \quad (122)$$

The parameters give $\alpha - m - a \geq 3d/32$. Integrating the tail of $(R/\text{dist}(S, V))^a$, and using (122) for values of this variable above one, gives

$$\mathbb{E}_\nu(R/\text{dist}(S, V))^a \leq KC^d. \quad (123)$$

Indeed the tail integral is bounded by $1 + KC^d a/(\alpha - m - a)$. The tube bound also proves that V has zero ν -mass.

Draw $(S_1, \dots, S_p) \sim \nu^{\otimes p}$. Let F be the matrix with columns $S_i - S_1$, $2 \leq i \leq p$, and set $v = \sqrt{\det(F^\top F)}$. Gram–Schmidt writes v as the product of the distances from S_i to $\text{aff}(S_1, \dots, S_{i-1})$. The preceding zero-mass statement shows successively that these distances are positive almost surely. Integrating the last point first and repeating (123) proves

$$\mathbb{E}_{\nu^{\otimes p}}(R^q/v)^a \leq (KC^d)^q. \quad (124)$$

We use this moment in a normalized comparison of posterior replicas. Temporarily add an independent $N(0, \sigma^2 I_k)$ vector η to the labels, where $0 < \sigma \leq R$, and set $b = AS + \eta$. Apply the

given message kernel to (A, b) and call its output Z_σ . Write P_A for the Gaussian matrix law. With φ_σ denoting the noise density, write

$$f_\sigma(b | A) = \int \varphi_\sigma(b - As) d\nu(s).$$

Sample (A, b, Z_σ) from this experiment. Conditional on these variables, sample p independent points from the posterior of S given (A, b) ; their conditional law does not depend further on Z_σ . Let P_σ be the resulting marginal on $(S_1, \dots, S_p, A, b)$ and Q_σ its point-tuple marginal. Each pair (S_i, Z_σ) has the experiment's original marginal. Put $D_Q = D(Q_\sigma \| \nu^{\otimes p})$, initially allowing the value $+\infty$. Once D_Q is finite, relative entropy to a product bounds the sum of the marginal relative entropies: their difference is the relative entropy of the joint law to the product of its own marginals. Apply this after conditioning on the common message and use the matching single-point marginals. The conditional chain rule then gives

$$\begin{aligned} pI(S; Z_\sigma) &\leq \mathbb{E}_{Z_\sigma} D(Q_{\sigma, S_1, \dots, S_p | Z_\sigma} \| \nu^{\otimes p}) \\ &= D_Q + I((S_1, \dots, S_p); Z_\sigma) \leq D_Q + \log w. \end{aligned} \quad (125)$$

It therefore suffices to prove $D_Q/p \leq C(d + \log K)$, uniformly in σ . We will prove finiteness and this bound by comparing the posterior tuple law with a normalized reference whose tuple marginal is exactly $\nu^{\otimes p}$.

Relative to $\nu^{\otimes p} \otimes P_A \otimes db$, the density of P_σ is

$$\frac{\prod_{i=1}^p \varphi_\sigma(b - AS_i)}{f_\sigma(b | A)^q}. \quad (126)$$

For a fixed tuple, the integral of the numerator over (A, b) is

$$c_\sigma(F) = (2\pi)^{-kq/2} \det(F^\top F + \sigma^2(I_q + \mathbf{1}\mathbf{1}^\top))^{-k/2}. \quad (127)$$

Here $\mathbf{1} \in \mathbb{R}^q$ is the all-ones column. To verify this formula, take p independent noisy labels with the same matrix A . Integrating their product density over their common value b evaluates at zero the density of the q differences from the first label. In one row, the projected signal differences have covariance $F^\top F$, and the noise differences have covariance $\sigma^2(I_q + \mathbf{1}\mathbf{1}^\top)$. The rows are independent, which proves (127). On full-rank tuples,

$$c_\sigma(F) \leq (2\pi)^{-kq/2} v^{-k}. \quad (128)$$

Define a reference probability law R_σ by the density

$$\frac{\prod_{i=1}^p \varphi_\sigma(b - AS_i)}{c_\sigma(F)} \quad \text{relative to } \nu^{\otimes p} \otimes P_A \otimes db. \quad (129)$$

The denominator is exactly the integral for each fixed tuple. Thus the tuple marginal of this reference law is $\nu^{\otimes p}$, rather than an unnormalized multiple of that law.

All logarithms in this comparison are integrable at fixed $\sigma > 0$. The tuple is supported in $B(c, R)^p$, and

$$\sigma^2 I_q \preceq F^\top F + \sigma^2(I_q + \mathbf{1}\mathbf{1}^\top) \preceq (4qR^2 + p\sigma^2)I_q.$$

Hence $\log c_\sigma$ is bounded as the tuple varies. Also $\log f_\sigma$ is at most $-\frac{k}{2} \log(2\pi\sigma^2)$ and at least this value minus

$$\frac{(\|b - Ac\| + \|A\|_{\text{op}} R)^2}{2\sigma^2}.$$

The latter term is integrable under the noisy experiment. Data processing therefore proves $D_Q < \infty$ and gives

$$D_Q \leq D(P_\sigma \| R_\sigma) = \mathbb{E}_{P_\sigma} \log c_\sigma(F) - q \mathbb{E}_{P_\sigma} \log f_\sigma(b | A). \quad (130)$$

To bound the second term, compare $f_\sigma(\cdot | A)$ with the $N(Ac, 2R^2 I_k)$ density. Since $\mathbb{E} \|b - Ac\|^2 \leq kR^2 + k\sigma^2 \leq 2kR^2$, nonnegative relative entropy yields

$$-\mathbb{E} \log f_\sigma(b | A) \leq k \log R + \frac{k}{2} \log(4\pi e).$$

Together with (128), this gives, initially allowing the expectation on the right to be extended,

$$D_Q \leq k \mathbb{E}_{Q_\sigma} \log(R^q/v) + Ckq. \quad (131)$$

Here the needed expectation is in fact finite. Put $X = a \log(R^q/v)$. Hadamard's inequality gives $v \leq (2R)^q$, so $X \geq -aq \log 2$. For bounded X' the probability law proportional to $e^{X'} \nu^{\otimes p}$ and nonnegative relative entropy give

$$\mathbb{E}_{Q_\sigma} X' \leq D_Q + \log \mathbb{E}_{\nu^{\otimes p}} e^{X'}.$$

Apply this to $X' = \min(X, n)$ and let $n \rightarrow \infty$. Monotone convergence after adding the fixed lower bound, together with (124), proves both finiteness and

$$a \mathbb{E}_{Q_\sigma} \log(R^q/v) \leq D_Q + q(\log K + Cd).$$

Because $k/a \leq 1/4$, substitution into (131) and absorption of $(k/a)D_Q$ give

$$D_Q/p \leq C(d + \log K). \quad (132)$$

Combining (132) with (125) proves (121) for the noisy experiment, uniformly over $0 < \sigma \leq R$.

It remains to pass this inequality to the given exact-label rule. This step uses the rule itself, rather than a comparison of the statistical difficulties of noisy and noiseless experiments. For every rank- k matrix with $k < d$, the image of μ has a Lebesgue density. One direct proof uses a normalized standard Gaussian: the squared length of the first k coordinates is a beta variable with parameters $k/2, (d-k)/2$, obtained from the ratio of independent chi-square variables, and its direction is uniform. This gives a density on the unit k -ball. Orthogonal changes of coordinates and an invertible range map give the claim for any rank- k matrix. Absolute continuity $\nu \ll \mu$ preserves it. Since a Gaussian matrix has full row rank almost surely and itself has a Lebesgue density, (A, AS) has a joint Lebesgue density.

Let $g_z(A, b)$ be the output probability of message z . For any bounded measurable function g and a random vector X with a Lebesgue density,

$$\mathbb{E} |g(X+h) - g(X)| \longrightarrow 0 \quad (h \rightarrow 0). \quad (133)$$

To see this, restrict the input density to a bounded cube and to values below a fixed bound, losing arbitrarily little probability. For small h , cut g off on a larger cube. The remaining integral is bounded by a constant times the Lebesgue L^1 distance between this cutoff and its translate. That distance tends to zero, since continuous compactly supported functions are dense in L^1 and their translates converge uniformly on a common compact set. This proves (133). Apply it to shifts in the label coordinates, then to $h = (0, \sigma G)$ with G standard Gaussian by bounded convergence. The finite alphabet gives

$$\sum_z \mathbb{E} |g_z(A, AS + \eta) - g_z(A, AS)| \longrightarrow 0. \quad (134)$$

Couple the two message probability vectors at each (S, A, η) by their common masses, so that their disagreement probability δ is half the expected sum in (134). This is a measurable coupling because the alphabet is finite. Write $h_2(t) = -t \log t - (1-t) \log(1-t)$, with $0 \log 0 = 0$, for binary entropy. The disagreement indicator bounds each of the marginal and conditional entropy differences by $h_2(\delta) + \delta \log w$: on disagreement there are at most w possible values, and binary entropy is concave when conditional disagreement probabilities are averaged. Consequently

$$|I(S; Z_\sigma) - I(S; Z)| \leq 2\{h_2(\delta) + \delta \log w\} \longrightarrow 0.$$

For $w = 1$ both messages are constant and the assertion is immediate. The uniform noisy bound therefore proves (121) for the exact labels AS . $\square$

Lemma 9.2 (Adaptive cells predicted by independent side data). *Let ν be a probability measure with $d\nu/d\mu \leq B < \infty$. Draw $S \sim \nu$, an independent standard Gaussian k -by- d matrix A with $1 \leq k \leq k_*$, and a message Z in an alphabet of size $w \geq 1$ from a measurable kernel of (A, AS) . Independently of the signal, the matrix, and the message randomness, draw a standard Gaussian ℓ -by- d matrix W , and put $D_* = (W, WS)$. Then*

$$I(S; Z \mid D_*) \leq C \left[d + \log \left(C \left(2 + \frac{\log B}{d} \right) \right) \right] + \frac{\log w}{p}. \quad (135)$$

Proof. Since ν is a probability, $B \geq 1$. Put $m_s(r) = \nu(B(s, r))$ and $R_j = 2^{1-j}$ for integers $j \geq 0$. Choose $J(s)$ to be the least index maximizing $2^{\alpha j} m_s(R_j)$, and write $J = J(S)$. The score at $j = 0$ is one because $R_0 = 2$ covers the sphere. The second cap bound in (119) gives

$$2^{\alpha j} m_s(R_j) \leq B 2^{2(d-1)-j(d-1-\alpha)} \longrightarrow 0.$$

The ball masses are measurable in s , by integrating the Borel indicator of $\|s - u\| \leq R_j$. Thus the least maximizer is measurable and

$$0 \leq J(s) \leq \frac{\log_2 B + 2(d-1)}{d-1-\alpha}, \quad H(J) \leq \log \left(C \left(2 + \frac{\log B}{d} \right) \right). \quad (136)$$

The entropy bound follows from the displayed finite range. For $R = R_{J(s)}$, compare the maximum with the least dyadic radius at least r when $r \leq 2$, and with $R_0 = 2$ when $r > 2$. This proves

$$m_s(r) \leq 2^\alpha m_s(R) (r/R)^\alpha \quad (r > 0). \quad (137)$$

At each scale choose a finite maximal R_j -separated set of centers on the sphere. Its R_j -balls cover the sphere. Enumerate the centers and assign a point to the first center within distance R_j ; this is a measurable rule. Let $U = (J, i)$ record the scale and assigned center. Its range is finite by (136) and the finiteness of each net. For each cell $u = (j, i)$ of positive probability, write c_u for its center, $b_u = \mathbb{P}(U = u)$, and $m_u^+ = \nu(B(c_u, 2R_j))$.

At one scale, a point lies in at most 5^d of the enlarged balls $B(c_u, 2R_j)$. Indeed the disjoint interiors of the radius- $R_j/2$ balls about those centers lie in a ball of radius $5R_j/2$ about the point. Volume comparison proves the count, and integration gives $\sum_{u=(j,i)} m_u^+ \leq 5^d$. Write $p_j = \mathbb{P}(J = j)$. Concavity of the logarithm with weights b_u/p_j at each positive-mass scale yields

$$\sum_u b_u \log \frac{m_u^+}{b_u} \leq \sum_{j:p_j>0} p_j \log \frac{5^d}{p_j} = d \log 5 + H(J). \quad (138)$$

The conditional prior ν_u is supported in $B(c_u, R_j)$ and remains absolutely continuous with respect to μ . If s is in that cell, then $m_s(R_j) \leq m_u^+$. Dividing (137) by b_u gives, for ν_u -almost every s and all $r > 0$,

$$\nu_u(B(s, r)) \leq \frac{2^\alpha m_u^+}{b_u} (r/R_j)^\alpha.$$

Here $m_u^+ \geq b_u$, so the factor is at least one. The matrix A remains independent of (S, U) , and the same message kernel is used. Apply Lemma 9.1 for each u , average, and use (138). We obtain

$$I(S; Z | U) \leq C(d + H(J)) + \frac{\log w}{p}. \quad (139)$$

We have bounded the information inside each cell. We now pay for its label in the presence of the side data. For a known scale j , predict the center index from (W, y) by the normalized probabilities

$$\pi_{j,i}(W, y) = \frac{b_{(j,i)} \exp(-\|W c_{(j,i)} - y\|^2 / (2R_j^2))}{\sum_{i'} b_{(j,i')} \exp(-\|W c_{(j,i')} - y\|^2 / (2R_j^2))}. \quad (140)$$

Discard zero-probability cells. At an occurring scale the denominator is positive and the set of indices is finite. Nonnegative relative entropy between the conditional center distribution and these probabilities bounds $H(i | D_*, J)$ by the expected negative logarithm of the true cell's prediction.

Fix a true point s with $J(s) = j$ and put $R = R_j$. For its center,

$$\mathbb{E}_W \frac{\|W(c_U - s)\|^2}{2R^2} \leq \ell/2.$$

For any candidate center c , direct Gaussian integration gives

$$\mathbb{E}_W \exp(-\|W(c - s)\|^2 / (2R^2)) = \left(1 + \frac{\|c - s\|^2}{R^2}\right)^{-\ell/2}.$$

Centers within distance $2R$ of s have total cell probability at most $m_s(3R)$, since their cells are within an additional distance R . Centers at distances in $(2^t R, 2^{t+1} R]$, $t \geq 1$, have total cell probability at most $m_s(2^{t+2} R)$ and expected kernel at most $2^{-t\ell}$ per unit cell probability. The expected denominator in (140), evaluated at $y = Ws$, is therefore at most

$$\begin{aligned} m_s(3R) + \sum_{t \geq 1} 2^{-t\ell} m_s(2^{t+2} R) &\leq m_s(R) \left(6^\alpha + 2^{3\alpha} \sum_{t \geq 1} 2^{-t(\ell-\alpha)}\right) \\ &\leq e^{Cd} m_s(R). \end{aligned} \quad (141)$$

The first inequality uses (137); the series converges because $\ell - \alpha \geq 1$ for large d .

Jensen's inequality for the logarithm of this denominator, together with the true-center squared term, bounds the expected prediction loss at s by $Cd + \log(m_s(R)/b_U)$. The logarithms are integrable: the denominator is at least the true term, and its negative logarithm is bounded by $-\log b_U$ plus a Gaussian squared term. Averaging and using $m_S(R_J) \leq m_U^+$ in (138) gives

$$H(U | D_*) \leq H(J) + Cd + \mathbb{E} \log \frac{m_S(R_J)}{b_U} \leq Cd + 2H(J). \quad (142)$$

Finally U is a function of S , and Z is conditionally independent of D_* given (S, U) . Since Z is finite,

$$\begin{aligned} I(S; Z \mid D_*, U) &= H(Z \mid D_*, U) - H(Z \mid S, U) \\ &\leq H(Z \mid U) - H(Z \mid S, U) = I(S; Z \mid U). \end{aligned}$$

The chain rule then gives $I(S; Z \mid D_*) \leq H(U \mid D_*) + I(S; Z \mid U)$. Combining (136), (139), and (142) proves (135). The cell has been revealed only inside this calculation, with its full conditional entropy charged; no local regularity of the original state posterior was assumed. $\square$

9.1 The full state transcript and the precision endpoint

We now give a second proof of Corollary 8.4. The information bound will be conditional on the same side data D_* at every block, and its endpoint will account for the residual sphere left by those data.

Proof of Corollary 8.4 using side data. Take $S \sim \mu$ and use the reductions following Lemma 2.2. First fix the shared rule randomness with uniform-prior success greater than $3/5$, apply Lemma 2.1, and then fix the independent initial state with success at least $3/5$. Fresh transition and terminal output randomness may remain: the two lemmas permit finite message kernels, and the calculation below integrates the terminal output law.

Put $L = \log(1/\epsilon)$ with $0 < \epsilon \leq 1/10$. If $T = 0$, the fixed initial state and terminal rule give an output independent of S , whose success is at most $\epsilon^{d-1} < 3/5$ by (119). Thus the successful runs considered below have $T \geq 1$.

It suffices to consider $T < dL$. There are at most $(T+1)2^M$ terminal pairs consisting of the stopping index and state. For a fixed pair h , the joint subprobability law λ_h of S and the event of reaching h satisfies $\lambda_h \leq \mu$. The unit output conditional on h uses only its terminal law and fresh randomness, so is independent of S . Angular success implies Euclidean error at most ϵ , and (119) applies since $\epsilon \leq 1/10$. Integrating the terminal output law and summing over pairs gives

$$\frac{3}{5} \leq (T+1)2^M \epsilon^{d-1}, \quad (d-1)L \leq M \log 2 + \log(T+1) + O(1). \quad (143)$$

Using $T < dL$ and $\log(dL+1) \leq \log(d+1) + L$ yields $(d-2)L \leq M \log 2 + O(\log d)$. Along a sequence $M = o(d^2)$ this implies $L = o(d)$ in the present short-run case and eventually $T \leq d^2$.

After an early stop, preserve its state together with its terminal index and ignore the remaining fresh samples. The active states and all terminal tags together give the global width bound

$$w = (T+2)2^M, \quad \log w = M \log 2 + \log(T+2) = o(d^2). \quad (144)$$

The output law at a terminal tag is the original terminal law. Split the T padded samples into $N = \lceil T/k_* \rceil$ consecutive blocks of length at most k_* . Let Π_i record every state at the first i block ends, with Π_0 empty. This is the full state transcript, not just its final state.

Independently draw one matrix W with ℓ Gaussian rows and retain the same $D_* = (W, WS)$ for all blocks. For every positive-mass history h of Π_{i-1} , its conditional signal law satisfies

$$\frac{d\nu_h}{d\mu}(s) = \frac{\mathbb{P}(\Pi_{i-1} = h \mid S = s)}{\mathbb{P}(\Pi_{i-1} = h)} \leq B_h := \frac{1}{\mathbb{P}(\Pi_{i-1} = h)}. \quad (145)$$

Given h , the next block matrix is still fresh Gaussian and independent of S , and W remains independent of both. The incoming state is known from h . The persistent-memory restriction makes

the successor state a kernel of that state and the block's complete exact data, with alphabet size at most w . It has no access to D_* . Lemma 9.2 therefore applies conditionally on each history.

The posterior-density cost is small after averaging. Since Π_{i-1} is a tuple of at most N states from an alphabet of size w ,

$$\mathbb{E}_h \log B_h = H(\Pi_{i-1}) \leq N \log w.$$

Jensen's inequality gives

$$\mathbb{E}_h \log \left(C \left(2 + \frac{\log B_h}{d} \right) \right) \leq \log \left(C \left(2 + \frac{N \log w}{d} \right) \right) = O(\log d).$$

Here $N = O(d)$ follows from $T \leq d^2$ and $k_* \geq d/64$ for large d ; also $(\log w)/p = o(d)$ by (144). Summing the conditional information bounds for successive states by the chain rule gives an absolute C_1 such that

$$I(S; \Pi_N \mid D_*) \leq C_1 d N. \quad (146)$$

The local cell U used in each application is not an uncharged part of Π_N : its revelation was paid by (142) inside that block's conditional bound.

For a lower bound on the same information, condition on D_* . Almost surely W has rank ℓ . Its labels specify $P_W S$, where P_W is the orthogonal projector onto its row space. The remaining component has uniform direction in the nullspace, of dimension $d - \ell$, and radius

$$\rho = \sqrt{1 - \|P_W S\|^2}.$$

This conditional law follows by writing S as a normalized standard Gaussian: the nullspace direction is independent of its length and of the row-space component. The event $\mathcal{G} = \{\rho \geq 1/2\}$ is determined by D_* , and Markov's inequality gives

$$\mathbb{P}(\mathcal{G}^c) \leq \frac{4}{3} \mathbb{E} \|P_W S\|^2 = \frac{4\ell}{3d} \leq \frac{4}{9}. \quad (147)$$

On $\mathcal{G}$, the mass of the conditional sphere within Euclidean distance ϵ of any fixed estimate is at most $(4\epsilon)^{d-\ell-1}$. If the ball meets that sphere, choose a point of intersection. The rest of the intersection lies within 2ϵ of it. After rescaling the residual sphere by $\rho \geq 1/2$, the radius is at most $4\epsilon \leq 2/5 < 1/2$, so the first bound in (119) applies in ambient dimension $d - \ell$.

The final output $\hat{S}$ is conditionally independent of S given (Π_N, D_*) , because the transcript contains its terminal tag. For each value of D_* , compare the actual joint law of $(S, \Pi_N, \hat{S})$ with the product of its conditional S marginal and conditional $(\Pi_N, \hat{S})$ marginal. On $\mathcal{G}$ the success probability under this product is at most $(4\epsilon)^{d-\ell-1}$. Binary relative entropy is at least the true event probability times the negative logarithm of its product-law probability, minus $\log 2$. Integrating over $\mathcal{G}$, and using nonnegativity elsewhere, gives

$$\begin{aligned} I(S; \Pi_N \mid D_*) &= I(S; \Pi_N, \hat{S} \mid D_*) \\ &\geq \left(\frac{3}{5} - \frac{4}{9} \right) (d - \ell - 1) \log \frac{1}{4\epsilon} - \log 2 \geq c_0 d L \end{aligned} \quad (148)$$

for an absolute $c_0 > 0$ and sufficiently large d . The coefficient $3/5 - 4/9$ accounts for the success mass outside the failed-radius event. For the last inequality, $d - \ell - 1$ is a positive multiple of d and $L - \log 4 \geq (1 - \log 4 / \log 10) L$.

Combining (146) and (148), and using $N = \lceil T/k_* \rceil \leq T/k_* + 1$, gives the exact intermediate bound

$$T \geq k_* \left(\frac{c_0}{C_1} L - 1 \right). \quad (149)$$

We retain a separate linear endpoint to absorb its additive constant. If $T < d/3$, reveal all T pre-generated rows and exact labels, even after an early stop. The same normalized-Gaussian argument gives a uniform conditional sphere of ambient dimension $d - T$. Its radius is at least $1/2$ except on an event of probability at most $4T/(3d) < 4/9$. The preceding conditional cap argument therefore bounds the success of even a full-data estimator by

$$\frac{4}{9} + (4\epsilon)^{d-T-1} < \frac{3}{5}$$

for sufficiently large d , a contradiction. Thus $T \geq d/3$. When $L \geq 2C_1/c_0$, (149) implies $T \geq cdL$; for smaller L , the linear bound gives the same conclusion after reducing the absolute constant. The case $T \geq dL$ needed no reduction. Every learner observation throughout this argument is the exact inner product; the auxiliary noise appeared only in the proved passage to the exact-message inequality. $\square$

10 Localization by separated tuples

This argument measures concentration after an additional fresh block of exact observations. On an actual block, an auxiliary revelation has one of two effects: it places the signal in a smaller ball, or it produces several well-separated signals compatible with the same data. In the second case an exact comparison on the equal-label constraints divides the cost of the learner's finite message by the number of signals. The radius decrease pays for the revelations needed to reach that case.

Let S be uniform on S^{d-1} . In this section σ denotes unnormalized surface area, and

$$k = \ell = \lfloor d/16 \rfloor, \quad q = \ell - 1, \quad n = d - 1 - k.$$

We take d sufficiently large that $\ell \geq 2$. A block is $B = (X_B, Y_B)$, where X_B has k independent standard Gaussian rows and $Y_B = X_B S$. A block is fresh relative to a variable U when its matrix is independent of the pair (S, U) . We use h_j for entropy relative to unnormalized area on a j -dimensional sphere, including spheres in affine subspaces, and h for Lebesgue entropy. Conditional entropies are averaged. All logarithms are natural.

10.1 Spherical slicing and a concentration potential

For a full-row-rank matrix X , let $V = \text{row}(X)$ and define

$$R_X(s) = \sqrt{1 - \|P_V s\|^2}, \quad j_X(s) = \sqrt{\det(XX^\top)} R_X(s).$$

At $R_X(s) > 0$, the fiber $\{z \in S^{d-1} : Xz = Xs\}$ is an n -sphere of radius $R_X(s)$. Write $\sigma_{X,y}$ for its unnormalized area measure.

The normal Jacobian in the coarea formula is $j_X(s)$; compare Federer's Definition 2.10 and Theorem 3.1 [6, pp. 423 and 426–427]. We give the coordinate calculation, including the normalization needed here. Choose orthonormal coordinates in V and $V^\perp$, and write $s = (u, R\theta)$ with $R = \sqrt{1 - \|u\|^2}$ and $\theta \in S^n$. The θ -directions scale area by R^n . The Gram matrix in the u -directions is $I + uu^\top/R^2$, of determinant $1/R^2$, and these two tangent groups are orthogonal. The surface element is consequently $R^{n-1} du d\sigma_n(\theta)$. The map from u to $y = Xs$ has absolute determinant $\sqrt{\det(XX^\top)}$, whereas the area element on the fiber is $R^n d\sigma_n(\theta)$. Let $\mathcal{E}_X = \{y : y^\top(XX^\top)^{-1}y < 1\}$. Thus for every nonnegative Borel g ,

$$\int_{S^{d-1}} g(s) \sigma(ds) = \int_{\mathcal{E}_X} \int_{\{Xs=y\}} \frac{g(s)}{j_X(s)} \sigma_{X,y}(ds) dy. \quad (150)$$

The right side uses only positive-radius fibers. The omitted critical set $R_X = 0$ has zero full-sphere area. The nonnegative formula follows directly from the coordinates and Tonelli, including by exhaustion when an integral is infinite.

Suppose the relevant conditional log densities are integrable and put $Y = XS$. If $p(s | X, U)$ is the full-sphere density and $p_Y(y | X, U)$ the label density, then the conditional slice density is

$$\frac{p(s | X, U)}{j_X(s) p_Y(Xs | X, U)}.$$

Taking its logarithm gives the separate entropy identity

$$h_n(S | X, Y, U) = h_{d-1}(S | X, U) - h(Y | X, U) + \mathbb{E} \log j_X(S). \quad (151)$$

The area identity (150) and the entropy identity (151) will be used at different points below.

Here is a finiteness check for every later use. For a fresh Gaussian matrix, $R_X(S)^2$ has the law $U/(U+V)$, where $U \sim \chi_{d-k}^2$ and $V \sim \chi_k^2$ are independent. The chi-square density at zero and its exponential tail imply $\mathbb{E} |\log R_X(S)| < \infty$. If $Z_j = \sum_{i=1}^j g_i^2$ with standard Gaussians g_i , then arithmetic-geometric mean and Jensen give

$$\log j + \mathbb{E} \log g_1^2 \leq \mathbb{E} \log Z_j \leq \log j.$$

It follows that $\mathbb{E} \log R_X(S) \geq -C$. Gram-Schmidt on the rows of X expresses $\det(XX^\top)$ as a product of chi-squares in dimensions $d, d-1, \dots, d-k+1$. Hence

$$\mathbb{E} |\log \det(XX^\top)| < \infty, \quad \mathbb{E} \log \det(XX^\top) \geq k \log d - Cd.$$

Given an unconditioned block, S is uniform on its fiber. Its slice area is $|S^n| R_X(S)^n$, and its label density is that area divided by $j_X(S) |S^{d-1}|$. The sampled log densities are therefore integrable by the preceding estimates.

Adding a countable variable U with finite Shannon entropy preserves this property. Indeed, the average divergence of a law conditioned further on U from the preceding conditional law is at most $H(U)$. For a likelihood ratio v , its negative log part has numerator expectation at most $\int v(\log v)_- dQ \leq 1/e$. The likelihood log is consequently integrable, and so are the new log densities. This reasoning also works after conditioning on X , even if U depends on X . We establish finite Shannon entropy for every auxiliary string before using (151) with that string.

Let K be a countable transcript with $H(K) < \infty$. Suppose K specifies a closed ball $B(z, r)$ containing S , where $0 < r \leq 1$, and put

$$\Lambda = \mathbb{E}[-\log r] < \infty.$$

For an absolute C_v , every affine n -sphere Σ obeys

$$\text{area}_n(\Sigma \cap B(z, r)) \leq |S^n| (C_v r)^n. \quad (152)$$

If the radius R of Σ is at most $4r$, use its entire area. Otherwise, choose a point of the intersection. The intersection lies in its chordal cap of radius $2r$, whose polar angle is at most C_r/R . Polar integration bounds that area by $|S^{n-1}| R^n (C_r/R)^n / n$. The sphere-area formula, or the same polar integral, gives $|S^{n-1}|/n \leq C|S^n|$; increasing C_v proves (152).

For a fresh block B , define

$$b_n = \log |S^n| + n \log C_v, \quad P = b_n - n\Lambda - h_n(S | K, B). \quad (153)$$

The block B is an auxiliary probe used only to evaluate this potential. After the transcript changes, the updated potential is evaluated using a new independent fresh block. The entropy of a probability density supported on a set of area v is at most $\log v$. Applying this fact on each fiber and using (152) gives $P \geq 0$. The same conclusion holds with a nonfresh slicing block if the conditioning still specifies the containing ball. With $r = 1$ and a transcript independent of the signal and block,

$$P_0 = n \log C_v - n \mathbb{E} \log R_X(S) = O(d).$$

Lemma 10.1 (Information controlled by the sliced potential). *For the preceding K, r and a fresh block,*

$$I(Y_B; K \mid X_B) \leq \frac{k}{d-1} I(S; K) + Cd, \quad h(Y_B \mid X_B) \geq -Cd, \quad (154)$$

$$I(S; K) \leq C(P + d\Lambda + d), \quad -h(Y_B \mid X_B, K) \leq C(P + d\Lambda + d). \quad (155)$$

Proof. The subset averaging below is the entropy-cover mechanism used in the Product Theorem of Chung, Graham, Frankl, and Shearer [2, Section V, Equation (22), pp. 33–34]. We give the conditional differential-entropy form required here. Let Q be an independent Haar orthogonal frame and set $Z_i = \langle Q_i, S \rangle$. Put $N = d - 1$. For a k -subset $E \subset \{1, \dots, N\}$, the entropy chain rule and conditioning give

$$h(Z_E \mid Q, K) \geq \sum_{i \in E} h(Z_i \mid Q, K, Z_1, \dots, Z_{i-1}).$$

Every index occurs in the same number of k -subsets. Averaging over the subsets and over Q , whose law is unchanged by column permutations, yields

$$h(Z_1, \dots, Z_k \mid Q, K) \geq \frac{k}{N} h(Z_1, \dots, Z_N \mid Q, K). \quad (156)$$

This uses no rotation invariance of the conditional law given K . All projected entropies are finite: the uniform projection density has an integrable logarithm for every number of coordinates at most $d - 1$, and finite-entropy conditioning preserves this as above.

Let h_j^0 be the entropy of j fixed orthogonal coordinates of a uniform sphere point. The Gaussian maximum-entropy bound and the covariance I_j/d give

$$h_k^0 \leq \frac{k}{2} \log(2\pi e/d).$$

For $j = N$, the two hemispherical graphs give

$$h_N^0 = \log(|S^{d-1}|/2) + \mathbb{E} \log |S_d| \geq -\frac{N}{2} \log d - Cd.$$

For the last inequality, use the sphere-area formula and Stirling's bound, and write $S = g/\|g\|$; Jensen bounds $\mathbb{E} \log \|g\|$ above by $\frac{1}{2} \log d$, whereas $\mathbb{E} \log |g_d|$ is a finite absolute constant. Data processing gives

$$h(Z_1, \dots, Z_N \mid Q, K) \geq h_N^0 - I(S; K).$$

Thus (156) bounds the information in k frame coordinates by $\frac{k}{N} I(S; K) + Cd$.

Given X_B , its label is an invertible linear transform of the orthogonal coordinates in its row space. That row space is uniform, independent of (S, K) ; extending an orthonormal row frame gives the Q experiment above. Hence the same information bound holds for $I(Y_B; K \mid X_B)$. Without

K , (156) gives $h_k^0 \geq (k/N)h_N^0$. Passing from orthogonal coordinates to Y_B adds $\frac{1}{2}\mathbb{E} \log \det(X_B X_B^\top)$, which cancels the leading $-\frac{k}{2} \log d$ term. This proves both inequalities in (154).

By the definition of P ,

$$I(S; K | B) = P - P_0 + n\Lambda.$$

Freshness and $Y_B = X_B S$ give the chain-rule identity

$$I(S; K) = I(S; K | B) + I(Y_B; K | X_B).$$

Since $k/(d-1) \leq 1/4$, the first inequality of (154) can be absorbed on the left. This proves the information estimate in (155). Finally $h(Y_B | X_B, K) = h(Y_B | X_B) - I(Y_B; K | X_B)$; the two bounds just proved give the negative-entropy estimate. $\square$

10.2 A terminating revelation and a separated tuple

Let $A = (X_A, Y_A)$ be the actual next block, fresh relative to K . Let W be its outgoing message, with at most e^b values. The transcript includes the incoming learner state, so the law of W is a kernel of (K, A) . The next construction uses its own auxiliary randomness. It does not change the learner's rule.

For a tuple $\mathbf{s} = (s_1, \dots, s_\ell)$, define

$$D_{\mathbf{s}} = [s_2 - s_1, \dots, s_\ell - s_1], \quad V(\mathbf{s}) = \sqrt{\det(D_{\mathbf{s}}^\top D_{\mathbf{s}})}.$$

This is $q!$ times the simplex volume and is symmetric in its points. Gram–Schmidt expresses it as the product of successive distances to predecessor affine spans.

Lemma 10.2 (Entropy cost of obtaining a separated tuple). *There are fixed absolute constants $a \geq 1$, $\delta > 0$, and $b_0 > 0$ with the following property. Put $\eta = e^{-a}$, $p_0 = 1/(4\ell)$, and $u_d = \log(1/p_0) + 1$. There is a countable string C_* , generated from (K, A, S) and separate randomness, and an integrable integer $N \geq 0$, such that K, C_* specifies a ball of radius $r_* = \eta e^{-aN}$ containing S . The string is conditionally independent of W given (K, A, S) , and in fact W is independent of (S, C_*) given (K, A) . Moreover,*

$$H(C_* | K, A) \leq (b_0 d + \ell a + u_d) \mathbb{E} N + u_d, \quad (157)$$

$$H(C_* | K) \leq C_a d (1 + \mathbb{E} N), \quad (158)$$

$$\begin{aligned} 0 \leq P_{\text{mid}} &:= b_n + n \mathbb{E} \log r_* - h_n(S | K, A, C_*) \\ &\leq P - \delta d a \mathbb{E} N + u_d. \end{aligned} \quad (159)$$

Conditional on (K, A, C_*) , there is a symmetric law Q of ℓ points, each with the conditional signal marginal, such that

$$V(\mathbf{s}) \geq (\eta r_*)^q, \quad D(Q \| \mathcal{L}(S | K, A, C_*)^{\otimes \ell}) \leq \log 2. \quad (160)$$

Proof. We describe a decision tree conditional on K, A . At a node let ν be the signal law after the recorded prefix, supported in the current ball $B(z_0, r_0)$. Enumerate once and for all a countable dense family of affine subspaces of dimensions at most $\ell - 2$, for example those generated by rational points. If the first subspace E with

$$\nu\{s : \text{dist}(s, E) < 2\eta r_0\} > p_0$$

exists, record whether the signal belongs to this tube. On a negative answer, continue with the conditional law and the same ball. On a positive answer, cover the current ball by a fixed ordered maximal ηr_0 -separated net, choose the first center within ηr_0 of S , record its index, and replace the ball by the radius- ηr_0 ball about that center. Then continue.

The full ambient net has at most $(1+2/\eta)^d$ points. Given the selected subspace, only $\exp(b_0 d + \ell a)$ indices can occur, where b_0 is independent of $a \geq 1$. To see this, an eligible center lies within $3\eta r_0$ of E . Its projection to E lies in a radius- r_0 ball, coverable by $(1+2/\eta)^v$ radius- ηr_0 balls when $v = \dim E$. Over each covering ball, the eligible centers lie in an ambient radius- $4\eta r_0$ ball. Their disjoint radius- $\eta r_0/2$ balls fit in its radius- $9\eta r_0/2$ enlargement, giving at most 9^d centers. Since $v \leq \ell - 2$, the claimed exponential bound follows after enlarging b_0 .

If no selected subspace exists, every affine subspace of dimension at most $\ell - 2$ has mass at most p_0 in its open ηr_0 -tube. Indeed its distance function on the bounded current ball can be approximated uniformly within ηr_0 by that of a subspace from the countable family. Mass greater than p_0 in the smaller open tube would therefore trigger the larger tube test for a member of the family.

In this case, draw $\ell - 1$ additional independent points from ν , using S as the first point, and record whether

$$V(s_1, \dots, s_\ell) \geq (\eta r_0)^q.$$

On failure continue at the same scale with the updated signal law; on success stop. Before the outcome, the tuple is i.i.d. from ν . If the volume is smaller than $(\eta r_0)^q$, one of the q successive distances is smaller than ηr_0 . Conditionally on its predecessors, each such event has probability at most p_0 . The pass probability is therefore at least $1 - qp_0 \geq 1/2$.

Record the decision type, its outcome, and any ambient net index, but not the extra test points. Given (K, A) and the prefix, the type and selected subspace are already determined. The first successful subspace is a measurable choice because the family is countable and its tube probabilities are kernel integrals of Borel indicators. Fixed scaled nets and ordered ties give measurable centers. Regular conditional laws at nodes, with arbitrary defaults at null nodes, complete a recursion by probability kernels on standard Borel spaces. Independent auxiliary draws make W independent of (S, C_*) given (K, A) .

First truncate the tree after u decisions, and write C_u, N_u for its string and number of refinements. At every active node the progress outcome, meaning refinement or a successful test, has conditional probability $p \geq p_0$. Its binary entropy is at most

$$p\{\log(1/p_0) + 1\} = pu_d,$$

because $-p \log p \leq p \log(1/p_0)$ and $-(1-p) \log(1-p) \leq p$. The expected number of progress outcomes is at most $\mathbb{E}N_u + 1$. Summing the node entropies and the eligible index costs gives

$$H(C_u \mid K, A) \leq (b_0 d + \ell a + u_d) \mathbb{E}N_u + u_d. \quad (161)$$

The expected number of active decisions is at most $p_0^{-1}(\mathbb{E}N_u + 1)$. For an encoding that is decoded given K alone, use two bits for the type and outcome at each active node and a fixed-length full ambient net index at a refinement. The decoder knows the truncation depth u ; the recorded type and outcome determine any earlier stop. The ambient index recovers the next center without knowing the selected subspace. This prefix encoding has expected length at most $C_a d(1 + \mathbb{E}N_u)$ nats, so

$$H(C_u \mid K) \leq C_a d(1 + \mathbb{E}N_u). \quad (162)$$

At finite depth all strings have finite entropy. The area bound and the conditional mutual-information identity give

$$\begin{aligned} 0 \leq P_{\text{mid},u} &= P - na\mathbb{E}N_u + I(S; C_u \mid K, A) \\ &\leq P - \{(n - \ell)a - b_0d - u_d\}\mathbb{E}N_u + u_d. \end{aligned}$$

For large d , $n - \ell > d/2$ and $u_d = O(\log d)$. Choose a once and for all large enough that the coefficient in braces is at least δda with an absolute $\delta > 0$. It follows that $\sup_u \mathbb{E}N_u < \infty$, and then that the expected number of active decisions is bounded uniformly in u . By monotone convergence the full number of decisions is integrable, so the tree terminates almost surely. Denote its string and refinement count by C_*, N .

The same prefix encoding at termination has finite expected length, so $H(C_* \mid K) < \infty$. The truncated prefixes increase to the full string and $N_u \uparrow N$. Conditional information in the prefixes increases to the information in the full countable string; this also follows from the chain rule and $H(C_* \mid C_u, K, A) \rightarrow 0$, using its finite entropy. Passing to the limit in the preceding bounds proves (157)–(159). Thus the stopping argument did not presume entropy finiteness for an infinite procedure.

Finally condition on a terminal prefix just before its successful test, with signal law ν . Let E_{pass} be its symmetric volume event and set $Q = \nu^{\otimes \ell}(\cdot \mid E_{\text{pass}})$. The symmetry makes all marginals equal, and the first marginal is exactly the signal law after recording the pass. Call it μ_* . If $p = \nu^{\otimes \ell}(E_{\text{pass}}) \geq 1/2$, then the product-reference decomposition gives

$$D(Q \parallel \mu_*^{\otimes \ell}) = D(Q \parallel \nu^{\otimes \ell}) - \ell D(\mu_* \parallel \nu) \leq \log(1/p) \leq \log 2.$$

All terms are finite because $dQ/d\nu^{\otimes \ell} = \mathbf{1}_{E_{\text{pass}}}/p$. The volume bound holds throughout the conditioned law. This proves (160). $\square$

10.3 An exact comparison on the equal-label constraints

Use the coupling of Lemma 10.2 conditional on (K, A, C_*) , and draw a common W from the learner's kernel given (K, A) . Every (S_i, K, A, C_*, W) has the true signal marginal. Write

$$h_f = h_{d-1}(S \mid K, C_*), \quad h_{\text{mid}} = h_n(S \mid K, A, C_*),$$

and let $D_{\text{coup}} \leq \log 2$ be the averaged divergence in (160). We call divergence of a tuple law from the product of its marginals total correlation, following Watanabe [15]; the probability-law formulation and product decompositions are given by [1, Section 5, Equation (26) and Proposition 5.1].

Conditional on (K, C_*) , form a reference law by drawing ℓ independent points from $\mathcal{L}(S \mid K, C_*)$, then drawing each row of X as a standard Gaussian in $(\text{col } D_s)^\perp$. Let $\mathcal{E}$ be the averaged relative entropy of the actual coupled tuple and matrix X_A from this reference.

Lemma 10.3 (The tuple comparison). *The divergence $\mathcal{E}$ is finite and*

$$\begin{aligned} \mathcal{E} &= I(X_A; C_* \mid S, K) + q\{h_f - h_{\text{mid}} + \mathbb{E} \log j_{X_A}(S)\} \\ &\quad - \frac{kq}{2} \log(2\pi) - k\mathbb{E} \log V(\mathbf{S}) + D_{\text{coup}}. \end{aligned} \tag{163}$$

Furthermore,

$$\ell I(S; W \mid K, C_*) \leq b + \mathcal{E}. \tag{164}$$

Proof. We first identify a common measure for the two singular laws. Let $\mathcal{R}$ be the open region in the ambient tuple–matrix space where $D_{\mathbf{s}}$ has rank q , X has rank k , and $j_X(s_i) > 0$ for every i . On the incidence constraint $XD_{\mathbf{s}} = 0$, let $\gamma_{\mathbf{s}}$ be the Gaussian matrix law conditioned on that constraint. The ambient Gaussian density of $XD_{\mathbf{s}}$ at zero is

$$\phi_{\mathbf{s}} = (2\pi)^{-kq/2} \mathbf{V}(\mathbf{s})^{-k}.$$

Indeed each row has covariance $D_{\mathbf{s}}^{\top} D_{\mathbf{s}}$. Write p_X for ambient Gaussian matrix density. On the regular incidence set the locally finite measure

$$\sigma^{\otimes \ell}(d\mathbf{s}) \phi_{\mathbf{s}} \gamma_{\mathbf{s}}(dX) \quad (165)$$

also has the integration description

$$p_X(X) dX \sigma(ds_1) \prod_{i=2}^{\ell} \frac{\sigma_{X, Xs_1}(ds_i)}{j_X(s_i)}. \quad (166)$$

To verify this equality, take a continuous test $G(\mathbf{s}, X)$ compactly supported in $\mathcal{R}$. In its integral under $\sigma^{\otimes \ell}(d\mathbf{s}) p_X(X) dX$, insert the normalized indicator of a radius- ε ball in $\mathbb{R}^{kq}$, evaluated at $XD_{\mathbf{s}}$. Fixing the tuple first, the conditional law given $XD_{\mathbf{s}} = Z$ is $\gamma_{\mathbf{s}}$ translated by $Z(D_{\mathbf{s}}^{\top} D_{\mathbf{s}})^{-1} D_{\mathbf{s}}^{\top}$. This translation and the Gaussian image density vary continuously at $Z = 0$. On the tuple projection of the compact support, the smallest singular value of $D_{\mathbf{s}}$ is bounded away from zero, so the image densities are uniformly bounded. Dominated convergence gives (165).

In the other order, fix X, s_1 and apply the area identity (150) separately to $s_2, \dots, s_{\ell}$, using labels $X(s_i - s_1)$. The affine fibers near the zero label are continuous translations and scalings of unit spheres. On the compact regular support their radii and $j_X(s_i)$ are bounded away from zero, so the fiber integrals of G are continuous at zero and uniformly bounded. Dominated convergence in the label vector gives (166). Equality on all continuous compactly supported tests identifies the locally finite Borel measures on the regular set. Thus it also holds for nonnegative measurable integration there. This is a regular-region identity; no global finite mass for the unweighted incidence measure has been assumed.

Both probability laws in the comparison give this region full mass. For the reference, the one-point law has a density relative to sphere area, so its independent tuple is affinely independent almost surely. Its constrained rows live in dimension $d - q > k$ and have full row rank almost surely. On the constraint, all points have the same row-space projection. That projection has norm strictly less than one: if s_1 were in the row space, then $\langle s_1, s_i - s_1 \rangle = 0$ would give $\langle s_1, s_i \rangle = 1$, forcing $s_i = s_1$ and contradicting full affine rank. For the actual law, the volume lower bound gives full affine rank, and the true (S_1, X_A) marginal gives full row rank and positive residual radius almost surely. The quantities $j_X(s_i)$ agree on the constraint because their row-space projections agree.

Condition now on (K, C_*) . Jointly measurable density versions can be obtained by taking Radon–Nikodym derivatives of the joint laws with their conditioning variables and then disintegrating; absolute continuity follows from the finite-entropy conditioning above and from the coupling bound. Let p_c be the full-sphere marginal density, $p_{1,X}$ the density of the true (S_1, X_A) law relative to $\sigma \otimes dX$, and $p_{\text{sl}}(\cdot | A)$ the conditional slice density. Let $t(\mathbf{s} | A)$ be the coupling density relative to the product of these slice marginals. The reference density relative to (165) is

$$\frac{\prod_{i=1}^{\ell} p_c(s_i)}{\phi_{\mathbf{s}}}.$$

The actual density relative to the same measure, using (166), is

$$\frac{p_{1,X}(s_1, X)}{p_X(X)} \prod_{i=2}^{\ell} \{p_{\text{sl}}(s_i | A) j_X(s_i)\} t(\mathbf{s} | A).$$

Here $A = (X, X_{S_1})$ is determined by X, s_1 . Each $p_c(S_i)$ is finite and positive almost surely under the actual law, since each coordinate has that marginal after averaging A . These densities therefore establish absolute continuity of the actual law with respect to the reference, rather than transferring arbitrary null sets between the singular laws.

The first-coordinate contribution to the expected log ratio is

$$\begin{aligned} \mathbb{E} \log \frac{p_{1,X}(S_1, X_A)}{p_X(X_A) p_c(S_1)} &= I(S; X_A | K, C_*) + I(X_A; C_* | K) \\ &= I(X_A; C_* | S, K). \end{aligned}$$

The last equality uses independence of X_A from (S, K) at block entry. Each of the other q coordinates contributes $h_f - h_{\text{mid}} + \mathbb{E} \log j_{X_A}(S)$. The term $\log \phi_{\mathbf{S}}$ contributes $-\frac{kq}{2} \log(2\pi) - k\mathbb{E} \log \mathbf{V}$, and $\log t$ contributes D_{coup} . These are finite log integrals. The signal and slice log densities are integrable because $H(K, C_*) < \infty$; the first-coordinate divergence is at most $H(C_* | K)$; the Jacobian has its original true marginal; and

$$q(\log \eta + \log r_*) \leq \log \mathbf{V}(\mathbf{S}) \leq q \log 2$$

with $\mathbb{E}[-\log r_*] < \infty$. The coupling log likelihood is integrable by its finite divergence and the usual $1/e$ negative-part bound. Taking the log ratio now proves (163).

Discard X_A by data processing. The divergence of the tuple from the product of its (K, C_*) -conditional marginals is at most $\mathcal{E}$. Adding W and using an independent (K, C_*) -conditional copy of W in that reference adds at most $H(W | K, C_*) \leq b$. Decomposing the same finite divergence after conditioning on W gives

$$\ell I(S; W | K, C_*) + \mathbb{E} D(\mathcal{L}(\mathbf{S} | K, C_*, W) \parallel \mathcal{L}(S | K, C_*, W)^{\otimes \ell}).$$

The equality of every signal–message marginal, established before the lemma, is what makes the first term ℓ times the same information. The second term is nonnegative. This proves (164). $\square$

10.4 Cancellation and iteration

Set $K_+ = (K, C_*, W)$, and let B now be fresh relative to K_+ . Write

$$h_+ = h_n(S | K_+, B), \quad \Lambda_+ = \Lambda + a\mathbb{E}N.$$

The entropy identity (151) gives

$$\begin{aligned} h_+ &= h_f - I(S; W | K, C_*) - h(Y_B | X_B, K_+) + \mathbb{E} \log j_{X_B}(S), \\ h_{\text{mid}} - h_f - \mathbb{E} \log j_{X_A}(S) &= -h(Y_A | X_A, K, C_*) - I(S; X_A | K, C_*). \end{aligned}$$

The expected Jacobians agree because, before conditioning, each matrix has the standard Gaussian law independent of the uniform signal. Subtract the first identity from h_{mid} , then use (163) and (164). The coefficient of the mid-versus-full entropy difference is $1 - q/\ell = 1/\ell$. The chain rule and entry independence give

$$I(X_A; C_* | S, K) - I(S; X_A | K, C_*) = I(X_A; C_* | K).$$

The resulting inequality is

$$h_{\text{mid}} - h_+ \leq h(Y_B | X_B, K_+) - \frac{h(Y_A | X_A, K, C_*)}{\ell} - \frac{kq}{2\ell} \log(2\pi) - \frac{k}{\ell} \mathbb{E} \log \mathbf{V}(\mathbf{S}) + \frac{I(X_A; C_* | K) + b + D_{\text{coup}}}{\ell}. \quad (167)$$

In particular, the selection of the actual rows has left the explicit cost $I(X_A; C_* | K)$; those rows have not been treated as fresh after selection.

Let z_* be the decoded final center. Compare the fresh label law to $N(X_B z_*, r_*^2 I_k)$. Since X_B is independent of (S, K_+) and $\|S - z_*\| \leq r_*$,

$$h(Y_B | X_B, K_+) \leq k \mathbb{E} \log r_* + \frac{k}{2} \log(2\pi e).$$

Together with $\log \mathbf{V} \geq q(\log \eta + \log r_*)$, the scale-dependent terms in (167) are at most

$$k \mathbb{E} \log r_* - \frac{kq}{\ell} \mathbb{E} \log r_* + O_a(k) = \frac{k}{\ell} \mathbb{E} \log r_* + O_a(k) \leq O_a(k).$$

The residual coefficient is positive and $\log r_* \leq 0$. For the actual labels,

$$-h(Y_A | X_A, K, C_*) \leq -h(Y_A | X_A, K) + H(C_* | K), \quad I(X_A; C_* | K) \leq H(C_* | K).$$

Apply Lemma 10.1 and (158). Since $\ell \geq d/32$ for large d , when $b \leq d^2$ we obtain

$$h_{\text{mid}} - h_+ \leq C_a d + \frac{C'}{d} (P + d\Lambda) + C_a \mathbb{E} N. \quad (168)$$

Proposition 10.4 (The controlled-horizon recurrence). *Suppose the outgoing message has at most e^b values, with $b \leq d^2$. For the localization update above, there are absolute $\gamma, C_1, C_2 > 0$ such that*

$$P_+ + \gamma d \Lambda_+ \leq (1 + C_1/d)(P + \gamma d \Lambda) + C_2 d. \quad (169)$$

Starting with radius one and signal-independent initial information, after $m \leq d$ blocks whose messages have at most e^{d^2} values,

$$P_m + \gamma d \Lambda_m \leq C d(m + 1), \quad I(S; K_m) \leq C d(m + 1). \quad (170)$$

Proof. By definition, $P_+ = P_{\text{mid}} + h_{\text{mid}} - h_+$. Combine (159) and (168), and take $\gamma = \delta/2$. For sufficiently large d , the coefficient $-\delta d a + C_a + \gamma d a$ of $\mathbb{E} N$ is nonpositive. The term $u_d = O(\log d)$ is absorbed in $C_2 d$. Since $P, \Lambda \geq 0$, the remaining $C'(P + d\Lambda)/d$ is bounded by $(C_1/d)(P + \gamma d \Lambda)$. This proves (169).

Every new transcript is countable with finite entropy, specifies the new containing ball, and has $\Lambda_+ < \infty$. Fresh future matrices can therefore be generated and the construction repeated. Starting from $P_0 = O(d)$ and $\Lambda_0 = 0$, iteration through $m \leq d$ steps has multiplier $(1 + C_1/d)^m \leq e^{C_1}$, which proves the potential bound in (170). Lemma 10.1 then gives the information bound. No linear-growth claim for unrestricted numbers of blocks is made. $\square$

10.5 The full accuracy range

Theorem 10.5 (The sample bound by separated-tuple localization). *Let $M(d) = o(d^2)$ be integer-valued and let $0 < \epsilon(d) \leq 1/10$. Every learner of Section 2 with uniform-sphere angular success probability at least $2/3$ must satisfy*

$$T(d) \geq c d \log(1/\epsilon(d))$$

for all sufficiently large d , with an absolute $c > 0$. The eventual dimension threshold may depend on the memory sequence. Success at least $2/3$ for every fixed signal implies the same conclusion by averaging.

Proof. Set $L = \log(1/\epsilon)$. For almost every realization of shared rule randomness, Lemmas 2.1 and 2.2 give Borel rules with the same uniform-prior experiment. All bounds below are numerical bounds on that conditional success probability, with constants independent of the realization, so they may be averaged. No jointly selected family of Borel versions is needed. The joint independence of the shared randomness and initial state from the signal and rows ensures the required signal-independent initial transcript for almost every value of the shared randomness.

We first explain why the restriction $m \leq d$ suffices. An angular cap of radius ϵ has uniform mass at most $(2\epsilon)^{d-1}$. For example, its polar-area numerator is at most $\epsilon^{d-1}/(d-1)$, while its denominator is at least $(\pi/3)(\sqrt{3}/2)^{d-2}$; the displayed bound follows for large d . There are at most $(T+1)2^M$ terminal time–state pairs. For each pair, integration over its fixed fresh output law contributes at most one cap mass, even before multiplying by its reach probability. Summing and averaging over shared randomness gives

$$(d-1)L \leq M \log 2 + \log(T+1) + (d-1) \log 2 + \log(3/2). \quad (171)$$

At indices with $T < dL$, $\log(T+1) \leq \log(d+1) + L$. Since $M = o(d^2)$, (171) implies $L = o(d)$ along those indices. Also

$$b = \log((T+2)2^M) \leq d^2$$

eventually, and after padding to a multiple of k ,

$$m = \lceil T/k \rceil \leq 32L + 1 = o(d) \leq d$$

eventually. Thus every potentially short run lies within the proved horizon.

At indices with $T \geq dL$, the claimed conclusion holds with constant one. At all remaining indices, the horizon and width bounds just established apply eventually. Pad an early-stopping program by retaining its terminal state and stopping index, then ignore later observations and any suffix added to complete the last block. At boundaries its state is a message with at most e^b values. The augmented transcript retains all earlier auxiliary strings, but the learner uses only its state. Future matrices remain fresh, so Proposition 10.4 applies with $m = \lceil T/k \rceil \leq d$. For each shared seed, $I(S; K_m) \leq C_0 d(m+1)$, with an absolute C_0 .

Let π be the success probability for that seed. The output uses information contained in K_m and fresh randomness. Under the product of its marginal and the uniform signal law, success has probability at most $(2\epsilon)^{d-1}$. By data processing and (1),

$$I(S; K_m) \geq \pi(d-1) \log \frac{1}{2\epsilon} - \log 2.$$

Combining the two bounds gives a numerical inequality in π ; averaging it over the shared seed replaces π by the original success probability, at least $2/3$. Thus, for absolute $L_0, c_0 > 0$, whenever $L \geq L_0$,

$$c_0 d L \leq C_0 d(m+1) \leq C_0 d(T/k + 2),$$

so $T \geq k(c_0 L / C_0 - 2)$. Choose $L_1 = \max\{L_0, 4C_0/c_0\}$. Since $k \geq d/32$, this gives

$$T \geq \frac{c_0}{64C_0} dL \quad (L \geq L_1).$$

For $L < L_1$, the local full-data endpoint in Lemma 2.3 gives $T > d/4$, hence $T \geq dL/(4L_1)$. The minimum of these absolute constants and one works for every allowed accuracy sequence, including nonmonotone ones. $\square$

Remark 10.6 (Vanishing success with sublinear samples). The full-data calculation also gives a stronger small-sample conclusion. For $0 \leq T \leq d-3$, reveal all T rows and labels. As in Lemma 2.3, the conditional signal is uniform on a residual sphere of radius R , with $\Pr\{R < 1/2\} \leq 4T/(3d)$. On $R \geq 1/2$, a radius- ϵ error ball meeting that sphere is contained in a chordal cap of radius 2ϵ about a point of the intersection. For $\epsilon \leq 1/10$, its polar angle is at most $\theta_0 = 2\arcsin(1/5)$. Since $\sin \theta_0 < 2/5$, polar integration bounds its conditional mass by

$$\frac{\int_0^{\theta_0} \sin^{d-T-2} \theta d\theta}{\int_0^\pi \sin^{d-T-2} \theta d\theta} \leq C \left(\frac{4}{5\sqrt{3}} \right)^{d-T-2}.$$

Consequently every estimate based on these data and independent randomness has success probability at most

$$\frac{4T}{3d} + C \left(\frac{4}{5\sqrt{3}} \right)^{d-T-2}.$$

This tends to zero when $T = o(d)$, uniformly for $0 < \epsilon \leq 1/10$, and therefore also applies to early-stopping finite-state learners.

11 Localization near a predecessor span

Several signals drawn from the posterior of one exact block satisfy the same linear equations. Their affine volume records how close each new signal is to the span of its predecessors. We use this volume in two steps. First, a relative-entropy calculation divides the information cost of the block's finite message by the number of posterior draws. A draw close to the span of its predecessors incurs an additional cost. Second, a short discrete description locates that draw in a smaller ball, and the decrease in radius pays for the additional cost.

Throughout this section, μ is uniform probability on S^{d-1} , and all logarithms are natural. The signal and all auxiliary variables live in standard Borel spaces. If the conditional signal law given A is $f_A \mu$, write

$$\mathcal{D}(A) = \mathbb{E} D(f_A \mu \| \mu).$$

The variable A is part of the analysis and may contain continuous information. The learner retains only the finite state specified in Section 2.

11.1 The exact equal-label input and an additional moment

For an integer $1 \leq r \leq d-2$, let γ_r be the law of an r -by- d standard Gaussian matrix. If X has full row rank, write $\rho_{0,r}(X, y)$ for the density of Xs , $s \sim \mu$, and $\mu_{X,y}$ for its conditional law. Put

$$X^\dagger = X^\top (X X^\top)^{-1}.$$

On the open ellipsoid $\|X^\dagger y\| < 1$, the conditional law is uniform on the sphere centered at $X^\dagger y$, of radius $\sqrt{1 - \|X^\dagger y\|^2}$, in the affine space parallel to $\ker X$. Its label density is

$$\rho_{0,r}(X, y) = \begin{cases} \frac{\Gamma(d/2)}{\pi^{r/2}\Gamma((d-r)/2)} \frac{(1 - \|X^\dagger y\|^2)^{(d-r-2)/2}}{\sqrt{\det(XX^\top)}}, & \|X^\dagger y\| < 1, \\ 0, & \text{otherwise.} \end{cases} \quad (172)$$

One way to verify the exponent is to parameterize the unit sphere in orthogonal row and null coordinates by $(u, \sqrt{1 - \|u\|^2} \theta)$. Scaling the null-space sphere contributes $(1 - \|u\|^2)^{(d-r-1)/2}$, and the Gram determinant of the u -derivatives contributes $(1 - \|u\|^2)^{-1/2}$. The two tangent groups are orthogonal. Normalizing this density and changing row coordinates to y gives (172). A jointly measurable fiber kernel can be obtained by projecting an extra Gaussian onto $\ker X$ and normalizing it; fixed values are assigned on unused fibers. On rank-deficient matrices set the density to zero and use a fixed fiber probability.

For $p \geq 2$ and $\mathbf{s} = (s_1, \dots, s_p)$, set

$$B_{\mathbf{s}} = [s_2 - s_1, \dots, s_p - s_1], \quad F_{\mathbf{s}} = \text{col}(B_{\mathbf{s}}), \quad J(\mathbf{s}) = \sqrt{\det(B_{\mathbf{s}}^\top B_{\mathbf{s}})}.$$

When $B_{\mathbf{s}}$ has full column rank, let $\gamma_{r, B_{\mathbf{s}}}$ denote the probability law of a matrix whose rows are independent standard Gaussians in $F_{\mathbf{s}}^\perp$.

The geometric input is the finite-measure theorem for equal labels in [8, Theorem 3.3]. We use its density-one spherical instance, stated here in our notation.

Proposition 11.1 (Finite measure for equal labels). *Let $r \geq 1$, $p \geq 2$, and $r + p - 1 \leq d - 2$. For every nonnegative Borel H ,*

$$\begin{aligned} & \int \gamma_r(dX) \int_{\mathbb{R}^r} \rho_{0,r}(X, y)^p \int H(X, y, \mathbf{s}) \mu_{X,y}^{\otimes p}(d\mathbf{s}) dy \\ &= (2\pi)^{-r(p-1)/2} \int J(\mathbf{s})^{-r} \mu^{\otimes p}(d\mathbf{s}) \int H(X, Xs_1, \mathbf{s}) \gamma_{r, B_{\mathbf{s}}}(dX). \end{aligned} \quad (173)$$

These are finite measures before inserting H . They are supported on $Xs_1 = \dots = Xs_p = y$, full ranks of X and $B_{\mathbf{s}}$, and positive-radius spherical fibers. The input uses the exact density and fiber versions above; its ball-average and Gaussian approximation versions agree at the constrained label outside a null set for the finite measure. The more general theorem permits bounded nonnegative densities over spherical or Gaussian base probability. Only the density-one spherical instance is used below.

Proof. Apply [8, Theorem 3.3] with row count r , difference count $p - 1$, tuple size p , and base and reweighted measure both equal to μ . Its dimension condition is exactly $r + p - 1 \leq d - 2$, and its determinant and constrained Gaussian law are J and $\gamma_{r, B_{\mathbf{s}}}$ above. $\square$

We shall need one more inverse power than appears in the $r = k$ identity. The strict margin in the next lemma makes that use explicit.

Lemma 11.2 (The additional inverse-volume moment). *Let $k \geq 1$, $p \geq 2$, and $k + p + 1 < d$. Then*

$$\int J^{-(k+1)} d\mu^{\otimes p} < \infty.$$

Proof. A new sphere point avoids the proper affine span of its predecessors almost surely, so $J > 0$ for $\mu^{\otimes p}$ -almost every tuple. Apply (173) separately with $r = k + 1$. Its dimension condition is

$$r + (p - 1) = k + p \leq d - 2,$$

which follows from the integer inequality $k + p + 1 < d$. We check directly that the total mass on the label side is finite. The exponent in (172) is $(d - k - 3)/2 \geq 0$, and hence

$$\int \rho_{0,k+1}(X, y)^p dy \leq c_{d,k+1}^{p-1} \det(XX^\top)^{-(p-1)/2}.$$

Here $c_{d,k+1}$ is the positive normalizing constant in (172). Gram–Schmidt on the Gaussian rows expresses $\sqrt{\det(XX^\top)}$ as the product of their successive perpendicular lengths. Conditional on the preceding rows, these lengths have chi laws in dimensions $d, d - 1, \dots, d - k$. A chi variable in dimension v has a finite inverse moment of order $p - 1$ whenever $v > p - 1$, as is seen by integrating its density $c_v t^{v-1} e^{-t^2/2}$ at zero and infinity. The smallest dimension here is $d - k > p - 1$, so iterated conditioning makes the displayed determinant bound integrable.

Set $H = 1$ in the $r = k + 1$ instance. Its right side is the positive constant $(2\pi)^{-(k+1)(p-1)/2}$ times $\int J^{-(k+1)} d\mu^{\otimes p}$, because the constrained matrix law is a probability. The finite label-side mass proves the claim. This argument is separate from the later logarithmic estimates. $\square$

11.2 An unbounded finite-entropy prior

Suppose A contains the state at the beginning of a block. Assume that the conditional signal law $f_A \mu$ is supported in an A -measurable closed ball $B(c_A, r_A)$, where $0 < r_A \leq 1$, and

$$\mathcal{D}(A) < \infty, \quad \mathbb{E}|\log r_A| < \infty. \quad (174)$$

There is no boundedness assumption on f_A . To obtain a jointly measurable version, take the Radon–Nikodym derivative of the joint law of (A, S) with respect to $\mathcal{L}(A) \otimes \mu$; the required absolute continuity follows by integrating the assumed conditional absolute continuity over A . Disintegration shows that this derivative is f_A for almost every A . Replacing the derivative by zero on its infinite-value set gives a finite-valued Borel version without changing those conditional laws. All conditional formulas below are asserted only for this full-measure set of values of A ; arbitrary values may be used elsewhere.

Fix $1 \leq k$ with $k + p + 1 < d$. Draw $X \sim \gamma_k$ independently of (A, S) , put $Y = XS$, and draw the block destination w by the learner’s kernel of the incoming state and (X, Y) . Assume that w has at most W values. Given (A, X, Y, w) , draw $s_1, \dots, s_p$ independently from the conditional law of S given (A, X, Y) . The destination provides no further information once these data are known. Thus every (s_i, A, X, Y, w) has the original (S, A, X, Y, w) law.

We will replace S by a signal having the same joint law with (A, w) , and enlarge (A, w) to an analysis variable A' that determines a supporting ball $B(c', r')$, with $0 < r' \leq r_A$. For any fixed coefficient $a \geq k + p$, our one-block objective is

$$\mathcal{D}(A') + a\mathbb{E} \log r' \leq \mathcal{D}(A) + a\mathbb{E} \log r_A + \frac{\log W}{p} + \frac{k}{2} + Cp + 2a \log 2,$$

where C is absolute. We obtain this bound by retaining a uniformly selected posterior draw and its predecessors, then discretely locating that draw near their affine span. First we derive the exact likelihood of the tuple and prove its logarithms integrable, so that the entropy chain rule can be applied.

For fixed A and full-row-rank X , define

$$\rho_A(X, y) = \begin{cases} \rho_{0,k}(X, y) \int f_A(s) \mu_{X,y}(ds), & \rho_{0,k}(X, y) > 0 \text{ and } \int f_A d\mu_{X,y} < \infty, \\ 0, & \text{otherwise.} \end{cases}$$

Set $\rho_A = 0$ on rank failures. Kernel integration gives a jointly measurable function of (A, X, y) . Disintegration shows that it is a conditional label density; the zero assignment on infinite fiber means changes it only on a Lebesgue-null set for almost every (A, X) . The piecewise definition also avoids a product $0 \cdot \infty$ on unused fibers. The actual label has $0 < \rho_A(X, Y) < \infty$ almost surely, and on that set its posterior is

$$\frac{\rho_{0,k}(X, Y)}{\rho_A(X, Y)} f_A \mu_{X,Y}.$$

This definition fixes the version used below.

Let π_A be the law of $(\mathbf{s}, X)$ in this experiment, with w discarded. Define the probability measure

$$\lambda(d\mathbf{s}, dX) = \mu^{\otimes p}(d\mathbf{s}) \gamma_{k,B_{\mathbf{s}}}(dX),$$

with an arbitrary definition on the $\mu^{\otimes p}$ -null set where $J = 0$.

Lemma 11.3 (Replica likelihood with finite logarithms). *Under (174),*

$$\frac{d\pi_A}{d\lambda}(\mathbf{s}, X) = (2\pi)^{-k(p-1)/2} J(\mathbf{s})^{-k} \frac{\prod_{i=1}^p f_A(s_i)}{\rho_A(X, X_{s_1})^{p-1}}. \quad (175)$$

The multiplier is defined as zero unless $J > 0$ and its denominator is finite and positive. Under the replica experiment, each $\log f_A(s_i)$, $\log \rho_A(X, Y)$, and $\log J$ is absolutely integrable after averaging over A .

Proof. For each such value of A , on the label side of (173) with $r = k$, insert the everywhere-defined nonnegative Borel multiplier

$$\begin{cases} \frac{\prod_i f_A(s_i)}{\rho_A(X, y)^{p-1}}, & 0 < \rho_A(X, y) < \infty, \\ 0, & \text{otherwise.} \end{cases}$$

The identity applies first to bounded truncations and then to this multiplier by monotone convergence. The label-side measure becomes the actual label law $\rho_A(X, y) dy$ followed by p normalized posterior draws. The other side is the right side of (175) relative to λ . This proves the likelihood formula. In particular, no bounded posterior theorem has been applied to f_A ; the only geometric input was the identity for μ . Values on invalid label fibers are handled by the multiplier itself, so no unproved transfer of a Lebesgue-null set to the singular reference is being used.

We justify logarithmic integrability before taking any entropy difference. Since $s_i \mid A$ has density f_A ,

$$\int f_A(\log f_A)_- d\mu \leq 1/e.$$

Together with $\mathcal{D}(A) < \infty$, this gives averaged absolute integrability of $\log f_A(s_i)$. Let g be standard Gaussian density on $\mathbb{R}^k$. For almost every (A, X) ,

$$\int \rho_A(X, y) \left(\log \frac{g(y)}{\rho_A(X, y)} \right)_+ dy \leq \frac{1}{e} \int g(y) dy = \frac{1}{e}.$$

Moreover $\mathbb{E}\|Y\|^2 = k$, because X is fresh and $\|S\| = 1$. Since $-\log g(Y) = \frac{k}{2} \log(2\pi) + \frac{1}{2}\|Y\|^2$, the negative part of $\log \rho_A(X, Y)$ is integrable.

Put $c = (2\pi)^{-k(p-1)/2}$ and

$$Z = c \int J^{-k} d\mu^{\otimes p}.$$

The $r = k$ input shows $0 < Z < \infty$. Let Q be the probability with density cJ^{-k}/Z relative to λ . The likelihood formula gives $\pi_A \ll \lambda$. Since $0 < J < \infty$ λ -almost surely, the density of Q is finite and strictly positive there; thus Q and λ have the same null sets, and $\pi_A \ll Q$. We may therefore write

$$\log \frac{d\pi_A}{dQ} = \log Z + \sum_{i=1}^p \log f_A(s_i) - (p-1) \log \rho_A(X, Y). \quad (176)$$

Its positive part satisfies

$$\left(\log \frac{d\pi_A}{dQ} \right)_+ \leq |\log Z| + \sum_i (\log f_A(s_i))_+ + (p-1)(\log \rho_A(X, Y))_+,$$

so it is integrable by the two bounds just proved, without using $\log J$. Its negative part has integral at most $1/e$: if $v = d\pi_A/dQ$, then $\int v(\log v)_- dQ \leq 1/e$. Hence $\mathbb{E}D(\pi_A||Q) < \infty$, and the log likelihood in (176) is absolutely integrable on average. Rearranging that identity now proves full integrability of $\log \rho_A(X, Y)$. This step has not used $\log J$.

By Lemma 11.2,

$$\int e^{(\log J)_-} dQ \leq \frac{c}{Z} \left(\int J^{-k} d\mu^{\otimes p} + \int J^{-(k+1)} d\mu^{\otimes p} \right) < \infty.$$

For probabilities P, Q and nonnegative F , exponential tilting gives

$$\mathbb{E}_P F \leq D(P||Q) + \log \mathbb{E}_Q e^F$$

first for bounded F , and then for nonnegative F by truncation. This is the bounded variational inequality of [5, Equations (1.1)–(1.2)], with the limiting step given here. Apply it to $F = (\log J)_-$ conditionally on A and average. Its right side is finite. The positive part of $\log J$ is at most $(p-1) \log 2$ by Hadamard's inequality, since every chord has length at most two. This completes the integrability proof. $\square$

11.3 Information in one selected draw

For $i \geq 2$, let

$$\delta_i = \text{dist}(s_i, \text{aff}(s_1, \dots, s_{i-1})), \quad \Delta_i = \max\{0, \log(r_A/\delta_i)\}, \quad \Delta_1 = 0.$$

Gram–Schmidt gives $J = \prod_{i=2}^p \delta_i$. The likelihood formula implies absolute continuity of the tuple marginal with respect to $\mu^{\otimes p}$, so every $\delta_i > 0$ almost surely. Also $\delta_i \leq 2$ and $\Delta_i \leq (-\log \delta_i)_+$. Thus

$$(-\log \delta_i)_+ \leq (-\log J)_+ + (p-2) \log 2,$$

and Lemma 11.3 implies $\mathbb{E}\Delta_i < \infty$.

Choose an independent uniform $I \in \{1, \dots, p\}$, use s_I as the signal for the next block, and set

$$A_* = (A, w, I, s_1, \dots, s_{I-1}).$$

The triple (s_I, A, w) has the original (S, A, w) law, since each draw has that marginal and I is independent. Uniform selection turns the entropy-chain average into information about the selected draw; the following bound is not asserted for every fixed draw.

Lemma 11.4 (Information of a selected draw). *The conditional signal law given A_* is absolutely continuous with respect to μ , and*

$$\mathcal{D}(A_*) \leq \mathcal{D}(A) + \frac{\log W}{p} + \frac{k}{2} + k \mathbb{E} \Delta_I. \quad (177)$$

Proof. Taking logarithms in (175) is legitimate by Lemma 11.3. Discard X by data processing, then condition on w . The latter step adds the conditional mutual information of the tuple and w , at most $\log W$. The product-reference chain rule, in the finite form discussed in [1, Section 3.3, Equations (13)–(18)], therefore yields

$$\begin{aligned} \mathbb{E} D(\mathcal{L}(\mathbf{s} \mid A, w) \parallel \mu^{\otimes p}) &\leq \log W + p \mathcal{D}(A) + (p-1) \mathbb{E}[-\log \rho_A(X, Y)] \\ &\quad - \frac{k(p-1)}{2} \log(2\pi) - k \mathbb{E} \log J. \end{aligned} \quad (178)$$

The label entropy is bounded by its cross entropy with $N(Xc_A, r_A^2 I_k)$. Freshness and the containing ball give $\mathbb{E}(\|Y - Xc_A\|^2 / r_A^2 \mid A) \leq k$, whence

$$\mathbb{E}[-\log \rho_A(X, Y)] \leq \frac{k}{2} \log(2\pi) + k \mathbb{E} \log r_A + \frac{k}{2}.$$

All terms are finite by (174) and the preceding lemma. Substitute this estimate and $\log J = \sum_{i=2}^p \log \delta_i$ into (178). Since $\log(r_A / \delta_i) \leq \Delta_i$, the result is

$$\mathbb{E} D(\mathcal{L}(\mathbf{s} \mid A, w) \parallel \mu^{\otimes p}) \leq \log W + p \mathcal{D}(A) + \frac{k(p-1)}{2} + k \mathbb{E} \sum_{i=1}^p \Delta_i.$$

This finite relative entropy also implies that, almost surely, each conditional coordinate law given its predecessors is absolutely continuous with respect to μ . The product-reference chain rule and the uniform choice of I give the exact identity

$$\mathcal{D}(A_*) = \frac{1}{p} \mathbb{E} D(\mathcal{L}(\mathbf{s} \mid A, w) \parallel \mu^{\otimes p}).$$

Dividing the preceding bound by p and using $(p-1)/p \leq 1$ gives (177). $\square$

11.4 A radius credit from the predecessor span

We next encode a smaller ball when the selected draw is close to its predecessor span. The code is discrete even though the predecessor coordinates, already in A_* , are continuous.

Lemma 11.5 (A discrete description of a smaller ball). *There is a discrete code C , determined by (A_*, s_I) , and an $A' = (A_*, C)$ -measurable ball $B(c', r')$ supporting the conditional signal law, for which*

$$0 < r' \leq r_A, \quad \log r' \leq \log r_A - \Delta_I + 2 \log 2, \quad (179)$$

$$H(C \mid A_*) \leq p \mathbb{E} \Delta_I + C_1 p, \quad \mathcal{D}(A') \leq \mathcal{D}(A_*) + H(C \mid A_*). \quad (180)$$

Here C_1 is absolute. Both $\mathcal{D}(A')$ and $\mathbb{E}|\log r'|$ are finite.

Proof. If $I = 1$, set $j = 0$. For $I \geq 2$, set

$$j = \max\{0, \lfloor \log_2(r_A/\delta_I) \rfloor\}.$$

When $j = 0$, keep the old ball. When $j \geq 1$, the predecessor span has dimension $v = I - 2$ almost surely. Use orthonormal coordinates in that affine span based at s_1 . The projection of $s_I - s_1$ has norm at most $2r_A$, because both points lie in the old ball.

For every integer pair (j, v) , fix an ordered 2^{-j} -net of the radius-two ball in $\mathbb{R}^v$ with at most $(1 + 4 \cdot 2^j)^v$ points. A maximal 2^{-j} -separated set has this size by comparing the volumes of its disjoint radius- 2^{-j-1} balls with the containing ball. In dimension zero use one point. Record the first net point within 2^{-j} of the projection coordinates divided by r_A . After scaling and translating back, let c' be that point of the predecessor span. The in-span error and perpendicular error are both at most $r_A 2^{-j}$. Hence $B(c', 2r_A 2^{-j})$ contains s_I ; set $r' = 2r_A 2^{-j}$. It is at most r_A for $j \geq 1$.

Figure 1 depicts this step, writing $\eta = r_A 2^{-j}$ and $F = \text{aff}(s_1, \dots, s_{I-1})$ within the illustration. Only the v -dimensional projection must be encoded; the perpendicular error is absorbed by the decoded radius. The radius does not increase, but the balls need not be nested.

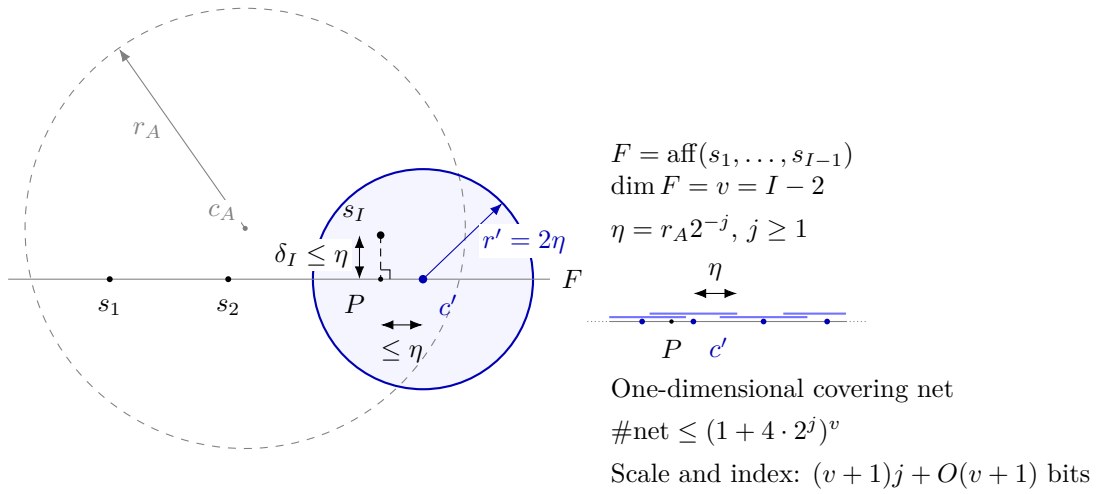


Figure 1: Predecessor-span localization, shown schematically in two dimensions. Only the projection of s_I onto F is quantized: the projected point P lies in the radius- $2r_A$ ball about s_1 in the v -dimensional span. A fixed η -cover supplies the first qualifying ordered net point c' ; the inset illustrates part of such a cover in one dimension, with omitted stretches dotted. Both errors are at most η , so $B(c', 2\eta)$ contains s_I . The scale and net index have the displayed code-length bound, with dimension v . The new radius does not increase, but the new ball need not lie inside the old one.

For $j \geq 1$, $j \log 2 \leq \Delta_I < (j + 1) \log 2$. Thus

$$\log r_A - \Delta_I \leq \log r' \leq \log r_A - \Delta_I + 2 \log 2.$$

For $j = 0$, either $\Delta_I = 0$ or $\Delta_I < \log 2$, and the same upper bound holds with $r' = r_A$. These inequalities give (179) and, using the finite expectations already proved, $\mathbb{E}|\log r'| < \infty$.

Encode j by the unary prefix code of length $j + 1$ bits. For $j \geq 1$, append a fixed-length net index of $\lceil v \log_2(1 + 4 \cdot 2^j) \rceil$ bits. Given A_* , the dimension v is known, so the combined code is prefix-free. Since $1 + 4 \cdot 2^j \leq 5 \cdot 2^j$, its length is at most $(1 + v)j + Cv + 2$. Now $1 + v \leq p$ and $j \log 2 \leq \Delta_I$, so its expected length in nats is at most $p\mathbb{E}\Delta_I + C_1p$. We give the countable conditional entropy step

explicitly. For a conditioning value a_* with finite conditional expected length, let $\ell_{a_*}(c)$ be the bit length and let $p_{a_*}(c)$ be the conditional code distribution. Kraft's inequality [7], in the finite or countable form stated in [4, Section 2.4.2, Theorem 2.4.2], gives

$$0 < K_{a_*} := \sum_c 2^{-\ell_{a_*}(c)} \leq 1.$$

Normalize $q_{a_*}(c) = 2^{-\ell_{a_*}(c)} / K_{a_*}$. Its cross entropy is

$$\sum_c p_{a_*}(c) \log \frac{1}{q_{a_*}(c)} = (\log 2) \mathbb{E}[\ell_{a_*}(C) \mid a_*] + \log K_{a_*} \leq (\log 2) \mathbb{E}[\ell_{a_*}(C) \mid a_*] < \infty.$$

For a finite partition of the code alphabet with one remaining tail atom, nonnegative relative entropy bounds the partition entropy by its q_{a_*} cross entropy, and that cross entropy is at most the displayed full cross entropy. Refining such partitions gives the countable entropy by monotone convergence. Hence $H(C \mid a_*) \leq (\log 2) \mathbb{E}[\ell_{a_*}(C) \mid a_*]$. Finite conditional expected length holds for almost every a_* because its average is finite. Averaging proves the first inequality in (180).

Gram-Schmidt with fixed conventions supplies measurable orthonormal coordinates from the predecessor differences, and the fixed ordering resolves net ties. Define the code arbitrarily on the null degeneracy set. The decoded ball is therefore measurable. The pointwise construction puts s_I in that decoded ball outside the null degeneracy set. The event $\{\|s_I - c'(A')\| \leq r'(A')\}$ is Borel and has probability one. Disintegrating it over A' shows that its conditional probability is one for almost every A' . Thus the decoded ball supports the conditional signal law. Conditioning an absolutely continuous law on a discrete code keeps absolute continuity almost surely. The conditional entropy chain rule gives the second inequality in (180), and its finite right side proves the remaining finiteness claim. $\square$

11.5 The sample bound

The preceding two lemmas now give a complete potential argument. The result uses the local model and the local cap and full-data calculation of Section 2; the only companion input remains (173).

Proposition 11.6 (A predecessor-span block bound). *For sufficiently large d , put $K = p = \lfloor d/8 \rfloor$. Consider a learner of Section 2 with horizon T and M persistent bits, and set $W = (T + 2)2^M$. If $\log W \leq 2d^2$, then its uniform-sphere angular success probability π at $0 < \epsilon \leq 1/10$ satisfies*

$$\pi(d - 1) \log(1/\epsilon) \leq Cd(\lceil T/K \rceil + 1) \quad (181)$$

for an absolute C . In particular, success at least $2/3$ implies

$$\lceil T/K \rceil \geq c \log(1/\epsilon) - C'$$

with absolute $c > 0, C' > 0$.

Proof. Condition on the shared rule randomness. For almost every value, the joint initialization independence in Section 2 leaves the initial state independent of the signal and all sample rows. By Lemmas 2.1 and 2.2, the conditional uniform-prior experiment has Borel rule versions. We prove the displayed numerical bound for each such value, with its conditional success probability. Averaging that bound at the end handles shared randomness without requiring jointly selected Borel versions.

Pad an early-stopping program to T observations by retaining its terminal state and stopping index, and then ignoring the remaining pairs. At each index the padded state has at most W values,

and the final output law uses only that state. Divide the observations into blocks of length K , allowing a shorter final block. Set

$$a = K + p, \quad \Phi(A, r_A) = \mathcal{D}(A) + a\mathbb{E} \log r_A.$$

For every nonempty block length $1 \leq k \leq K$, the inequalities $k + p + 1 < d$ and $a \leq (d - 1)/2$ hold once d is sufficiently large. Initially A is the signal-independent initial state and the containing ball is $B(0, 1)$, so $\Phi = 0$.

At a block, select a replica and encode its new ball as above. Combining (177)–(180) gives

$$\begin{aligned} \Phi(A', r') &\leq \Phi(A, r_A) + \frac{\log W}{p} + \frac{k}{2} + C_1 p + 2a \log 2 + (k + p - a)\mathbb{E} \Delta_I \\ &\leq \Phi(A, r_A) + C_2 d. \end{aligned} \tag{182}$$

Indeed $k + p - a \leq 0$, while $p \geq d/16$ and $\log W \leq 2d^2$ make the width charge $O(d)$. The cancellation has a direct explanation: the likelihood charges $k\Delta_I$, the code charges at most $p\Delta_I$, and the radius credit is $a\Delta_I = (K + p)\Delta_I$.

Every posterior draw has the correct joint marginal with the block destination. Uniformly selecting one draw therefore preserves the learner's state–signal marginal, and recording its code changes no marginal law. Generate the next Gaussian block independently of the pair consisting of the current signal and augmented analysis variables. Since the learner's transition uses the past only through its state, induction preserves the actual state–signal marginal at every boundary. The predecessor points and codes are never given to the learner. All required finiteness properties propagate by Lemma 11.5. Thus after $b = \lceil T/K \rceil$ blocks,

$$\Phi \leq C_2 db.$$

For $T = 0$ this holds directly with $b = 0$.

Write $n = d - 1$. Lemma 2.3 gives $\mu(B(c_A, r_A)) \leq (2r_A)^n$. A density supported in a set E has relative entropy at least $-\log \mu(E)$, by comparison with $\mu(\cdot | E)$. Therefore

$$D(f_A \mu \| \mu) \geq -n \log(2r_A)$$

almost surely, and

$$\Phi \geq \left(1 - \frac{a}{n}\right) \mathcal{D}(A) - a \log 2 \geq \frac{1}{2} \mathcal{D}(A) - a \log 2. \tag{183}$$

The unconditional signal law remains μ , so $\mathcal{D}(A) = I(S; A)$. The final A includes the padded state w , and hence $\mathcal{D}(A) \geq I(S; w)$.

Under the product of μ and the output marginal, angular success has probability at most ϵ^n , by Lemma 2.3. Data processing through the terminal output and then through the success event gives

$$I(S; w) \geq I(S; \hat{S}) \geq \pi n \log(1/\epsilon) - \log 2$$

by (1). This holds with the success probability for the fixed shared seed. Combining it with the potential upper bound and (183) proves (181) with constants independent of that seed. Averaging proves the asserted bound for the original shared randomness as well. $\square$

Theorem 11.7 (The full accuracy range by predecessor-span localization). *Let $M(d) = o(d^2)$ be integer-valued and let $0 < \epsilon(d) \leq 1/10$. A family of learners in Section 2 whose angular success probability is at least $2/3$ for a uniform spherical signal must use*

$$T(d) \geq c d \log(1/\epsilon(d))$$

for all sufficiently large d , where $c > 0$ is absolute. The eventual threshold may depend on the memory sequence. The same conclusion follows from success at least $2/3$ for every fixed signal.

Proof. Put $L = \log(1/\epsilon) \geq \log 10$. The last statement follows by averaging over the uniform signal, so it suffices to prove the average statement. There are at most $(T+1)2^M$ terminal time–state pairs. For each fixed pair its fresh output law succeeds on uniform mass at most ϵ^{d-1} , by the cap bound and integration over that output law. More explicitly, for a terminal pair h let $a_h(s)$ be that fixed output law’s success probability at signal s , and let $t_h(s) = \Pr(h \text{ is reached} \mid S = s)$. The pair’s contribution is

$$\int t_h(s) a_h(s) \mu(ds) \leq \int a_h(s) \mu(ds) \leq \epsilon^{d-1},$$

because $0 \leq t_h(s) \leq 1$. Summing over all pairs, for almost every shared seed and hence after averaging, yields

$$\frac{2}{3} \leq (T+1)2^M \epsilon^{d-1}.$$

Equivalently,

$$(d-1)L \leq M \log 2 + \log(T+1) + \log(3/2). \quad (184)$$

At indices where $T \geq dL$ the conclusion is immediate. At the other indices, (184) gives

$$(d-1)L \leq M \log 2 + \log(dL+1) + \log(3/2).$$

For all sufficiently large d , uniformly for $L \geq \log 10$, $\log(dL+1) \leq dL/6$, while $(d-1)L \geq dL/2$. Consequently $dL \leq 3M \log 2 + 3 \log(3/2) = o(d^2)$ along such short indices. In particular $\log((T+2)2^M) \leq 2d^2$ eventually, so Proposition 11.6 applies. It gives

$$T \geq K\{c_0 L - C_0 - 1\}$$

with absolute $c_0 > 0, C_0$. Above a sufficiently large absolute threshold for L , use $K \geq d/16$ and absorb the constants to obtain $T \geq c_1 dL$.

For completeness, the local full-data argument supplies the remaining constant-accuracy range with explicit constants. If $T \leq d/8$, reveal every row and label. The conditional unknown component is uniform on a residual sphere of radius R . By isotropy and Markov’s inequality,

$$\Pr\{R < 1/2\} \leq \frac{4T}{3d} \leq \frac{1}{6}.$$

On $R \geq 1/2$, antipodal residual points correspond to signals separated by at least one, so a radius- ϵ error ball with $\epsilon \leq 1/10$ contains at most one point of each pair. Conditional success is at most $1/2$, also for a randomized estimate. Total success is at most

$$\frac{1}{6} + \frac{5}{6} \cdot \frac{1}{2} = \frac{7}{12} < \frac{2}{3}.$$

Thus $T > d/8$. This proves $T \geq c_2 dL$ throughout the bounded range of L left above. Taking the minimum of the absolute constants completes the proof, including nonmonotone accuracy sequences and early stopping. $\square$

12 A convex potential from inverse-distance kernels

Explicit revelations are one way to pay for a concentrated posterior. This section uses a different accounting method. From the relative entropy of a posterior we subtract the logarithm of an inverse-distance average. The resulting functional is convex, so after a block we may forget the entering state without increasing it. A mixed projection estimate then charges only the rows that the learner actually saw.

Throughout this section, σ is uniform probability on S^{d-1} , $n = d - 1$, $k = \lfloor d/16 \rfloor$, and $p = 2k$. We work in sufficiently large dimension that $k \geq 1$ and $2p < n$. For a probability measure $\nu \leq H\sigma$, $H < \infty$, put

$$J_\nu(s) = \int \|u - s\|^{-p} d\nu(u), \quad B(\nu) = \int \log J_\nu(s) d\nu(s), \quad \Phi(\nu) = D(\nu\|\sigma) - B(\nu). \quad (185)$$

The sphere has diameter two, so $J_\nu \geq 2^{-p}$. The centered ball bound of Lemma 2.3, summed on dyadic annuli, bounds J_ν above uniformly in s because $p < n$. Thus every term in (185) is finite.

Lemma 12.1 (Convexity and coercivity). *The functional Φ is convex under finite mixtures of bounded-density probability measures and*

$$\Phi(\nu) \geq \frac{1}{2}D(\nu\|\sigma) - Cd, \quad \Phi(\sigma) \leq p \log 2$$

for an absolute C .

Proof. Writing $f = d\nu/d\sigma$, we have $\Phi(\nu) = \int f \log(f/J_\nu) d\sigma$, with value zero where $f = 0$. Both f and J_ν are linear in ν . For $x > 0, y > 0$, the Hessian of $x \log(x/y)$ has quadratic form $(a - bx/y)^2/x$; continuity at $x = 0$ gives joint convexity on $x \geq 0, y > 0$. Integration proves the mixture assertion.

By Jensen and rotational invariance,

$$\int J_\nu(s)^2 d\sigma(s) \leq \int \int \|u - s\|^{-2p} d\sigma(s) d\nu(u) \leq e^{Cd}.$$

For the last bound, distances greater than one contribute at most one. On $2^{-i-1} < \|u - s\| \leq 2^{-i}$, the contribution is at most $2^{2p}2^{-i(n-2p)}$; its sum converges since $n - 2p > 0$. Apply the entropy inequality (2), equivalently $\mathbb{E}_\nu F \leq D(\nu\|\sigma) + \log \mathbb{E}_\sigma e^F$, to $F = 2 \log J_\nu$. This logarithm is bounded for fixed ν . It gives $2B(\nu) \leq D(\nu\|\sigma) + Cd$. Finally, $D(\sigma\|\sigma) = 0$ and $J_\sigma \geq 2^{-p}$. $\square$

12.1 Exact projection densities and the kernel entropy

We will use differential entropy of exact labels. The following elementary density statement fixes both its version and its integrability before the information identity is used.

Lemma 12.2 (Projection versions and logarithms). *Let $1 \leq \ell < d - 1$, let G have ℓ independent standard Gaussian rows, and let $\nu \leq H\sigma$ be a probability measure. For each full-row-rank g , the image $g_\# \nu$ has a Lebesgue density. The limit, where finite and existent, of*

$$p_{\nu,a}(g, y) = (2a)^{-\ell} \nu\{u : \|gu - y\|_\infty \leq a\}, \quad a = 1, 1/2, 1/3, \dots,$$

and zero elsewhere is a jointly Borel density version $p_\nu(g, y)$. Under the independent law of G and $S \sim \nu$, $\log p_\nu(G, GS)$ is integrable.

If instead G and $S \sim \sigma$ are marginally independent and W is finite with any joint law with them, the actual conditional law of GS given (G, W) has a Lebesgue density. Its logarithm and $\log p_{\nu_W}(G, GS)$, where $\nu_w = P_{S|W=w}$, are integrable under the actual law. The same assertion holds after first conditioning on a finite entering variable V , provided the fresh G is jointly independent of (S, V) . Equivalently, conditional on each positive-probability value of V , the joint law of (G, S) must be the product of the original Gaussian row law and the conditional law of S .

Proof. Writing $H_g = (gg^\top)^{1/2}$, an orthogonal change of coordinates and a nonsingular change of label coordinates give

$$p_\sigma(g, y) = \frac{c_{d,\ell}}{\det(gg^\top)^{1/2}} \left(1 - \|H_g^{-1}y\|^2\right)^{(d-\ell-2)/2} \mathbf{1}_{\{\|H_g^{-1}y\| < 1\}}.$$

The spherical coordinate formula follows by splitting a normalized standard Gaussian into its ℓ row-space coordinates and the remaining coordinates: the squared projected length is Beta($\ell/2, (d-\ell)/2$). Domination proves absolute continuity for ν . Lebesgue differentiation proves the claim about cube limits, and their defining integrals are jointly Borel.

Under $G \otimes \sigma$, the logarithm in the displayed formula is integrable. The beta density has integrable logarithms at its two endpoints. Gram–Schmidt expresses $\det(GG^\top)$ as a product of chi-square variables of positive degrees, each with integrable logarithm. Domination by H preserves this baseline integrability for $S \sim \nu$. The projected likelihood ratio p_ν/p_σ has relative entropy at most $D(\nu\|\sigma) \leq \log H$ by data processing. Its negative logarithmic part has expectation at most $1/e$, since $t \log^-(t) \leq 1/e$ under the probability reference. Its positive part is then finite as well. This proves the first integrability assertion.

For a positive-probability w , the actual law of (G, S) given w is dominated by $P(W = w)^{-1}$ times $G \otimes \sigma$. Its image therefore has a density, and the baseline logarithm is integrable; averaging over w restores its original finite expectation. The log ratio of the actual image density to p_σ has relative entropy $I(W; GS \mid G) \leq H(W)$ and negative part at most $1/e$. Thus the actual log density is integrable. Moreover,

$$\mathbb{E}D(P_{S|G,W}\|\nu_W) = I(S; G \mid W) \leq H(W),$$

where the last inequality uses $I(S; G) = 0$ and the chain rule. Data processing under $s \mapsto Gs$ gives integrability of the log ratio between the actual image density and p_{ν_W} ; subtracting it proves the remaining claim. If a finite entering variable is first fixed, its signal law is bounded relative to σ , and the same argument applies to each positive-probability case and then to their finite average. $\square$

Lemma 12.3 (The kernel controls projection entropy). *Let X have p independent standard Gaussian rows and be independent of $S \sim \nu \leq H\sigma$, $H \geq 1$. For a sufficiently large absolute C_0 , put $b_H = \lceil C_0(d + \log(2H)) \rceil$. Then*

$$h(XS \mid X) \leq -B(\nu) + Cd + 2\log(b_H + 2).$$

Proof. For $r_i = 2^{1-i}$, let $m_i(s) = \nu(B(s, r_i))$. Dyadic shells give

$$J_\nu(s) \leq 2^p \sum_{i \geq 0} r_i^{-p} m_i(s).$$

Since $m_i(s) \leq H(2r_i)^n$ and $n - p \geq 1$, choosing C_0 large makes the sum over $i > b_H$ at most 2^{-p} . The term $i = 0$ is itself 2^{-p} . Choose the first maximizer $j = j(s) \leq b_H$. This is measurable and satisfies

$$\log(r_j^{-p} m_j(s)) \geq \log J_\nu(s) - Cd - \log(b_H + 2).$$

For fixed X , compare the actual projected density to

$$Q_X(y) = \frac{1}{b_H + 1} \sum_{i=0}^{b_H} \int \gamma_{r_i}(y - Xu) d\nu(u),$$

where γ_r is the $N(0, r^2 I_p)$ density. At $y = Xs$, retain the j -summand and restrict it to $B(s, r_j)$. Jensen's inequality for the logarithm on that normalized restriction and $\mathbb{E}_X \|X(s-u)\|^2 = p\|s-u\|^2 \leq pr_j^2$ give

$$\mathbb{E}_X \log Q_X(Xs) \geq \log(r_j^{-p} m_j(s)) - \log(b_H + 1) - \frac{p}{2} \log(2\pi) - \frac{p}{2}.$$

Integrate over s . The comparison density has finite cross entropy by this bound and its finite Gaussian-mixture upper bound. The actual differential entropy is finite by Lemma 12.2. Nonnegativity of relative entropy to $Q_X(y) dy$ proves the result. $\square$

12.2 The mixed estimate and the actual-row charge

The following is the positive-scale assertion of [9, Theorem 3.1]. We state its parameters explicitly because its application has only one unit of angular slack. Its proof is in that companion; the reweighting and change of row law below are proved here.

Theorem 12.4 (Mixed projection estimate). *Let $m, r, q \geq 1$ be integers, $\ell = m + r < n = d - 1$, and $\ell - m - (q - 1) \geq 1$. Let ρ be a finite measure on the sphere such that, for every ambient z and $t > 0$, $\rho(B(z, t)) \leq Bt^\ell$ and $\rho(B(z, t)) \leq Dt^n$, with $B, D > 0$. For independent Gaussian matrices X, Z with m, r rows, put $G = (X; Z)$ and*

$$p_{\rho, \delta}(G, y) = (2\delta)^{-\ell} \int \mathbf{1}_{\{\|Gu-y\|_\infty \leq \delta\}} d\rho(u).$$

For every fixed $s \in S^{d-1}$ and every $\delta > 0$,

$$[\mathbb{E}_X (\mathbb{E}_Z p_{\rho, \delta}(G, Gs))^q]^{1/q} \leq e^{Cd} B(1 + \log^+(D/B)).$$

The estimate is for positive cube averages and every fixed signal. In its use below, we first change the law of the actual rows while $\delta > 0$, and only then pass to exact labels under their actual law. No assertion about a fixed-signal value of an arbitrary density version is used.

Lemma 12.5 (Reweight by the potential). *For $\nu \leq H\sigma$, define $d\rho = d\nu/J_\nu$. Then*

$$\rho(B(z, r)) \leq (2r)^p, \quad \rho \leq 2^p H\sigma.$$

By Theorem 12.4, for independent k -row Gaussian matrices A, A' , $X = (A; A')$, every $s \in S^{d-1}$, and every $\delta > 0$,

$$[\mathbb{E}_A (\mathbb{E}_{A'} p_{\rho, \delta}(X, Xs))^k]^{1/k} \leq e^{Cd} (b_H + 2). \quad (186)$$

Proof. For $s \in B(z, r)$, $J_\nu(s) \geq (2r)^{-p} \nu(B(z, r))$. Integrating its reciprocal over the ball proves the first inequality; if the ball has zero ν -mass there is nothing to prove. The second uses $J_\nu \geq 2^{-p}$.

Apply Theorem 12.4 with $m = r = q = k$, $\ell = p = 2k$, $B = 2^p$, and $D = 2^{p+n} H$, using the ambient-center cap bound of Lemma 2.3. The angular margin is exactly $p - k - (k - 1) = 1$, and the radial margin is $n - p > 0$. Moreover $1 + \log^+(D/B) \leq C(d + 1 + \log H) \leq C(b_H + 2)$; the factor 2^p is absorbed in e^{Cd} . This proves (186). The local inverse factor behind this substitution is $\|u - s\|^{-k} \text{dist}(u - s, E)^{-k}$, where E is a linear subspace with $\dim E \leq k - 1$. $\square$

Proposition 12.6 (Drift of the kernel functional). *Let $S \sim \sigma$, let V have at most N values, and let a k -row Gaussian matrix A be independent of (S, V) . Suppose W has at most N values and is generated by a Borel probability kernel of (V, A, AS) and independent randomness. If $\log N \leq d^2$, then, for sufficiently large d ,*

$$\mathbb{E}\Phi(P_{S|W}) \leq \mathbb{E}\Phi(P_{S|V}) + Cd.$$

Proof. Condition first on $V = v$ with positive probability. Put $\nu = P_{S|v}$ and $H_v = P(V = v)^{-1}$. In this conditional experiment w always ranges over values with $P(V = v, W = w) > 0$; for these values put $\nu_w = P_{S|v,w}$ and $H_{vw} = P(V = v, W = w)^{-1}$. The two posteriors are bounded by $H_v\sigma$ and $H_{vw}\sigma$. Introduce a k -row Gaussian matrix A' independent of (S, V, A, W) , and put $X = (A; A')$, $Y = XS$. Within the conditional experiment X is independent of S , and W depends on S only through (X, Y) . The chain rule gives

$$I(S; W) = h(Y | X) - h(Y | X, W) - I(X; S | W). \quad (187)$$

For example, $I(S; W) = I(X, Y; W) - I(X; W | S)$, and $I(X; S) = 0$ converts the difference of the two row-information terms into $-I(X; S | W)$. All displayed differential entropies are finite by Lemma 12.2, and $I(X; S | W) \leq H(W)$.

For fixed w , use the finite nonzero measure $\rho_w = \nu_w / J_{\nu_w}$ as a reference, and write $a_w = \rho_w(S^{d-1})$. First normalize this reference. The relative entropy of the actual conditional law of (X, S) with respect to $P_{X|w} \otimes (\rho_w / a_w)$ is

$$I(X; S | w) + B(\nu_w) + \log a_w < \infty.$$

Indeed, $\log J_{\nu_w}$ is bounded for this v, w , and $I(X; S | w) < \infty$. Probability data processing under $(X, s) \mapsto (X, Xs)$ makes the corresponding post-map relative entropy finite. For probability laws $P \ll Q$, the elementary bound

$$\int_{\{dP/dQ < 1\}} -\log(dP/dQ) dP \leq 1/e$$

shows that a finite relative entropy has an integrable log density ratio. The post-map ratio is the actual conditional label density divided by $p_{\rho_w}(X, Y)/a_w$. Its logarithm is therefore integrable. Lemma 12.2 makes the logarithm of the actual conditional label density integrable as well, so $\log p_{\rho_w}(X, Y)$ is integrable under the actual conditional law. Here p_{ρ_w} is the cube-limit density version; the same construction applies to a finite measure.

We may now expand both finite-reference divergences and subtract $\log a_w$ from probability data processing. Before the map the divergence is $I(X; S | w) + B(\nu_w)$, while after the map it is $-h(Y | X, w) - \mathbb{E}[\log p_{\rho_w}(X, Y) | w]$. Consequently,

$$-h(Y | X, w) - I(X; S | w) \leq B(\nu_w) + \mathbb{E}[\log p_{\rho_w}(X, Y) | w]. \quad (188)$$

This is precisely the finite-reference data-processing statement proved in Section 2, including when ρ_w has mass greater than one.

We now bound the last logarithm under the selected actual rows. Let Q be the original Gaussian law of A , still conditional on v , and write $\kappa_w(a, as)$ for the Borel block probability of message w . For $r_w(s) = \int \kappa_w(a, as) Q(da) > 0$, choose the conditional row law

$$Q_{s,w}(da) = \frac{\kappa_w(a, as)}{r_w(s)} Q(da);$$

when $r_w(s) = 0$, assign $Q_{s,w} = Q$. The normalizer is positive for almost every (s, w) under the actual law, and this is a conditional version of the row law there. In particular $Q_{s,w} \ll Q$. Also

$$\mathbb{E}_{s,w} D(Q_{s,w} \| Q) = I(A; W | S) \leq H(W) \leq \log N,$$

so the divergence is finite for almost every (s, w) under that law. The fresh A' retains its independent Gaussian law after conditioning on (A, s, w) . For almost every such pair and for $\delta > 0$, Jensen in A' , followed by (2) with exponent k , gives

$$\begin{aligned} & \mathbb{E}[\log(1 + p_{\rho_w, \delta}(X, Xs)) \mid s, w] \\ & \leq \frac{D(Q_{s,w} \| Q)}{k} + \frac{1}{k} \log \mathbb{E}_{A \sim Q} (1 + \mathbb{E}_{A'} p_{\rho_w, \delta}(X, Xs))^k \\ & \leq \frac{D(Q_{s,w} \| Q)}{k} + Cd + \log(b_{H_{vw}} + 2). \end{aligned} \tag{189}$$

The last step uses the L^k triangle inequality and (186), whose fresh moment holds for every fixed $s \in S^{d-1}$. The displayed row-charge inequality is used only for almost every (s, w) under the actual law, while δ is positive.

Here is the exact null-set transfer in that passage. For each full-rank X , Lebesgue differentiation makes $p_{\rho_w, 1/a}(X, y)$ converge to $p_{\rho_w}(X, y)$ for Lebesgue-almost every y . The failure set is jointly Borel by the cube definitions. Under the actual conditional law given (v, w) , (X, S) is dominated by a finite multiple of $\gamma_p \otimes \sigma$; pushing forward shows that (X, Y) is absolutely continuous with respect to $\gamma_p(dX) dy$. Consequently it avoids this failure set, including after the actual rows have been tilted by w . Fatou applied to the nonnegative $\log(1 + p_{\rho_w, 1/a}(X, Y))$, after integrating (189) over s, w , proves the same integrated upper bound for $\log(1 + p_{\rho_w}(X, Y))$, and hence for $\log p_{\rho_w}$. This argument does not invoke an unsmoothed moment at ρ_w -almost every fixed signal.

The preceding estimate supplies the quantitative positive-log bound; integrability of both signs was established before (188). Substituting Lemma 12.3 and (188)–(189) into (187), and using $I(S; W) = \mathbb{E}_w D(\nu_w \| \sigma) - D(\nu \| \sigma)$, yields

$$\mathbb{E}_w \Phi(\nu_w) \leq \Phi(\nu) + Cd + 2 \log(b_{H_v} + 2) + \mathbb{E}_w \log(b_{H_{vw}} + 2) + \frac{\log N}{k}.$$

Restore the average over v . $\mathbb{E} \log H_V = H(V) \leq \log N$ and $\mathbb{E} \log H_{VW} = H(V, W) \leq 2 \log N$. Jensen bounds the averaged cutoff terms by $C \log(d + 1 + 2 \log N)$, while $\log N/k \leq Cd$. For fixed w , $P_{S|W=w}$ is the finite mixture of $\nu_w = P_{S|v,w}$ over v . Lemma 12.1 permits forgetting v , proving the drift. $\square$

12.3 Precision and the global stopping index

Corollary 12.7. *Let $M(d) = o(d^2)$ and $0 < \epsilon(d) \leq 1/10$. A learner in the model of Section 2 with uniform-sphere average angular success at least $2/3$ requires $T(d) \geq cd \log(1/\epsilon(d))$ for an absolute $c > 0$ in sufficiently large dimension. The threshold may depend on the memory sequence.*

Proof. Fix the shared rule randomness with uniform-prior success at least $2/3$, and apply Lemmas 2.2 and 2.1. Keep the internal transition and output randomness. Put $L = \log(1/\epsilon)$. Under the product of the marginal laws of $(S, \hat{S})$, angular success has probability at most ϵ^{d-1} , by Lemma 2.3. Equation (1) therefore gives

$$I(S; \hat{S}) \geq \frac{2}{3}(d-1)L - \log 2. \tag{190}$$

Assume $T < dL$. The other case is already sufficient. The terminal state and stopping index have at most $(T+1)2^M$ values, so the left side of (190) is at most $M \log 2 + \log(T+1)$. Since $\log(T+2) \leq C + \log d + \log(L+1) \leq C + \log d + L$, absorption gives

$$L \leq C \frac{M + 1 + \log d}{d}, \quad T = O(d + M).$$

For $M = o(d^2)$, pad stopped states with their original terminal labels and indices and add unused rows to complete the last block. Every padded layer has at most $N = (T + 2)2^M$ states and $\log N = o(d^2)$.

The initial conditional signal law is σ , because the model makes initialization independent of the signal conditional on the fixed shared rule randomness. Iterate Proposition 12.6 for $\lceil T/k \rceil$ blocks and apply Lemma 12.1. For the final padded state F ,

$$I(S; F) = \mathbb{E}D(P_{S|F} \| \sigma) \leq Cd(1 + \lceil T/k \rceil).$$

The output is generated from F and fresh randomness. Data processing with (190) gives $L \leq C(1 + T/d)$. The 2/3 instance of Lemma 2.3 gives $T > d/4$ for this same prior and exact observation model. Thus $1 + T/d \leq 5T/d$, proving the claimed sample bound. The global-clock calculation controls only the padded width; the precision-dependent conclusion comes from the block drift. $\square$

13 Maximum cell mass and inverse volumes at all scales

This section gives an independent proof of the uniform-sphere sample lower bound in Corollary 12.7. We track what the complete boundary-state history reveals about the signal, subtracting a correction built from local cell masses. The correction accounts for concentration already present in the conditional signal law. Its geometric input controls inverse simplex volumes without any bounded-support assumption.

After the model and stopping reductions below, Section 13.1 establishes coercivity of the correction, Section 13.2 proves the inverse-volume bound without a support restriction, Section 13.3 compares incoming and outgoing exact-observation fibers, and Section 13.4 completes the information-growth and decoding estimates.

The information identities below are the conditional relative-entropy chain rule; see [1, Section 3.3] for its standard Borel formulation. All fiber comparisons needed here are proved explicitly.

We use uniform probability σ on S^{d-1} for the signal, written U when random. All logarithms are natural except in dyadic scale definitions. Set $L = \log(1/\epsilon)$, and suppose $M = M(d) = o(d^2)$ and $0 < \epsilon \leq 1/10$. Work in sufficiently large dimension that $M \leq d^2$.

Assume uniform-prior angular success at least 2/3 in the model of Section 2. Fix the shared rule randomness with success strictly greater than 3/5. Lemmas 2.2 and 2.1 replace those fixed rules by everywhere Borel rules for this prior. Then fix the initial state and the independent transition and output seeds with success at least 3/5. The resulting deterministic program has the same state bound. Its output is determined by its terminal state and index.

We may assume $T < dL$. Otherwise the desired lower bound is immediate. There are at most $(T + 1)2^M$ possible output labels. By the cap bound of Lemma 2.3, the union of their successful signals has mass at most $(T + 1)2^M \epsilon^{d-1}$. Hence

$$\frac{3}{5} \leq (T + 1)2^M \epsilon^{d-1}.$$

Since $L \geq \log 10$ and $T < dL$, $\log(T + 1) \leq \log(2d) + L$. Taking logarithms gives $(d - 2)L \leq M \log 2 + \log(2d) + \log(5/3)$. Thus $L = O(d)$ and $T = O(d^2)$ in the case under consideration. The 3/5 endpoint in Lemma 2.3 also gives $T > d/20$. These reductions will bound the entropy of the complete boundary history; they do not supply the precision-dependent block estimate.

13.1 Selected cells and their entropy cost

Use the parameters

$$n = k = \lfloor d/100 \rfloor, \quad \alpha = (d-1)/2.$$

Group sample layers into blocks of size n , completing the last block with independent unused samples. In each block denote by A the n -by- d matrix of sample rows and by $b = AU$ the responses. We keep, for the analysis, a history consisting of the messages at all the block boundaries so far, starting with an empty history. Write V for the incoming history and $W = (V, Z)$ for the history after the next message.

The message records the ending program state or, if the program halted within the block, the layer of that block and final state; it can be a dummy message in subsequent blocks. An initial state is fixed. This scheme also permits recording if halted at the start. Thus conditional on V there are at most $(n+2)2^M + 1$ messages. Given V, A, b the next message can be formed, and A is fresh and independent of V, U . The final history determines the output.

There are $O(d)$ blocks by the preceding reduction. The message alphabet therefore bounds the entropy of every history by $O(d^3)$. This bound will control the scale selectors below. The complete history need not be available in program memory.

For a history V , let $p_v = \Pr(V = v)$, and write μ_v for the conditional law of U given v , with density f_v with respect to σ . Throughout we ignore conditioning states of probability zero. These densities exist and can be taken bounded by $1/p_v$, since the marginal law is σ . Write $I_V = I(U; V) = \mathbb{E} \log f_V(U)$; use the analogous notations for W . We will track information with a correction for local concentration: a log cell-mass quantity J_V below will be subtracted from I_V in bounding its growth per block.

Define levels $j \geq 0$ of cells of scale $r_j = 2^{1-j}$. Level 0 consists of the whole sphere. For $j \geq 1$ use the sphere intersected with a partition into half-open grid cubes of side $r_j/\sqrt{d}$. Denote the cell containing u by $C_j(u)$; each cell has diameter at most r_j and σ -measure at most $(2r_j)^{d-1}$. For any of our probability laws $\mu = \mu_v$ define

$$m_\mu(u) = \max_{j \geq 0} \frac{\mu(C_j(u))}{r_j^\alpha}.$$

In fact $m_\mu \geq 2^{-\alpha}$ and the terms for large j are bounded by $2^{d-1}r_j^{d-1-\alpha}/p_v$. Let $j_v(u)$ choose the first level giving the maximum. It is bounded for each v by

$$1 + \frac{\log_2(1/p_v) + d - 1 + \alpha}{d - 1 - \alpha}.$$

Together the selector and the selected cell define a partition for each v , with parts D consisting of points having the same selector and selected cell. When speaking of the random part D we include its identification by this choice (given v). Denote by $p_D = \mu_v(D)$ its probability, by $j = j_v(U)$ the chosen level, and by $p_C = \mu_v(C_j(U))$ the original cell probability. Then

$$\mathbb{E} \log(p_C/p_D) \leq H(j | V). \tag{191}$$

Indeed conditional on v, j , p_D is the selector probability given v times the probability of the cell given v, j . The expected log of its ratio with p_C , in the order in (191), equals the surprisal of the selector given v minus a nonnegative relative entropy for the cell distributions (at level j).

We record some finiteness and entropy bounds. By the bound on the selector and $\mathbb{E} \log(1/p_V) = H(V)$, Jensen's inequality gives $H(j | V) = O(\log d)$. Also $H(D | V) < \infty$; the log of the number of cells at level j meeting the sphere is $O(d(1+j))$. For this count, the disjoint cubes are within the

ambient ball of radius $1 + r_j$; the ambient unit ball has volume $\leq (C/\sqrt{d})^d$, as seen for example by integrating $e^{-\|x\|^2/2}$ on the ball of radius $\sqrt{d}$. The same entropy bounds apply with W in place of V . In addition, the bin

$$t = \lfloor \log_2 m_{\mu_W}(U) \rfloor$$

has $H(t | W) = O(\log d)$: the range for each w has size $O(d + \log(1/p_w))$, by the selector bound, the lower bound $2^{-\alpha}$ and $\mu_w(C_j) \leq 1$. The use of $O(\log d)$ here and above uses $H(V), H(W) \leq O(d^3)$.

Set $J_V = \mathbb{E} \log m_{\mu_V}(U)$. All such expectations are finite by the preceding bounds. Our objective on each block is

$$(I_W - J_W) - (I_V - J_V) \leq Cd.$$

We first show that the correction removes at most half the information, up to $O(d)$. Once the block estimate is proved, this coercivity will turn its iteration into a bound on the information in the final history.

Lemma 13.1 (Coercivity of the cell correction). *For the conditional laws and selected-cell partition just defined,*

$$J_V \leq \frac{\alpha}{d-1} I_V + O(d). \quad (192)$$

Proof. For the selected cell C ,

$$\log m_{\mu_V}(U) \leq \frac{\alpha}{d-1} \log \frac{p_C}{\sigma(C)} + \alpha \log 2,$$

using $\sigma(C) \leq (2r_j)^{d-1}$ and $\log p_C \leq 0$. The log ratio here is at most $\log(p_D/\sigma(D)) + \log(p_C/p_D)$. The expected first term is at most I_V by relative entropy under the partition for each v ; (191) bounds the other term. Since $\alpha/(d-1) = 1/2$, the correction removes at most half the information, up to an additive $O(d)$. $\square$

We will group points according to bounds on m_μ when taking projections. For any restriction of μ to a set with $m_\mu \leq m_0$, the resulting subprobability η satisfies

$$\eta(B(x, r)) \leq C^d m_0 r^\alpha \quad (x \in \mathbb{R}^d, r > 0), \quad (193)$$

where here $B(x, r)$ is an ambient ball. For $r \leq 1$, take a level $j \geq 1$ with $r \leq r_j \leq 2r$; a ball of radius r meets at most C^d cubes, by their side length and the volume bound applied to radius $r + r_j$. Each cube contributing mass has μ -mass at most $m_0 r_j^\alpha$. For $r > 1$ use the mass bound 1 and $m_0 \geq 2^{-\alpha}$ if the set is nonempty.

13.2 Inverse volumes at arbitrary length scales

Lemma 13.2 (Inverse volume without a support restriction). *Let $n = k = \lfloor d/100 \rfloor$ and $\alpha = (d-1)/2$, with d sufficiently large. Suppose a Borel probability ν on $\mathbb{R}^d$ satisfies $\nu(B(x, r)) \leq r^\alpha$ for all x, r . Fix any u , draw $z_1, \dots, z_k$ independently from ν , and denote by S the k -volume of the parallelepiped of $z_1 - u, \dots, z_k - u$ (the square root of the Gram determinant). There is an absolute constant C such that*

$$\mathbb{E} S^{-n} \leq \exp(Cdk). \quad (194)$$

Proof. The volume is positive almost surely, since ν gives zero mass to affine spaces of dimension $i < \alpha$. This last fact follows by covering any bounded portion by $O(r^{-i})$ balls of radius r as $r \downarrow 0$, and suffices by conditioning successively on the previous draws.

We order the vectors $z_i - u$ greedily, taking at each step one of largest distance to the span of those preceding. Every tuple admits such an ordering. By exchangeability, it suffices to bound the expectation restricted to the original order being greedy and multiply by $k!$; this also allows ties. In this order let δ_i be the distance to the preceding span, with the norm used for the first vector. Then $S = \prod_i \delta_i$, and the distances are nonincreasing. Fix their length bins

$$2^{-\ell_i-1} < \delta_i \leq 2^{-\ell_i}, \quad \ell_1 \leq \dots \leq \ell_k, \quad \ell_i \in \mathbb{Z}.$$

We will first bound the probability of these bins and the greedy conditions, then sum their contributions to the inverse volume.

Given the first $i-1$ draws, let e_j , $j < i$, be their translated vectors' successive Gram-Schmidt axes. Group these axes according to $h = \ell_j$. Greediness forces the projection of $z_i - u$ onto each group to have norm at most 2^{-h} : before the first axis in that group was chosen, the residual norm of $z_i - u$ was no greater than the distance chosen at that step. Its residual norm after all preceding axes is at most $2^{-\ell_i}$.

In a group of dimension a_h , cover the ball of radius 2^{-h} by a net of radius $2^{-\ell_i} 2^{-(\ell_i-h)}$. Euclidean ball packing bounds its cardinality by $(3 \cdot 2^{2(\ell_i-h)})^{a_h}$. The distinct group indices are integers $h \leq \ell_i$, so their squared net errors sum to at most

$$2^{-2\ell_i} \sum_{h \leq \ell_i} 2^{-2(\ell_i-h)} = \frac{4}{3} 2^{-2\ell_i}.$$

Including the orthogonal residual, all allowed points z_i therefore lie in at most $3^{i-1} 2^{2 \sum_{j < i} (\ell_i - \ell_j)}$ ambient balls of radius $2 \cdot 2^{-\ell_i}$. These necessary constraints involve only the preceding draws and the fixed bins. The local ball bound gives their conditional probability at most

$$3^{i-1} 2^\alpha 2^{2 \sum_{j < i} (\ell_i - \ell_j) - \alpha \ell_i}.$$

We may multiply these bounds sequentially, imposing only the necessary constraints just described and never conditioning preceding draws on future greedy choices. Use the local estimate when $\ell_i \geq 0$ and probability one otherwise. Put

$$P = \sum_i \max(\ell_i, 0), \quad N = \sum_i \max(-\ell_i, 0).$$

The double sum in the product exponent satisfies

$$\sum_{i: \ell_i \geq 0} \sum_{j < i} (\ell_i - \ell_j) \leq \sum_{i: \ell_i \geq 0} \left(k \ell_i + \sum_{j: \ell_j < 0} (-\ell_j) \right) \leq k(P + N).$$

Indeed, each inner sum is $(i-1)\ell_i - \sum_{j < i} \ell_j$. Its first term is at most $k\ell_i$, subtracting nonnegative ℓ_j can only decrease it, and each negative index contributes at most k times. The probability of the greedy and length conditions is consequently at most

$$\exp(Cdk) 2^{-(\alpha-2k)P+2kN}.$$

On these bins, $S^{-n} \leq 2^{nk} 2^{n(P-N)}$. When negative bins dominate, the large distances already make this inverse volume small; otherwise the local probability bound supplies the needed decay. For $N \leq 2P$, combining the two bounds gives the exponent

$$-(\alpha - 2k - n)P + (2k - n)N \leq -(\alpha - 6k - n)P - nN \leq -(P + N),$$

since $\alpha - 6k - n \geq 3$ and $n \geq 3$ for large d . For $N > 2P$, use probability at most one: then $n(P - N) \leq -(P + N)$, because $(n + 1)P \leq (n - 1)N$ for $n \geq 3$. Thus either region contributes at most $\exp(Cdk)2^{-(P+N)}$. The sum of $2^{-(P+N)}$, even over all integer tuples, is 3^k . This factor and $k!$ are absorbed by $\exp(Cdk)$, proving (194) at arbitrary length scales. $\square$

The inverse-volume estimate now controls a projection density at a prescribed target, even when that target lies outside the support of the projected measure.

Lemma 13.3 (A fixed-target Gaussian moment). *For a subprobability $\eta \ll \sigma$ on the sphere satisfying (193), let $q = \eta(S^{d-1}) > 0$. For Gaussian A write $g_{\eta,A}$ for the Lebesgue density of Az under η . Such densities exist almost surely since A has rank $n < d - 1$ and they exist for σ . We may evaluate a version by the liminf of mass densities on shrinking cubes (say radii $1/l$, $l \rightarrow \infty$), which gives the Lebesgue density almost everywhere by differentiation. With this latter evaluation, for every fixed u we have*

$$\mathbb{E}_A g_{\eta,A}(Au)^k \leq \left[\exp(Cd) q^{1-n/\alpha} m_0^{n/\alpha} \right]^k. \quad (195)$$

Proof. Take k independent draws from η/q . By (193), scaling the space (also scaling u) by $(C^d m_0/q)^{1/\alpha}$ makes (194) applicable. In the original space the bound for $\mathbb{E}S^{-n}$ is multiplied by $(C^d m_0/q)^{kn/\alpha}$. For fixed draws, the probability that all $\|A(z_i - u)\|_\infty \leq a$ is bounded by $(2a)^{nk} (2\pi)^{-nk/2} S^{-n}$, by the maximum Gaussian density in k coordinates for each row. The A -expectation of the cube mass to the power k is q^k times the draws' average of this event probability. Thus we get (195) for the mass divided by $(2a)^n$ in place of the density. Fatou's lemma proves it for the liminf. Constants use that n/α is bounded. $\square$

13.3 Comparing incoming and outgoing fibers

We return to V, W, A, b on the block and write $\beta = n/\alpha$. We use the same incoming selected part in both conditional signal laws. Its change in probability can then be charged to the information gained by the block, as we spell out below. Thus all parts D in this comparison use the partition from V ; let $r = r_j$ be the selected scale. Use these two densities evaluated at b :

- $g_{v,D,A}$ for the projection by A of μ_v restricted to D ;
- $g_{w,D,t,A}$ for the projection of μ_w restricted to $D \cap \{z : \lfloor \log_2 m_{\mu_w}(z) \rfloor = t\}$.

As random quantities the indices here take their actual values. Put $q_D = \mu_w(D)$, and denote by q_{Dt} this last restriction's mass. They are positive almost surely for the actual indices. Because D is determined by (V, U) and V is retained in $W = (V, Z)$, its probability change satisfies

$$\mathbb{E} \log(q_D/p_D) = I(D; Z \mid V) \leq I(U; Z \mid V) = I_W - I_V. \quad (196)$$

The projection bounds will attach the coefficient $1 - \beta$ to this term, leaving a positive fraction of the information gain to control.

At fixed full rank A , let $g_{0,A}$ be the projected density from σ , and let $\lambda_{A,b}$ be the corresponding conditional probability on the fiber at b . In orthonormal row-space coordinates the σ -projection has density $c_{d,n}(1 - \|x\|^2)^{(d-n-2)/2}$ in the unit ball, and the remaining component is conditionally uniform on its sphere. These facts follow, for example, by normalizing a standard Gaussian to generate σ and splitting it into the two orthogonal spaces; the squared lengths before normalization are independent chi-squares. Transforming the row-space coordinates to b multiplies the density by $\det(AA^\top)^{-1/2}$. The fiber probabilities can be taken measurably by adding to the fixed row-space component a suitably scaled, normalized Gaussian projection in the null space.

For a restricted density $f\mathbf{1}_E$ relative to σ , a projected density version is

$$g_{0,A}(b) \int f\mathbf{1}_E d\lambda_{A,b}$$

where $g_{0,A} > 0$, and zero outside. We use this version for conditional probability formulas. It agrees with the differentiation version almost everywhere for Lebesgue b at almost every A . The versions also agree when needed in the actual variables: the law of (A, b) is absolutely continuous with respect to Gaussian A times Lebesgue measure, and there are finitely many history choices and countably many sets used here.

We first lower bound the expected log of the incoming restricted projection density:

$$\mathbb{E} \log g_{v,D,A}(b) \geq (1 - \beta) \mathbb{E} \log p_D + \beta J_V - \beta H(j | V) - O(n). \quad (197)$$

Given V, D, A , the density of b is $g_{v,D,A}/p_D$, since A is independent of U, V . Take c to be the grid cube center (0 at level 0), so $\|U - c\| \leq r$. Compare this conditional density with a Gaussian of center Ac and covariance $r^2 I_n$. Nonnegativity of relative entropy gives

$$\mathbb{E} \log g_{v,D,A}(b) \geq \mathbb{E} [\log p_D - n \log r] - O(n).$$

Here $\mathbb{E} [\|A(U - c)\|^2 / r^2 | V, D] \leq n$. The identity

$$\log p_D - n \log r = (1 - \beta) \log p_D + \beta \log m_{\mu_v}(U) - \beta \log(p_C/p_D)$$

and the selector entropy bound (191) give (197).

Lemma 13.4 (Comparison on the exact observation fiber). *With the incoming selected part D , outgoing mass bin t , and density versions specified above,*

$$I_W - I_V \leq \mathbb{E} \log g_{w,D,t,A}(b) - \mathbb{E} \log g_{v,D,A}(b) + H(t | W, D). \quad (198)$$

Proof. Conditional on V, D, A, b, W , the density of U relative to $\lambda_{A,b}$ is

$$f_v \mathbf{1}_D g_{0,A}(b) / g_{v,D,A}(b),$$

since the message given V, A, b no longer depends on U . Compare on this fiber with the subprobability density

$$\sum_{t'} \frac{q_{Dt'}}{q_D} f_w \mathbf{1}_{D, t'} g_{0,A}(b) / g_{w,D,t',A}(b)$$

where zero-denominator terms are omitted, and the indicator restricts to the specified bin of m_{μ_w} as well as to D . Each included fiber density before its mixing weight has integral one. Omitting bins can only reduce the total mass, so the displayed mixture is a subprobability.

On the actual draw, the term of its bin has positive denominator almost surely. Indeed, a zero denominator with $g_{0,A}(b) > 0$ makes $f_w \mathbf{1}_{D, t'}$ zero $\lambda_{A,b}$ -almost everywhere. The actual conditional law is absolutely continuous on the fiber, and $f_w(U) > 0$ almost surely. The log ratio of the true density to the comparison density is therefore

$$\log \frac{f_v(U)}{f_w(U)} + \log \frac{g_{w,D,t,A}(b)}{g_{v,D,A}(b)} - \log \frac{q_{Dt}}{q_D}.$$

The last term averages to $H(t | W, D)$. Once integrability is verified, the nonnegative expectation of this log ratio gives (198), by Jensen on each fiber against the subprobability.

We verify the logarithmic integrability needed here and in (197). The log densities $\log f_V(U)$ and $\log f_W(U)$ are integrable by the state probability bounds and $x \log^- x \leq 1/e$ for densities relative to a probability. Both projection densities have integrable positive log parts: they are bounded above by $g_{0,A}/p_v$ or $g_{0,A}/p_w$, and

$$g_{0,A} \leq c_{d,n} \det(AA^\top)^{-1/2}$$

has integrable positive log part. For this last assertion, Gram–Schmidt on Gaussian rows gives squared successive distances with conditional chi-square laws and integrable logarithms.

For the incoming density $g_{v,D,A}$, the Gaussian comparison also bounds the negative log part. A log density ratio under its numerator law has expected negative part at most $1/e$ when compared to a probability. The Gaussian comparison density has integrable negative log part by $r \leq 2$ and the squared norm estimate, while $-\log p_D$ has finite expectation. Thus $\log g_{v,D,A}(b)$ is integrable.

The displayed fiber log ratio now has integrable positive part, using the outgoing positive log bound, the incoming integrability, and the finite expected surprisal for $t \mid W, D$. Its expected negative part is at most $1/e$, which remains true for comparison with a subprobability. The log ratio is therefore integrable, and its displayed decomposition also gives integrability of $\log g_{w,D,t,A}(b)$. This justifies (198). $\square$

We have compared the two fiber laws. The remaining step is to charge the correlation between the outgoing history and this fresh Gaussian block; the earlier blocks will carry no additional charge here.

To upper bound the target projection term, change the joint law Q of U, W, A to a reference R that keeps the joint U, W but makes A independent Gaussian. The relative entropy cost satisfies

$$D_{\text{KL}}(Q\|R) = I(A; U, W) = I(A; Z \mid U, V) \leq M \log 2 + O(\log d),$$

by freshness and the message alphabet. Under R , condition on U, W to fix all indices. Apply (195) to the indicated restriction of μ_w , using (193) with $m_0 = 2^{t+1} \leq 2m_{\mu_w}(U)$ and mass $q_{Dt} \leq q_D$. Thus the variable

$$X = \frac{g_{w,D,t,A}(AU)}{q_D^{1-\beta} m_{\mu_w}(U)^\beta}$$

has $\mathbb{E}_R X^k \leq \exp(Cdk)$. Here one can use the differentiation version in (195), which suffices also by version agreement under Q ; $\log X$ is integrable under Q by the bounds above (also $H(D \mid W) \leq H(D \mid V)$). The entropy bound $k\mathbb{E}_Q \log X \leq D_{\text{KL}}(Q\|R) + \log \mathbb{E}_R X^k$ follows from Jensen's inequality with the likelihood ratio. This is the locally justified entropy variational inequality (2); its unbounded logarithm is integrable by the preceding fiber calculation. Since k is proportional to d and $M \leq d^2$, we have proved

$$\mathbb{E} \log g_{w,D,t,A}(b) \leq (1 - \beta) \mathbb{E} \log q_D + \beta J_W + O(d). \quad (199)$$

Combining (197)–(199) and the selector and bin entropy bounds gives

$$I_W - I_V \leq (1 - \beta) \mathbb{E} \log(q_D/p_D) + \beta(J_W - J_V) + O(d).$$

Since $0 < \beta < 1$, the shared-part bound (196) can be absorbed on the left:

$$\beta(I_W - I_V) \leq \beta(J_W - J_V) + O(d).$$

The ratio $\beta = n/\alpha$ tends to $1/50$, so it is bounded away from zero. Dividing by it proves the block estimate

$$(I_W - J_W) - (I_V - J_V) \leq O(d). \quad (200)$$

Using incoming cells both in the incoming projection bound and in restricting the target is useful here: their probability change costs only a fraction of the information gain itself. Binning the target by local cell mass allows the random projection estimate without assuming an isotropic or single-scale conditional law.

13.4 Information growth and decoding

Initially $I_V = 0$ and $-J_V \leq \alpha \log 2$. By (200), and (192) with $\alpha/(d-1) = 1/2$, the final history has information at most

$$Cd(1 + \lceil T/n \rceil) \leq C'(d + T).$$

On the other hand compare its joint law with U to the product of their marginals on the event of chord error at most ϵ for its output. The joint probability is at least $3/5$, while the product probability is at most ϵ^{d-1} by the cap bound. Information, by relative entropy on this binary event, is at least $(3/5)(d-1)L - \log 2$. It follows that $T \geq cdL - C''d$ for absolute constants with $c > 0$. For large L this yields the desired bound directly, and for the remaining bounded L it follows by $T > d/20$ shown earlier.

All block projections and fiber laws used here are for the exact, noiseless data. The cost constraint used on a block message follows from the bit-state bound even when computation on the data within transitions is unrestricted. This completes this independent information proof of the sample lower bound.

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